
Part I: Linear Algebra[†]

Hao Jiang^{*}

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^{*}All remaining errors are my own.

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Introduction

Linear algebra is one of the basic mathematical languages of economics. Equilibrium conditions, linear approximations, least-squares projections, dynamic systems, Markov processes, and quadratic optimization all rely on linear-algebraic structure. For this reason, linear algebra will appear repeatedly throughout graduate microeconomics, macroeconomics, and econometrics.

Most of you have already worked with matrices, systems of linear equations, determinants, and eigenvalues. Our objective here is not simply to review these calculations. Instead, we will develop the structural ideas that explain why the calculations work and how the different parts of linear algebra fit together.

A central theme is the distinction between a mathematical object and its representation in coordinates. Vectors need not be columns of numbers, and linear maps need not be matrices. Once a basis is chosen, however, vectors acquire coordinates and linear maps acquire matrix representations. Thus we will repeatedly move between

abstract linear structure $\longleftrightarrow$ matrix representation.

This perspective allows many familiar matrix operations to be understood as manifestations of more general properties of linear maps.

We begin with vector spaces, linear combinations, linear independence, bases, and dimension. We then introduce linear maps and their matrix representations, followed by kernels, images, rank, and systems of linear equations. From there we study determinants, invariant subspaces, eigenvalues, eigenvectors, and diagonalization. Finally, inner products add geometric structure—length, orthogonality, and projection—and lead to the spectral theorem and quadratic forms.

1 Vector Spaces and Linear Relations of Vectors

1.1 Vector Spaces

The central object of linear algebra is the concept of vector spaces.

Definition 1.1 — Vector spaces

Let $\mathbb{K}$ be a field. A **vector space over $\mathbb{K}$** (or a **$\mathbb{K}$ -vector space**) is a set V equipped with two operations,

$$+ : V \times V \rightarrow V, \quad \cdot : \mathbb{K} \times V \rightarrow V,$$

called **vector addition** and **scalar multiplication**, respectively, such that for all $u, v, w \in V$ and all $\lambda, \mu \in \mathbb{K}$, the following eight axioms hold:

(1) **Associativity of vector addition:**

$$(u + v) + w = u + (v + w).$$

(2) **Commutativity of vector addition:**

$$u + v = v + u.$$

(3) **Existence of a zero vector:** there exists an element $0_V \in V$ such that

$$u + 0_V = 0_V + u = u.$$

(4) **Existence of additive inverses:** for every $u \in V$, there exists an element $-u \in V$ such that

$$u + (-u) = (-u) + u = 0_V.$$

(5) **Compatibility of scalar multiplication with multiplication in $\mathbb{K}$:**

$$(\lambda\mu)u = \lambda(\mu u).$$

(6) **Identity element of scalar multiplication:**

$$1_{\mathbb{K}}u = u,$$

where $1_{\mathbb{K}}$ denotes the multiplicative identity of $\mathbb{K}$.

(7) **Distributivity with respect to vector addition:**

$$\lambda(u + v) = \lambda u + \lambda v.$$

(8) **Distributivity with respect to scalar addition:**

$$(\lambda + \mu)u = \lambda u + \mu u.$$

We will not dive into the formal treatment of a field, for the interested audience, see the following remark.

Remark 1.2 — What is a field?

A **scalar field** is a set of elements (think of them as numbers) on which we can perform the usual arithmetic operations

$$+, \quad -, \quad \times, \quad \div,$$

and still remain within the set.

A field is usually denoted by $\mathbb{K}$ (from the German *Körper*) or by $\mathbb{F}$ (for *field*). For our purposes, it will suffice to take

$$\mathbb{K} = \mathbb{R} \quad \text{or} \quad \mathbb{K} = \mathbb{C}.$$

For a vector space $(V, +, \cdot)$, a subset W is called a **vector subspace** of $(V, +, \cdot)$ if $(W, +|_W, \cdot|_W)$ is a vector space, where $+|_W$ and $\cdot|_W$ are $+$ and $\cdot$ defined for V restricted in W .

Definition 1.3 — Vector subspaces

Let V be a vector space over a field $\mathbb{K}$, and let $W \subseteq V$. We say that W is a **vector subspace** of V , and write

$$W \leq V,$$

if W , equipped with the vector addition and scalar multiplication inherited from V , is itself a vector space over $\mathbb{K}$.

Equivalently, $W \leq V$ if and only if:

(1) $0_V \in W$;

(2) W is **closed under vector addition**:

$$u, v \in W \implies u + v \in W;$$

(3) W is **closed under scalar multiplication**:

$$u \in W, \lambda \in \mathbb{K} \implies \lambda u \in W.$$

Let us examine some examples of vector spaces.

Example 1.4 — Examples of vector spaces.

Let $\mathbb{K}$ be a field. The following are standard examples of vector spaces over $\mathbb{K}$.

(1) Finite-dimensional coordinate spaces.

The space

$$\mathbb{K}^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} : x_1, \dots, x_n \in \mathbb{K} \right\}$$

of n -dimensional column vectors is a vector space over $\mathbb{K}$.

Likewise, if we denote the space of row vectors by

$$\mathbb{K}_n = \left\{ (x_1 \ \cdots \ x_n) : x_1, \dots, x_n \in \mathbb{K} \right\},$$

then $\mathbb{K}_n$ is also a vector space over $\mathbb{K}$.

Addition and scalar multiplication are defined componentwise. For example,

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n),$$

and

$$\lambda(x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n).$$

(2) The space of sequences.

The set

$$\mathbb{K}^{\mathbb{N}} = \{(x_1, x_2, \dots) : x_i \in \mathbb{K} \text{ for every } i \in \mathbb{N}\}$$

of all sequences with values in $\mathbb{K}$ is a vector space over $\mathbb{K}$, with addition and scalar multiplication defined coordinatewise:

$$(x_i)_{i \in \mathbb{N}} + (y_i)_{i \in \mathbb{N}} = (x_i + y_i)_{i \in \mathbb{N}},$$

and

$$\lambda(x_i)_{i \in \mathbb{N}} = (\lambda x_i)_{i \in \mathbb{N}}.$$

(3) Polynomial spaces.

The set

$$\mathbb{K}[x] = \{a_0 + a_1x + \cdots + a_nx^n : n \in \mathbb{N}, a_0, \dots, a_n \in \mathbb{K}\}$$

of all polynomials with coefficients in $\mathbb{K}$ is a vector space over $\mathbb{K}$.

For any $n \in \mathbb{N}$, the subset

$$\mathbb{K}_n[x] = \{a_0 + a_1x + \cdots + a_nx^n : a_0, \dots, a_n \in \mathbb{K}\}$$

of polynomials of degree at most n is a vector subspace of $\mathbb{K}[x]$.

(4) Continuous functions.

The set

$$C([0, 1], \mathbb{K}) = \{f : [0, 1] \rightarrow \mathbb{K} : f \text{ is continuous}\}$$

is a vector space over $\mathbb{K}$, where

$$(f + g)(x) = f(x) + g(x)$$

and

$$(\lambda f)(x) = \lambda f(x).$$

(5) **Matrix spaces.**

The set

$$\mathbb{K}^{m \times n} = \{A = (a_{ij})_{i=1, \dots, m; j=1, \dots, n} : a_{ij} \in \mathbb{K}\}$$

of all $m \times n$ matrices with entries in $\mathbb{K}$ is a vector space over $\mathbb{K}$.

Addition and scalar multiplication are defined entrywise:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (\lambda A)_{ij} = \lambda a_{ij}.$$

Proposition 1.5 — Elementary properties of vector spaces

Let V be a vector space over a field $\mathbb{K}$. Then:

- (1) The zero vector 0_V is unique.
- (2) For every $v \in V$, the additive inverse $-v$ is unique.
- (3) For every $v \in V$ and $\lambda \in \mathbb{K}$,

$$0_{\mathbb{K}}v = 0_V, \quad \lambda 0_V = 0_V.$$

- (4) For every $v \in V$,

$$(-1_{\mathbb{K}})v = -v.$$

Hence vector subtraction may be defined by

$$u - v := u + (-v).$$

- (5) For every $\lambda \in \mathbb{K}$ and $v \in V$,

$$\lambda v = 0_V \implies \lambda = 0_{\mathbb{K}} \text{ or } v = 0_V.$$

Proof.

- (1) Suppose 0_V and $\tilde{0}_V$ are both zero vectors. Then

$$0_V = 0_V + \tilde{0}_V = \tilde{0}_V.$$

(2) Suppose w_1 and w_2 are both additive inverses of v . Then

$$w_1 = w_1 + (v + w_2) = (w_1 + v) + w_2 = 0_V + w_2 = w_2.$$

(3) By distributivity,

$$0_{\mathbb{K}}v = (0_{\mathbb{K}} + 0_{\mathbb{K}})v = 0_{\mathbb{K}}v + 0_{\mathbb{K}}v.$$

Cancelling $0_{\mathbb{K}}v$ gives

$$0_{\mathbb{K}}v = 0_V.$$

Similarly,

$$\lambda 0_V = \lambda(0_V + 0_V) = \lambda 0_V + \lambda 0_V,$$

and therefore

$$\lambda 0_V = 0_V.$$

(4) Since

$$v + (-1_{\mathbb{K}})v = (1_{\mathbb{K}} - 1_{\mathbb{K}})v = 0_{\mathbb{K}}v = 0_V,$$

the vector $(-1_{\mathbb{K}})v$ is the additive inverse of v . By uniqueness,

$$(-1_{\mathbb{K}})v = -v.$$

(5) Suppose $\lambda v = 0_V$. If $\lambda = 0_{\mathbb{K}}$, we are done. If $\lambda \neq 0_{\mathbb{K}}$, then λ^{-1} exists because $\mathbb{K}$ is a field. Hence

$$v = 1_{\mathbb{K}}v = \lambda^{-1}(\lambda v) = \lambda^{-1}0_V = 0_V.$$

Therefore either $\lambda = 0_{\mathbb{K}}$ or $v = 0_V$.

1.2 Linear Combinations of Vectors

The vector-space axioms allow us to add vectors and multiply them by scalars. By repeatedly applying these two operations, starting from a collection of vectors $v_1, \dots, v_n \in V$, we can construct vectors of the form

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n, \quad \lambda_1, \dots, \lambda_n \in \mathbb{K}.$$

Such vectors are called **linear combinations** of $v_1, \dots, v_n$.

Definition 1.6 — Linear combination

Let V be a vector space over a field $\mathbb{K}$, and let

$$v_1, \dots, v_n \in V.$$

A vector $v \in V$ is called a **linear combination** of $v_1, \dots, v_n$ if there exist scalars

$$\lambda_1, \dots, \lambda_n \in \mathbb{K}$$

such that

$$v = \lambda_1 v_1 + \dots + \lambda_n v_n = \sum_{i=1}^n \lambda_i v_i.$$

The scalars $\lambda_1, \dots, \lambda_n$ are called the **coefficients** of the linear combination.

We now ask what happens when the coefficients are allowed to vary over *all* possible values in $\mathbb{K}$. The resulting collection of vectors is called the **span** of $v_1, \dots, v_n$.

Definition 1.7 — Span

Let V be a vector space over a field $\mathbb{K}$, and let $v_1, \dots, v_n \in V$. The **span** of $v_1, \dots, v_n$ is the set of all their linear combinations:

$$\text{Span}(v_1, \dots, v_n) = \left\{ \lambda_1 v_1 + \dots + \lambda_n v_n = \sum_{i=1}^n \lambda_i v_i : \lambda_1, \dots, \lambda_n \in \mathbb{K} \right\}.$$

If

$$\text{Span}(v_1, \dots, v_n) = V,$$

we say that $v_1, \dots, v_n$ **span** V , or that they **generate** V .

The following proposition gives an equivalent definition of the span. Its proof uses a common technique used to establish set equality.

Proposition 1.8 — Span as the smallest subspace

Let V be a vector space over a field $\mathbb{K}$, and let $v_1, \dots, v_n \in V$. Then

$$\text{Span}(v_1, \dots, v_n) = \bigcap_{\substack{W \leq V \\ v_1, \dots, v_n \in W}} W,$$

where $W \leq V$ means that W is a vector subspace of V .

Equivalently, $\text{Span}(v_1, \dots, v_n)$ is the smallest vector subspace of V that contains $v_1, \dots, v_n$.

Proof.

Let

$$\mathcal{W} = \{W \leq V : v_1, \dots, v_n \in W\}$$

be the collection of all subspaces of V that contain $v_1, \dots, v_n$.

We prove the equality by showing both inclusions.

Left $\subseteq$ Right: Take

$$x \in \text{Span}(v_1, \dots, v_n).$$

By definition of span, there exist $\lambda_1, \dots, \lambda_n \in \mathbb{K}$ such that

$$x = \lambda_1 v_1 + \dots + \lambda_n v_n.$$

Now let $W \in \mathcal{W}$ be any subspace containing $v_1, \dots, v_n$. Since a subspace is closed under linear combinations,

$$\lambda_1 v_1 + \dots + \lambda_n v_n \in W.$$

Hence $x \in W$.

Because this is true for every $W \in \mathcal{W}$, we have

$$x \in \bigcap_{W \in \mathcal{W}} W.$$

Therefore,

$$\text{Span}(v_1, \dots, v_n) \subseteq \bigcap_{W \in \mathcal{W}} W.$$

Left $\supseteq$ Right: The span

$$\text{Span}(v_1, \dots, v_n)$$

is itself a subspace of V containing $v_1, \dots, v_n$. Therefore it is one of the subspaces in $\mathcal{W}$.

But the intersection of all sets in $\mathcal{W}$ must be contained in each particular set in $\mathcal{W}$. Hence

$$\bigcap_{W \in \mathcal{W}} W \subseteq \text{Span}(v_1, \dots, v_n).$$

Combining the two inclusions,

$$\text{Span}(v_1, \dots, v_n) = \bigcap_{W \in \mathcal{W}} W.$$

Equipped with the above proposition, it is clear that the span of $v_1, \dots, v_n$ is a vector subspace.

Let's see some examples.

Example 1.9 — Examples of spans

(1) The span of one nonzero vector in $\mathbb{R}^2$.

Let

$$v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Then

$$\text{Span}(v) = \left\{ \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix} : \lambda \in \mathbb{R} \right\}.$$

Thus $\text{Span}(v)$ is the line through the origin in the direction of v .

(2) Two vectors that span $\mathbb{R}^2$.

Let

$$v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then

$$\text{Span}(v_1, v_2) = \{ \lambda_1 v_1 + \lambda_2 v_2 : \lambda_1, \lambda_2 \in \mathbb{R} \}.$$

Since

$$\lambda_1 v_1 + \lambda_2 v_2 = \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix},$$

we obtain

$$\text{Span}(v_1, v_2) = \mathbb{R}^2.$$

(3) Two vectors that do not generate a larger space.

Let

$$v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 2 \\ 4 \end{pmatrix}.$$

Since

$$v_2 = 2v_1,$$

every linear combination satisfies

$$\lambda_1 v_1 + \lambda_2 v_2 = (\lambda_1 + 2\lambda_2)v_1.$$

Hence

$$\text{Span}(v_1, v_2) = \text{Span}(v_1).$$

So adding v_2 does not enlarge the span.

(4) Polynomial spaces.

In $\mathbb{K}[x]$,

$$\text{Span}(1, x, x^2) = \{ a_0 + a_1 x + a_2 x^2 : a_0, a_1, a_2 \in \mathbb{K} \}.$$

Therefore,

$$\text{Span}(1, x, x^2) = \mathbb{K}_2[x].$$

More generally,

$$\text{Span}(1, x, \dots, x^n) = \mathbb{K}_n[x].$$

(5) **A span of functions.**

In $C([0, 1], \mathbb{R})$, consider the functions

$$f_1(x) = 1, \quad f_2(x) = x.$$

Then

$$\text{Span}(f_1, f_2) = \{a + bx : a, b \in \mathbb{R}\}.$$

Thus the span consists of all affine functions on $[0, 1]$.

1.3 Linear Independence

The span of a collection of vectors tells us which vectors can be generated from them by taking linear combinations. A different question is whether those representations are *unique*.

Suppose that a vector v has two representations,

$$v = \lambda_1 v_1 + \dots + \lambda_n v_n = \mu_1 v_1 + \dots + \mu_n v_n.$$

Subtracting the two representations gives

$$(\lambda_1 - \mu_1)v_1 + \dots + (\lambda_n - \mu_n)v_n = 0_V.$$

Thus, uniqueness of the coefficients is determined entirely by the ways in which the zero vector can be represented as a linear combination of $v_1, \dots, v_n$. This motivates the notion of linear independence.

Definition 1.10 — Linear independence

Let V be a vector space over a field $\mathbb{K}$, and let $v_1, \dots, v_n \in V$.

The vectors $v_1, \dots, v_n$ are said to be **linearly independent** if

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0_V$$

implies

$$\lambda_1 = \dots = \lambda_n = 0_{\mathbb{K}}.$$

Otherwise, $v_1, \dots, v_n$ are said to be **linearly dependent**. Equivalently, they are linearly

dependent if there exist scalars $\lambda_1, \dots, \lambda_n \in \mathbb{K}$, not all zero, such that

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0_V.$$

Linear independence therefore means that none of the vectors carries information that can be reproduced by the others. The following proposition shows that it is also exactly the condition under which linear-combination representations are unique.

Proposition 1.11 — Uniqueness of representation

The vectors $v_1, \dots, v_n \in V$ are linearly independent if and only if every vector in

$$\text{Span}(v_1, \dots, v_n)$$

has a unique representation as a linear combination of $v_1, \dots, v_n$.

Proof.

Suppose first that $v_1, \dots, v_n$ are linearly independent, and that

$$\lambda_1 v_1 + \dots + \lambda_n v_n = \mu_1 v_1 + \dots + \mu_n v_n.$$

Moving everything to one side gives

$$(\lambda_1 - \mu_1)v_1 + \dots + (\lambda_n - \mu_n)v_n = 0_V.$$

By linear independence,

$$\lambda_i - \mu_i = 0_{\mathbb{K}} \quad \text{for every } i,$$

so

$$\lambda_i = \mu_i \quad \text{for every } i.$$

Hence the representation is unique.

Conversely, suppose every vector in the span has a unique representation. In particular, 0_V has the representation

$$0_V = 0_{\mathbb{K}}v_1 + \dots + 0_{\mathbb{K}}v_n.$$

If there were a nontrivial linear combination

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0_V,$$

then 0_V would have two different representations. Thus all λ_i must be zero, and $v_1, \dots, v_n$ are linearly independent.

There is another useful way to understand linear dependence: a dependent collection contains a

redundant vector.

Proposition 1.12 — Characterization of linear dependence

The vectors $v_1, \dots, v_n$ are linearly dependent if and only if at least one of them can be written as a linear combination of the others;

that is, for some i ,

$$v_i \in \text{Span}(v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_n).$$

Moreover, removing such a vector does not change the span.

Proof.

Suppose first that $v_1, \dots, v_n$ are linearly dependent. Then there exist $\lambda_1, \dots, \lambda_n \in \mathbb{K}$, not all zero, such that

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0_V.$$

Choose an index i for which $\lambda_i \neq 0_{\mathbb{K}}$. Because $\mathbb{K}$ is a field, λ_i^{-1} exists, and we can solve for v_i :

$$v_i = - \sum_{j \neq i} \frac{\lambda_j}{\lambda_i} v_j.$$

Thus v_i is a linear combination of the other vectors.

Conversely, suppose that for some i ,

$$v_i = \sum_{j \neq i} \lambda_j v_j.$$

Then

$$v_i - \sum_{j \neq i} \lambda_j v_j = 0_V$$

is a nontrivial linear relation, since the coefficient on v_i is 1. Hence the vectors are linearly dependent.

Finally, because v_i can already be generated by the remaining vectors, including v_i cannot enlarge their span. Therefore,

$$\text{Span}(v_1, \dots, v_n) = \text{Span}(v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_n).$$

The previous proposition formalizes the idea that linear dependence is *redundancy*. This leads to a fundamental counting principle: a collection generated by n vectors cannot contain more than n linearly independent directions.

Theorem 1.13 — Counting principle

Let

$$u_1, \dots, u_m, \quad v_1, \dots, v_n$$

be vectors in V . Suppose that

$$u_i \in \text{Span}(v_1, \dots, v_n) \quad \text{for every } i.$$

If

$$m > n,$$

then $u_1, \dots, u_m$ are linearly dependent.

Proof.

Suppose, toward a contradiction, that

$$u_1, \dots, u_m$$

are linearly independent.

Let

$$W = \text{Span}(v_1, \dots, v_n).$$

By assumption, every u_i belongs to W .

We now successively replace the vectors v_j by the vectors u_i without changing the span.

Because $u_1 \in W$, we can write

$$u_1 = a_1 v_1 + \dots + a_n v_n.$$

Since $u_1 \neq 0_V$, at least one coefficient is nonzero. Relabeling the v_j if necessary, suppose $a_1 \neq 0$.

We can solve for v_1 :

$$v_1 = \frac{1}{a_1} (u_1 - a_2 v_2 - \dots - a_n v_n).$$

Hence

$$W = \text{Span}(u_1, v_2, \dots, v_n).$$

Now consider u_2 . Since $u_2 \in W$,

$$u_2 = b_1 u_1 + b_2 v_2 + \dots + b_n v_n.$$

Because u_1, u_2 are linearly independent, u_2 cannot belong to $\text{Span}(u_1)$. Therefore at least one of

$$b_2, \dots, b_n$$

must be nonzero. We can solve for the corresponding v_j and replace it by u_2 . Thus, after

relabeling if necessary,

$$W = \text{Span}(u_1, u_2, v_3, \dots, v_n).$$

Continuing in this way, after n steps we obtain

$$W = \text{Span}(u_1, \dots, u_n).$$

But $m > n$, so u_{n+1} exists. Because $u_{n+1} \in W$,

$$u_{n+1} \in \text{Span}(u_1, \dots, u_n).$$

By Proposition 1.3.2, this means that

$$u_1, \dots, u_n, u_{n+1}$$

are linearly dependent, contradicting our assumption.

Therefore $u_1, \dots, u_m$ must be linearly dependent.

Remark 1.14

Theorem 1.3.1 can be summarized informally as:

“more independent vectors cannot be generated by fewer vectors.”

If m vectors are all generated by n vectors and $m > n$, some redundancy among the m vectors is unavoidable.

1.4 Basis

We have now developed two complementary ideas.

A collection of vectors that *spans* a space is large enough to generate every vector in that space. A collection of vectors that is *linearly independent* contains no redundancy: none of its vectors can be generated from the others.

A **basis** combines these two requirements. It is a collection that generates the whole space while containing no redundant vectors. As a consequence, a basis gives every vector in the space a unique system of coordinates.

Definition 1.15 — Basis

Let $W \leq V$ be a vector subspace. A finite collection

$$v_1, \dots, v_n \in W$$

is called a **basis** of W if

- (1) $v_1, \dots, v_n$ are linearly independent;
- (2) $v_1, \dots, v_n$ span W , that is,

$$W = \text{Span}(v_1, \dots, v_n).$$

Do different bases have the same size? The answer is yes for finite basis by the following theorem.

Theorem 1.16 — All finite bases have the same size

Let W be a vector space. If

$$u_1, \dots, u_m \quad \text{and} \quad v_1, \dots, v_n$$

are two finite bases of W , then

$$m = n.$$

Proof

Because

$$u_1, \dots, u_m$$

are linearly independent and

$$v_1, \dots, v_n$$

span W , [theorem 1.13](#) implies

$$m \leq n.$$

Reversing the roles of the two bases, $v_1, \dots, v_n$ are linearly independent and $u_1, \dots, u_m$ span W . Hence

$$n \leq m.$$

Therefore,

$$m = n.$$

[Theorem 1.16](#) shows that although a vector space may have many different bases, every finite basis contains the same number of vectors. This makes the following definition unambiguous.

Definition 1.17 — Dimension

Let W be a vector space that has a finite basis. The **dimension** of W , denoted by

$$\dim W,$$

is the number of vectors in any basis of W .

Thus, if

$$v_1, \dots, v_n$$

is a basis of W , then

$$\dim W = n.$$

If W has no finite basis, we say that W is **infinite-dimensional**.

There are two useful ways to recognize a basis. Starting from a linearly independent collection, a basis is one to which no new vector can be added without creating dependence. Starting from a spanning collection, a basis is one from which no vector can be removed without losing the spanning property.

Proposition 1.18 — Basis as a maximal linearly independent collection

Let

$$v_1, \dots, v_n \in W$$

be linearly independent. Then $v_1, \dots, v_n$ form a basis of W if and only if they are **maximal linearly independent**: no vector in W can be added to $v_1, \dots, v_n$ without making the resulting collection linearly dependent.

Proof

Suppose first that $v_1, \dots, v_n$ form a basis of W .

For every $w \in W$,

$$w \in \text{Span}(v_1, \dots, v_n).$$

Therefore

$$v_1, \dots, v_n, w$$

are linearly dependent. Hence no additional vector can be added while preserving linear independence.

Conversely, suppose $v_1, \dots, v_n$ are maximal linearly independent. If they did not span W , there would exist

$$w \in W \setminus \text{Span}(v_1, \dots, v_n).$$

But then

$$v_1, \dots, v_n, w$$

would still be linearly independent, contradicting maximality.

Therefore

$$W = \text{Span}(v_1, \dots, v_n),$$

so $v_1, \dots, v_n$ form a basis of W .

Proposition 1.19 — Basis as a minimal spanning collection

Let

$$v_1, \dots, v_n$$

span W . Then $v_1, \dots, v_n$ form a basis of W if and only if they are **minimal spanning**: removing any one of the vectors causes the remaining vectors to no longer span W .

Proof

Suppose first that $v_1, \dots, v_n$ form a basis of W .

If one vector, say v_i , could be removed without changing the span, then

$$v_i \in \text{Span}(v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_n).$$

Hence v_i would be a linear combination of the other vectors, contradicting linear independence.

Conversely, suppose $v_1, \dots, v_n$ span W and are minimal spanning.

If they were linearly dependent, one of the vectors could be expressed as a linear combination of the others. That vector could then be removed without changing the span, contradicting minimality.

Hence $v_1, \dots, v_n$ are linearly independent. Since they also span W , they form a basis.

Remark 1.20 — Maximal versus maximum

The word **maximal** in [proposition 1.18](#) means that the collection cannot be enlarged while retaining linear independence. It does not mean that we first compare the cardinalities of all linearly independent collections and choose one with the largest size.

Similarly, **minimal spanning** means that no vector can be removed while preserving the span.

Once the dimension of a vector space is known, checking that a collection is a basis becomes particularly easy. If we already have exactly the right number of vectors, it is enough to check either linear independence or spanning; we do not need to verify both.

Proposition 1.21 — The right number of vectors is enough

Suppose

$$\dim W = n.$$

Then:

- (1) any n linearly independent vectors in W form a basis of W ;
- (2) any n vectors that span W form a basis of W .

Proof

For (1), let

$$v_1, \dots, v_n$$

be linearly independent.

If they did not span W , there would exist

$$w \in W \setminus \text{Span}(v_1, \dots, v_n).$$

Then

$$v_1, \dots, v_n, w$$

would be $n + 1$ linearly independent vectors in W .

But any basis of W contains only n vectors, contradicting [theorem 1.13](#). Therefore

$$W = \text{Span}(v_1, \dots, v_n),$$

so the vectors form a basis.

For (2), let

$$v_1, \dots, v_n$$

span W .

If they were linearly dependent, one of them could be removed without changing the span. We would then have a spanning collection containing only $n - 1$ vectors.

But a basis of W contains n linearly independent vectors, again contradicting [theorem 1.13](#). Therefore $v_1, \dots, v_n$ are linearly independent and hence form a basis.

Remark 1.22 — Rank of a set

If one wishes to define the rank of an arbitrary subset $S \subseteq V$, the natural definition is

$$\text{Rank}(S) := \dim \text{Span}(S),$$

provided that $\text{Span}(S)$ is finite-dimensional.

Thus the rank measures the number of linearly independent directions generated by S .

Example 1.23 — Standard bases

(1) Coordinate space.

The vectors

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

form the **canonical basis** of $\mathbb{K}^n$.

Hence

$$\dim \mathbb{K}^n = n.$$

(2) Polynomial space.

The polynomials

$$1, x, x^2, \dots, x^n$$

form a basis of

$$\mathbb{K}_n[x].$$

Therefore,

$$\dim \mathbb{K}_n[x] = n + 1.$$

(3) Matrix space.

For

$$1 \leq i \leq m, \quad 1 \leq j \leq n,$$

let E_{ij} denote the $m \times n$ matrix whose (i, j) -entry is 1 and whose other entries are zero.

Then

$$\{E_{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$$

is a basis of $\mathbb{K}^{m \times n}$.

Hence

$$\dim \mathbb{K}^{m \times n} = mn.$$

(4) An infinite-dimensional space.

The polynomial space

$$\mathbb{K}[x]$$

is infinite-dimensional.

Indeed, for every n ,

$$1, x, x^2, \dots, x^n$$

are linearly independent. Thus $\mathbb{K}[x]$ contains linearly independent collections of arbitrarily large finite size, so it cannot have a finite basis.

1.5 Change of Basis

A basis is a coordinate system for a vector space. The coordinate is unique for each vector by proposition 1.11

Definition 1.24 — Coordinate vector.

Let

$$\mathcal{B} = (v_1, \dots, v_n)$$

be a basis of an n -dimensional vector space V over $\mathbb{K}$.

For every $x \in V$, there exist unique scalars $x_1, \dots, x_n \in \mathbb{K}$ such that

$$x = x_1 v_1 + \dots + x_n v_n.$$

The vector

$$[x]_{\mathcal{B}} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

is called the **coordinate vector of x with respect to $\mathcal{B}$** .

Now suppose that

$$\mathcal{E} = (e_1, \dots, e_n) \quad \text{and} \quad \mathcal{F} = (f_1, \dots, f_n)$$

are two bases of an n -dimensional vector space V .

Because $\mathcal{E}$ is a basis, each vector f_j can be written uniquely as a linear combination of $e_1, \dots, e_n$.

Thus there exist scalars $p_{ij} \in \mathbb{K}$ such that

$$\begin{cases} f_1 = p_{11}e_1 + p_{21}e_2 + \dots + p_{n1}e_n, \\ f_2 = p_{12}e_1 + p_{22}e_2 + \dots + p_{n2}e_n, \\ \vdots \\ f_n = p_{1n}e_1 + p_{2n}e_2 + \dots + p_{nn}e_n. \end{cases}$$

Collecting these equations into matrix form gives

$$(f_1, \dots, f_n) = (e_1, \dots, e_n) \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{pmatrix}.$$

Denote the matrix above by

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{pmatrix}.$$

Then the change of basis can be written simply as

$$(f_1, \dots, f_n) = (e_1, \dots, e_n)P.$$

Definition 1.25 — Transition matrix

Let

$$\mathcal{E} = (e_1, \dots, e_n) \quad \text{and} \quad \mathcal{F} = (f_1, \dots, f_n)$$

be two bases of V .

If

$$(f_1, \dots, f_n) = (e_1, \dots, e_n)P,$$

then P is called the **transition matrix from the basis $\mathcal{E}$ to the basis $\mathcal{F}$** .

The meaning of P is particularly simple. Its j -th column is the coordinate vector of f_j with respect to the basis $\mathcal{E}$:

$$P = ([f_1]_{\mathcal{E}}, [f_2]_{\mathcal{E}}, \dots, [f_n]_{\mathcal{E}}).$$

Thus, to find the transition matrix from $\mathcal{E}$ to $\mathcal{F}$, we express each vector of the new basis $\mathcal{F}$ in terms of the old basis $\mathcal{E}$, and place the resulting coordinate vectors into the columns of P .

Proposition 1.26 — The transition matrix is invertible

The transition matrix between two bases is invertible. Moreover, if

$$(f_1, \dots, f_n) = (e_1, \dots, e_n)P,$$

then

$$(e_1, \dots, e_n) = (f_1, \dots, f_n)P^{-1}.$$

Hence the transition matrix from $\mathcal{F}$ back to $\mathcal{E}$ is P^{-1} .

Proof

Since $f_1, \dots, f_n$ form a basis of V , each e_j can be written uniquely as a linear combination of

$f_1, \dots, f_n$. Therefore, there exists an $n \times n$ matrix Q such that

$$(e_1, \dots, e_n) = (f_1, \dots, f_n)Q.$$

Using

$$(f_1, \dots, f_n) = (e_1, \dots, e_n)P,$$

we obtain

$$(e_1, \dots, e_n) = (e_1, \dots, e_n)PQ.$$

Write

$$PQ = (a_{ij})_{i,j=1}^n.$$

Then the j -th column of the equation above gives

$$e_j = a_{1j}e_1 + \dots + a_{nj}e_n.$$

But since $e_1, \dots, e_n$ are linearly independent, the representation of e_j in this basis is unique. Hence

$$a_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Therefore,

$$PQ = I_n.$$

Similarly, substituting

$$(e_1, \dots, e_n) = (f_1, \dots, f_n)Q$$

into

$$(f_1, \dots, f_n) = (e_1, \dots, e_n)P$$

gives

$$(f_1, \dots, f_n) = (f_1, \dots, f_n)QP.$$

By uniqueness of representation with respect to the basis $f_1, \dots, f_n$,

$$QP = I_n.$$

Thus

$$Q = P^{-1}.$$

Hence P is invertible, and

$$(e_1, \dots, e_n) = (f_1, \dots, f_n)P^{-1}.$$

We now turn from the change of the *basis vectors* to the corresponding change of the *coordinates of a vector*.

Let $x \in V$. Suppose that the coordinates of x with respect to $\mathcal{E}$ and $\mathcal{F}$ are

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix},$$

respectively. Thus

$$x = x_1 e_1 + \cdots + x_n e_n = y_1 f_1 + \cdots + y_n f_n.$$

In matrix notation,

$$x = (e_1, \dots, e_n)X$$

and

$$x = (f_1, \dots, f_n)Y.$$

Since

$$(f_1, \dots, f_n) = (e_1, \dots, e_n)P,$$

we have

$$(e_1, \dots, e_n)X = (e_1, \dots, e_n)PY.$$

Because $e_1, \dots, e_n$ are linearly independent,

$$X = PY.$$

Equivalently,

$$Y = P^{-1}X.$$

Proposition 1.27 — Coordinate transformation formula

Suppose

$$(f_1, \dots, f_n) = (e_1, \dots, e_n)P.$$

If a vector $x \in V$ has coordinate vectors

$$X = [x]_{\mathcal{E}}, \quad Y = [x]_{\mathcal{F}},$$

then

$$X = PY, \quad Y = P^{-1}X.$$

Remark 1.28 — Basis vectors and coordinates move in opposite directions

It is important to distinguish the change of the basis from the change of coordinates.

The basis transformation is

$$(f_1, \dots, f_n) = (e_1, \dots, e_n)P,$$

whereas the coordinates satisfy

$$[x]_{\mathcal{E}} = P[x]_{\mathcal{F}}.$$

Therefore, when we change from the $\mathcal{E}$ -basis to the $\mathcal{F}$ -basis, the coordinate vector transforms by the inverse matrix:

$$[x]_{\mathcal{F}} = P^{-1}[x]_{\mathcal{E}}.$$

2 Linear Maps and Matrices

So far, we have studied the structure of a vector space itself: linear combinations, subspaces, linear independence, bases, and coordinates. A choice of basis allows us to describe abstract vectors by coordinate vectors in $\mathbb{K}^n$.

We now turn from studying *vectors* to studying *maps between vector spaces*.

Among all possible maps between vector spaces, the ones that are compatible with the vector-space structure are the **linear maps**. As we will see, once bases are chosen, every linear map can be represented by a matrix. Conversely, every matrix describes a linear map between coordinate spaces.

Thus there are two equivalent ways of describing the same object:



The bridge between these two descriptions is the choice of bases.

2.1 Linear Maps

We first define what it means for a map to preserve the linear structure of vector spaces.

Let V and W be vector spaces over the same field $\mathbb{K}$. A general map

$$T : V \rightarrow W$$

assigns to every vector $v \in V$ a vector $T(v) \in W$.

For such a map to respect the vector-space structure, it should be compatible with the two operations that define a vector space: vector addition and scalar multiplication.

Definition 2.1 — Linear map

Let V and W be vector spaces over the same field $\mathbb{K}$. A map

$$T : V \rightarrow W$$

is called a **linear map** if, for every $u, v \in V$ and every $\lambda \in \mathbb{K}$,

$$T(u + v) = T(u) + T(v)$$

and

$$T(\lambda v) = \lambda T(v).$$

Equivalently, T is linear if and only if

$$T(\lambda u + \mu v) = \lambda T(u) + \mu T(v)$$

for every $u, v \in V$ and every $\lambda, \mu \in \mathbb{K}$.

Example 2.2 — Examples of linear maps

Let V and W be vector spaces over the same field $\mathbb{K}$.

(1) **Zero map.**

$$0 : V \rightarrow W, \quad v \mapsto 0_W,$$

(2) **Identity map.**

$$I_V : V \rightarrow V, \quad v \mapsto v,$$

(3) **Scaling.**

Fix $c \in \mathbb{K}$. Define

$$T : V \rightarrow V, \quad T(v) = cv.$$

(4) **A linear map between coordinate spaces.**

Define

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ 2x - y \\ 3y \end{pmatrix}.$$

For

$$u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \quad v = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix},$$

we have

$$T(\lambda u + \mu v) = \lambda T(u) + \mu T(v),$$

so T is linear.

Remark 2.3 — Linear transformation

A linear map

$$T : V \rightarrow V$$

from a vector space to itself is called a **linear transformation** of V .

Thus the zero map $0 : V \rightarrow V$, the identity map I_V , and the scaling map

$$v \mapsto cv$$

are all linear transformations of V .

Proposition 2.4 — Linear maps preserve linear combinations

If $T : V \rightarrow W$ is linear, then for any $v_1, \dots, v_n \in V$ and $\lambda_1, \dots, \lambda_n \in \mathbb{K}$,

$$T(\lambda_1 v_1 + \dots + \lambda_n v_n) = \lambda_1 T(v_1) + \dots + \lambda_n T(v_n).$$

Proof

Follow the definition and use induction.

2.2 The Vector Space of Linear Maps

Once V and W are fixed, we can consider not just one linear map $T : V \rightarrow W$, but the collection of *all* linear maps from V to W . This collection itself has a natural vector-space structure.

Definition 2.5 — The space $\mathcal{L}(V, W)$

Let V and W be vector spaces over the same field $\mathbb{K}$. We denote by

$$\mathcal{L}(V, W)$$

the set of all linear maps from V to W :

$$\mathcal{L}(V, W) = \{T : V \rightarrow W : T \text{ is linear}\}.$$

For $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{K}$, define addition and scalar multiplication **pointwise** by

$$(S + T)(v) := S(v) + T(v), \quad v \in V,$$

and

$$(\lambda T)(v) := \lambda T(v), \quad v \in V.$$

The word *pointwise* means that to add two linear maps, we simply apply both maps to the same vector and add the resulting vectors in W .

Proposition 2.6 — $\mathcal{L}(V, W)$ is a vector space

With the operations defined above,

$$\mathcal{L}(V, W)$$

is a vector space over $\mathbb{K}$.

Proof

We first check that the operations stay inside $\mathcal{L}(V, W)$.

Let $S, T \in \mathcal{L}(V, W)$. For $u, v \in V$ and $\alpha, \beta \in \mathbb{K}$,

$$\begin{aligned} (S + T)(\alpha u + \beta v) &= S(\alpha u + \beta v) + T(\alpha u + \beta v) \\ &= \alpha S(u) + \beta S(v) + \alpha T(u) + \beta T(v) \\ &= \alpha(S + T)(u) + \beta(S + T)(v). \end{aligned}$$

Thus $S + T$ is linear.

Similarly, for $\lambda \in \mathbb{K}$,

$$\begin{aligned} (\lambda T)(\alpha u + \beta v) &= \lambda T(\alpha u + \beta v) \\ &= \alpha(\lambda T)(u) + \beta(\lambda T)(v), \end{aligned}$$

so λT is also linear.

The remaining vector-space axioms follow pointwise from the corresponding axioms in W . For example,

$$((R + S) + T)(v) = R(v) + S(v) + T(v) = (R + (S + T))(v)$$

for every $v \in V$.

The zero vector of $\mathcal{L}(V, W)$ is the **zero map**

$$0 : V \rightarrow W, \quad 0(v) = 0_W,$$

and the additive inverse of T is the map $-T$ defined by

$$(-T)(v) = -T(v).$$

Hence $\mathcal{L}(V, W)$ is a vector space.

Let's see an example.

Example 2.7 — Linear combinations of linear maps

Consider $S, T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$S(x, y) = (x, 0), \quad T(x, y) = (0, y).$$

Both are linear.

For $\alpha, \beta \in \mathbb{R}$,

$$(\alpha S + \beta T)(x, y) = \alpha S(x, y) + \beta T(x, y) = (\alpha x, \beta y).$$

Thus varying α and β generates a family of linear maps in $\mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$.

2.3 Composition of Linear Maps

Besides adding and scaling linear maps, we can also apply one linear map after another.

Suppose

$$T : U \rightarrow V, \quad S : V \rightarrow W$$

are linear maps. Starting from $u \in U$, we first apply T and then apply S :

$$u \xrightarrow{T} T(u) \xrightarrow{S} S(T(u)).$$

Graphically,

$$U \xrightarrow{T} V \xrightarrow{S} W.$$

ST

Definition 2.8 — Composition of linear maps

Let

$$T : U \rightarrow V, \quad S : V \rightarrow W$$

be linear maps. Their **composition** is the map

$$S \circ T : U \rightarrow W$$

defined by

$$(S \circ T)(u) = S(T(u)).$$

We will also use the product notation

$$ST := S \circ T.$$

Thus,

$$(ST)(u) = S(T(u)).$$

Proposition 2.9 — Composition preserves linearity

If

$$T : U \rightarrow V, \quad S : V \rightarrow W$$

are linear, then

$$ST : U \rightarrow W$$

is linear.

Proof

For $u, v \in U$ and $\lambda, \mu \in \mathbb{K}$,

$$\begin{aligned}(ST)(\lambda u + \mu v) &= S(T(\lambda u + \mu v)) \\ &= S(\lambda T(u) + \mu T(v)) \\ &= \lambda S(T(u)) + \mu S(T(v)) \\ &= \lambda(ST)(u) + \mu(ST)(v).\end{aligned}$$

Hence ST is linear.

Composition of linear maps behaves much like multiplication: it is associative and distributes over addition, but it is generally not commutative.

Proposition 2.10 — Properties of composition

Whenever the compositions are well-defined:

(1) **Associativity:** if

$$T : U \rightarrow V, \quad S : V \rightarrow W, \quad R : W \rightarrow X,$$

then

$$R(ST) = (RS)T.$$

(2) **Distributivity:**

$$(S_1 + S_2)T = S_1T + S_2T,$$

and

$$S(T_1 + T_2) = ST_1 + ST_2.$$

(3) **Compatibility with scalar multiplication:**

$$(\lambda S)T = \lambda(ST) = S(\lambda T).$$

(4) **Identity:**

$$I_V T = T = T I_U.$$

Remark 2.11 — Composition is not commutative

The order of composition matters.

For general maps

$$T : U \rightarrow V, \quad S : V \rightarrow W,$$

the composition

$$ST : U \rightarrow W$$

is defined, whereas TS need not even be defined.

Even when

$$S, T : V \rightarrow V,$$

so that both compositions exist, in general

$$ST \neq TS.$$

Example 2.12 — Composition need not commute

Define

$$S, T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

by

$$T(x, y) = (x, 0), \quad S(x, y) = (y, x).$$

Then

$$(ST)(x, y) = S(T(x, y)) = S(x, 0) = (0, x),$$

whereas

$$(TS)(x, y) = T(S(x, y)) = T(y, x) = (y, 0).$$

Hence

$$ST \neq TS.$$

The difference can be seen directly from the order in which the maps are applied:

$$(x, y) \xrightarrow{T} (x, 0) \xrightarrow{S} (0, x)$$

whereas

$$(x, y) \xrightarrow{S} (y, x) \xrightarrow{T} (y, 0).$$

Remark 2.13 — The algebra of linear transformations

When the domain and codomain are the same vector space,

$$\mathcal{L}(V, V),$$

usually abbreviated as

$$\mathcal{L}(V),$$

has two kinds of structure.

First, $\mathcal{L}(V)$ is a vector space under addition and scalar multiplication. Second, any two linear transformations

$$S, T \in \mathcal{L}(V)$$

can be composed, giving

$$ST \in \mathcal{L}(V).$$

Composition is associative and distributes over addition:

$$R(ST) = (RS)T,$$

$$(S + T)R = SR + TR, \quad R(S + T) = RS + RT.$$

It is also compatible with scalar multiplication, and the identity map I_V serves as a multiplicative identity.

Thus,

$$\mathcal{L}(V) \text{ is an associative algebra over } \mathbb{K}.$$

In general this algebra is not commutative, since one may have

$$ST \neq TS.$$

2.4 Matrix of a Linear Map

A linear map is completely determined by what it does to a basis.

Indeed, let

$$T : V \rightarrow W$$

be linear, and let

$$\mathcal{E} = (e_1, \dots, e_n)$$

be a basis of V . Every vector $v \in V$ can be written uniquely as

$$v = x_1 e_1 + \dots + x_n e_n.$$

By linearity,

$$T(v) = x_1 T(e_1) + \dots + x_n T(e_n).$$

Thus, once we know

$$T(e_1), \dots, T(e_n),$$

we know $T(v)$ for every $v \in V$.

Proposition 2.14 — A linear map is determined by its values on a basis

Let $\mathcal{E} = (e_1, \dots, e_n)$ be a basis of V .

If two linear maps

$$S, T : V \rightarrow W$$

satisfy

$$S(e_i) = T(e_i), \quad i = 1, \dots, n,$$

then

$$S = T.$$

Proof

Take any $v \in V$. Since $\mathcal{E}$ is a basis, there exist unique scalars $x_1, \dots, x_n \in \mathbb{K}$ such that

$$v = x_1 e_1 + \dots + x_n e_n.$$

By linearity,

$$S(v) = x_1 S(e_1) + \dots + x_n S(e_n),$$

and

$$T(v) = x_1 T(e_1) + \dots + x_n T(e_n).$$

Since $S(e_i) = T(e_i)$ for every i ,

$$S(v) = T(v).$$

Because this holds for every $v \in V$, we conclude that

$$S = T.$$

This observation suggests a simple way to record a linear map. We only need to record the images of the basis vectors of V .

Let

$$\mathcal{E} = (e_1, \dots, e_n)$$

be a basis of V , and let

$$\mathcal{F} = (f_1, \dots, f_m)$$

be a basis of W .

Since $T(e_j) \in W$, each $T(e_j)$ can be written uniquely in the basis $\mathcal{F}$. Thus

$$\begin{cases} T(e_1) = a_{11}f_1 + a_{21}f_2 + \cdots + a_{m1}f_m, \\ T(e_2) = a_{12}f_1 + a_{22}f_2 + \cdots + a_{m2}f_m, \\ \vdots \\ T(e_n) = a_{1n}f_1 + a_{2n}f_2 + \cdots + a_{mn}f_m. \end{cases}$$

Collecting the coefficients and taking the transpose gives

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

Equivalently, the entire system can be written compactly as

$$(T(e_1), \dots, T(e_n)) = (f_1, \dots, f_m)A.$$

Definition 2.15 — Matrix of a linear map

Let

$$T : V \rightarrow W$$

be linear, with

$$\mathcal{E} = (e_1, \dots, e_n)$$

a basis of V and

$$\mathcal{F} = (f_1, \dots, f_m)$$

a basis of W .

If

$$(T(e_1), \dots, T(e_n)) = (f_1, \dots, f_m)A,$$

then the $m \times n$ matrix A is called the **matrix of T with respect to the bases $\mathcal{E}$ and $\mathcal{F}$** .

We denote it by

$$[T]_{\mathcal{F} \leftarrow \mathcal{E}}$$

or simply by $[T]$ when the bases are clear from context.

The construction is worth remembering column by column:

$$j\text{-th column of } [T]_{\mathcal{F} \leftarrow \mathcal{E}} = [T(e_j)]_{\mathcal{F}}.$$

Thus the number of columns is determined by the dimension of the *domain*, while the number of rows is determined by the dimension of the *codomain*:

$$T : V^n \rightarrow W^m \quad \implies \quad [T]_{\mathcal{F} \leftarrow \mathcal{E}} \text{ is } m \times n.$$

Example 2.16 — A map from $\mathbb{R}^2$ to $\mathbb{R}^3$

Let

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

be defined by

$$T(x, y) = (x + 3y, 2x + 5y, 7x + 9y).$$

Take the standard bases

$$\mathcal{E} = (e_1, e_2)$$

of $\mathbb{R}^2$ and

$$\mathcal{F} = (f_1, f_2, f_3)$$

of $\mathbb{R}^3$.

We have

$$T(e_1) = T(1, 0) = (1, 2, 7) = f_1 + 2f_2 + 7f_3,$$

and

$$T(e_2) = T(0, 1) = (3, 5, 9) = 3f_1 + 5f_2 + 9f_3.$$

Therefore,

$$(T(e_1), T(e_2)) = (f_1, f_2, f_3) \begin{pmatrix} 1 & 3 \\ 2 & 5 \\ 7 & 9 \end{pmatrix},$$

and hence

$$[T]_{\mathcal{F} \leftarrow \mathcal{E}} = \begin{pmatrix} 1 & 3 \\ 2 & 5 \\ 7 & 9 \end{pmatrix}.$$

So far the matrix records what T does to the basis vectors. The important point is that, because T is linear, the same matrix tells us what T does to *every* vector.

Let

$$v = x_1 e_1 + \cdots + x_n e_n,$$

and write

$$X = [v]_{\mathcal{E}} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Then

$$T(v) = x_1 T(e_1) + \cdots + x_n T(e_n).$$

Using

$$(T(e_1), \dots, T(e_n)) = (f_1, \dots, f_m)A,$$

we obtain

$$\begin{aligned} T(v) &= (T(e_1), \dots, T(e_n)) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\ &= (f_1, \dots, f_m)A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \\ &= (f_1, \dots, f_m)[T]_{\mathcal{F} \leftarrow \mathcal{E}}[v]_{\mathcal{E}}. \end{aligned}$$

Therefore the coordinate vector of $T(v)$ in the basis $\mathcal{F}$ is

$$[T(v)]_{\mathcal{F}} = [T]_{\mathcal{F} \leftarrow \mathcal{E}}[v]_{\mathcal{E}}.$$

We consolidate the above discussion into a proposition.

Proposition 2.17 — A linear map acts by matrix multiplication

Let

$$T : V \rightarrow W$$

be linear, let $\mathcal{E}$ be a basis of V , and let $\mathcal{F}$ be a basis of W . Then, for every $v \in V$,

$$[T(v)]_{\mathcal{F}} = [T]_{\mathcal{F} \leftarrow \mathcal{E}} [v]_{\mathcal{E}}.$$

Remark 2.18 — What the matrix represents

The linear map T and its matrix are not the same object.

The map

$$T : V \rightarrow W$$

is defined independently of any choice of basis. The matrix

$$[T]_{\mathcal{F} \leftarrow \mathcal{E}}$$

is a coordinate representation of T after bases have been chosen for V and W .

Thus,

$$\text{linear map} + \text{choice of bases} \implies \text{matrix}.$$

Changing the bases generally changes the matrix, but does not change the underlying linear map.

2.5 Composition as Matrix Multiplication

We now ask how the matrix representation behaves under composition.

Let

$$T : U \rightarrow V, \quad S : V \rightarrow W$$

be linear maps, and choose bases

$$\mathcal{E} = (e_1, \dots, e_n)$$

of U ,

$$\mathcal{F} = (f_1, \dots, f_m)$$

of V , and

$$\mathcal{G} = (g_1, \dots, g_p)$$

of W .

Suppose the matrices of T and S with respect to these bases are

$$A \in \mathbb{K}^{m \times n}, \quad B \in \mathbb{K}^{p \times m}.$$

By definition,

$$(T(e_1), \dots, T(e_n)) = (f_1, \dots, f_m)A,$$

and

$$(S(f_1), \dots, S(f_m)) = (g_1, \dots, g_p)B.$$

Now consider the composition

$$ST : U \rightarrow W.$$

Applying S to the first relation gives

$$\begin{aligned} (ST(e_1), \dots, ST(e_n)) &= S(T(e_1), \dots, T(e_n)) \\ &= S((f_1, \dots, f_m)A). \end{aligned}$$

Because S is linear, the coefficients in A pass through S :

$$S((f_1, \dots, f_m)A) = (S(f_1), \dots, S(f_m))A.$$

Using

$$(S(f_1), \dots, S(f_m)) = (g_1, \dots, g_p)B,$$

we therefore obtain

$$\begin{aligned} (ST(e_1), \dots, ST(e_n)) &= (S(f_1), \dots, S(f_m))A \\ &= (g_1, \dots, g_p)BA. \end{aligned}$$

Hence the matrix of the composition ST is

$$[ST]_{\mathcal{G} \leftarrow \mathcal{E}} = [S]_{\mathcal{G} \leftarrow \mathcal{F}} [T]_{\mathcal{F} \leftarrow \mathcal{E}}.$$

In the shorter notation

$$[T] = A, \quad [S] = B,$$

this becomes

$$[ST] = BA.$$

Visually

$$\begin{array}{ccccc} U & \xrightarrow{T} & V & \xrightarrow{S} & W \\ & \searrow & & \nearrow & \\ & & ST & & \end{array}$$

$$\mathbb{K}^n \xrightarrow{A} \mathbb{K}^m \xrightarrow{B} \mathbb{K}^p$$

BA

What lives inside the product matrix? What do the elements of the product matrix look like?

Let

$$A = (a_{kj}) \in \mathbb{K}^{m \times n}, \quad B = (b_{ik}) \in \mathbb{K}^{p \times m}.$$

The j -th column of A gives the coordinates of $T(e_j)$ in the basis $\mathcal{F}$. Thus

$$T(e_j) = \sum_{k=1}^m a_{kj} f_k.$$

Applying S ,

$$ST(e_j) = S\left(\sum_{k=1}^m a_{kj} f_k\right) = \sum_{k=1}^m a_{kj} S(f_k).$$

But the k -th column of B gives the coordinates of $S(f_k)$ in the basis $\mathcal{G}$, so

$$S(f_k) = \sum_{i=1}^p b_{ik} g_i.$$

Therefore,

$$\begin{aligned} ST(e_j) &= \sum_{k=1}^m a_{kj} \left(\sum_{i=1}^p b_{ik} g_i \right) \\ &= \sum_{i=1}^p \left(\sum_{k=1}^m b_{ik} a_{kj} \right) g_i. \end{aligned}$$

Hence the coefficient of g_i in $ST(e_j)$ is

$$\sum_{k=1}^m b_{ik} a_{kj}.$$

Thus, if

$$C = [ST] = BA,$$

then

$$c_{ij} = \sum_{k=1}^m b_{ik} a_{kj}.$$

This motivates the definition of matrix multiplication.

If

$$B \in \mathbb{K}^{p \times m}, \quad A \in \mathbb{K}^{m \times n},$$

then their product

$$BA \in \mathbb{K}^{p \times n}$$

is defined by

$$(BA)_{ij} = \sum_{k=1}^m b_{ik} a_{kj}.$$

Thus the (i, j) -entry of BA is obtained by multiplying the i -th row of B by the j -th column of A :

$$(BA)_{ij} = \begin{pmatrix} b_{i1} & \cdots & b_{im} \end{pmatrix} \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}.$$

The dimensions also explain why the matrices can be multiplied:

$$U \xrightarrow{T} V \xrightarrow{S} W$$

with

$$\dim U = n, \quad \dim V = m, \quad \dim W = p$$

corresponds to

$$\mathbb{K}^n \xrightarrow{A} \mathbb{K}^m \xrightarrow{B} \mathbb{K}^p,$$

where

$$A : m \times n, \quad B : p \times m.$$

The two occurrences of m correspond to the intermediate space V . They are exactly the two dimensions that must match in order for the product

$$BA$$

to be defined.

2.6 Four Ways to Understand Matrix Multiplication

Let

$$A \in \mathbb{K}^{m \times n}, \quad B \in \mathbb{K}^{n \times p},$$

and write

$$C = AB \in \mathbb{K}^{m \times p}.$$

There are several equivalent ways to understand the product AB . Each viewpoint emphasizes a different part of the structure of matrix multiplication.

1. Row by column, element-wise

The most familiar interpretation computes the entries of C one at a time.

Write

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}, \quad B = \begin{pmatrix} b_{11} & \cdots & b_{1p} \\ \vdots & & \vdots \\ b_{n1} & \cdots & b_{np} \end{pmatrix}.$$

Then the (i, j) -entry of $C = AB$ is

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

Equivalently,

$$c_{ij} = \begin{pmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{pmatrix} \begin{pmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{pmatrix}.$$

Thus,

$$(i, j)\text{-entry of } AB = i\text{-th row of } A \times j\text{-th column of } B.$$

This viewpoint is useful when we want to compute a particular entry of a matrix product.

2. Matrix by columns

Instead of computing AB entry by entry, we may compute it one *column* at a time.

Write the columns of B as

$$B = \begin{pmatrix} | & | & \cdots & | \\ b_1 & b_2 & \cdots & b_p \\ | & | & & | \end{pmatrix}, \quad b_j \in \mathbb{K}^n.$$

Then

$$AB = \begin{pmatrix} | & | & \cdots & | \\ Ab_1 & Ab_2 & \cdots & Ab_p \\ | & | & & | \end{pmatrix}.$$

In other words,

$$j\text{-th column of } AB = A(j\text{-th column of } B).$$

Thus multiplying B on the left by A means that A acts separately on every column of B .

This is precisely the viewpoint that arises naturally when matrices are viewed as representations of linear maps.

3. Rows by matrix

There is a completely analogous row-wise interpretation.

Write the rows of A as

$$A = \begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{pmatrix}, \quad r_i \in \mathbb{K}^{1 \times n}.$$

Then

$$AB = \begin{pmatrix} r_1 B \\ r_2 B \\ \vdots \\ r_m B \end{pmatrix}.$$

Hence

$$i\text{-th row of } AB = (i\text{-th row of } A)B.$$

Thus multiplying A on the right by B means that each row of A is multiplied by B .

4. Column by row, then add

Finally, matrix multiplication can be viewed as a sum of *column-by-row products*.

Write A in terms of its columns,

$$A = \begin{pmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{pmatrix}, \quad a_k \in \mathbb{K}^m,$$

and write B in terms of its rows,

$$B = \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{pmatrix}, \quad s_k \in \mathbb{K}^{1 \times p}.$$

Then

$$AB = a_1 s_1 + a_2 s_2 + \cdots + a_n s_n = \sum_{k=1}^n a_k s_k.$$

Each product

$$a_k s_k$$

is an $m \times p$ matrix: it is obtained by multiplying one column of A by the corresponding row of B .

Indeed,

$$a_k s_k = \begin{pmatrix} a_{1k} \\ a_{2k} \\ \vdots \\ a_{mk} \end{pmatrix} \begin{pmatrix} b_{k1} & b_{k2} & \cdots & b_{kp} \end{pmatrix} = \begin{pmatrix} a_{1k}b_{k1} & \cdots & a_{1k}b_{kp} \\ a_{2k}b_{k1} & \cdots & a_{2k}b_{kp} \\ \vdots & & \vdots \\ a_{mk}b_{k1} & \cdots & a_{mk}b_{kp} \end{pmatrix}.$$

Adding these n matrices gives

$$AB = \sum_{k=1}^n \begin{pmatrix} a_{1k}b_{k1} & \cdots & a_{1k}b_{kp} \\ \vdots & & \vdots \\ a_{mk}b_{k1} & \cdots & a_{mk}b_{kp} \end{pmatrix}.$$

The (i, j) -entry of this sum is therefore

$$\sum_{k=1}^n a_{ik}b_{kj},$$

which is exactly the usual row-by-column formula.

These four viewpoints describe exactly the same matrix product. Which viewpoint is most useful depends on the question being asked. You will likely come back to this when you study the algebra of OLS in Econometrics I.

2.7 Inverse Maps and Inverse Matrices

We have seen that composition of linear maps corresponds to multiplication of their matrices. This immediately raises a natural question: when can the effect of a linear transformation be undone?

Let

$$T : V \rightarrow V$$

be a linear transformation. We seek another linear transformation that reverses the action of T .

Definition 2.19 — Invertible linear transformation

A linear transformation

$$T : V \rightarrow V$$

is called **invertible** if there exists a linear transformation

$$S : V \rightarrow V$$

such that

$$ST = I_V \quad \text{and} \quad TS = I_V,$$

where I_V denotes the identity transformation on V .

If such an S exists, it is called the **inverse** of T and is denoted by

$$T^{-1}.$$

Thus,

$$T^{-1}T = I_V, \quad TT^{-1} = I_V.$$

The meaning is simple. If

$$v \xrightarrow{T} T(v),$$

then applying T^{-1} brings us back to the original vector:

$$v \xrightarrow{T} T(v) \xrightarrow{T^{-1}} v.$$

Similarly,

$$v \xrightarrow{T^{-1}} T^{-1}(v) \xrightarrow{T} v.$$

Proposition 2.20 — The inverse is unique

If a linear transformation

$$T : V \rightarrow V$$

is invertible, then its inverse is unique.

Proof

Suppose that both S_1 and S_2 are inverses of T . Then

$$S_1T = I_V, \quad TS_2 = I_V.$$

Using associativity of composition,

$$\begin{aligned} S_1 &= S_1I_V \\ &= S_1(TS_2) \\ &= (S_1T)S_2 \\ &= I_VS_2 \\ &= S_2. \end{aligned}$$

Therefore the inverse of T is unique.

Hence it is unambiguous to write the inverse of T as

$$T^{-1}.$$

Now comes the matrix counterpart of the inverse map.

Definition 2.21 — Invertible matrix

A square matrix

$$A \in \mathbb{K}^{n \times n}$$

is called **invertible** if there exists a matrix

$$B \in \mathbb{K}^{n \times n}$$

such that

$$AB = I_n \quad \text{and} \quad BA = I_n,$$

where I_n is the $n \times n$ identity matrix.

If such a matrix B exists, it is called the **inverse** of A and is denoted by

$$A^{-1}.$$

Thus,

$$AA^{-1} = I_n, \quad A^{-1}A = I_n.$$

Proposition 2.22 — The matrix of an inverse transformation

Let $T : V \rightarrow V$ be an invertible linear transformation, and let $\mathcal{E}$ be a basis of the finite-dimensional vector space V .

If

$$A = [T]_{\mathcal{E}},$$

then A is invertible and

$$[T^{-1}]_{\mathcal{E}} = A^{-1}.$$

Equivalently,

$$[T^{-1}]_{\mathcal{E}} = ([T]_{\mathcal{E}})^{-1}.$$

Proof

Since

$$T^{-1}T = I_V \quad \text{and} \quad TT^{-1} = I_V,$$

composition as matrix multiplication gives

$$[T^{-1}]_{\mathcal{E}}[T]_{\mathcal{E}} = [I_V]_{\mathcal{E}} = I_n,$$

and

$$[T]_{\mathcal{E}}[T^{-1}]_{\mathcal{E}} = [I_V]_{\mathcal{E}} = I_n.$$

Therefore $[T^{-1}]_{\mathcal{E}}$ is the inverse of $[T]_{\mathcal{E}}$. Hence

$$[T^{-1}]_{\mathcal{E}} = ([T]_{\mathcal{E}})^{-1}.$$

Remark 2.23 — Inverse maps and inverse matrices

The inverse matrix is the coordinate representation of the inverse linear transformation.

If

$$T \longleftrightarrow A,$$

then

$$T^{-1} \longleftrightarrow A^{-1}.$$

Thus matrix inversion is not a separate operation introduced merely for computation. It arises naturally from reversing an invertible linear transformation.

We have talked about composition of linear maps, what about their inverses if such exist?

Proposition 2.24 — Inverse of a composition

Let

$$S, T : V \rightarrow V$$

be invertible linear transformations. Then ST is invertible and

$$(ST)^{-1} = T^{-1}S^{-1}.$$

Proof

We verify that $T^{-1}S^{-1}$ is an inverse of ST .

Using associativity of composition,

$$\begin{aligned}(T^{-1}S^{-1})(ST) &= T^{-1}(S^{-1}S)T \\ &= T^{-1}I_V T \\ &= T^{-1}T \\ &= I_V.\end{aligned}$$

Similarly,

$$\begin{aligned}(ST)(T^{-1}S^{-1}) &= S(TT^{-1})S^{-1} \\ &= SI_V S^{-1} \\ &= SS^{-1} \\ &= I_V.\end{aligned}$$

Therefore,

$$(ST)^{-1} = T^{-1}S^{-1}.$$

Translating this into the matrix world, we have the following

Corollary 2.25 — Inverse of a matrix product

If A and B are invertible $n \times n$ matrices, then BA is invertible and

$$(BA)^{-1} = A^{-1}B^{-1}.$$

Proof

Choose invertible linear transformations T and S represented by A and B , respectively. Since

$$[ST] = BA$$

and

$$(ST)^{-1} = T^{-1}S^{-1},$$

we obtain

$$(BA)^{-1} = A^{-1}B^{-1}.$$

Remark 2.26 — Why the order reverses

The reversal of order in

$$(ST)^{-1} = T^{-1}S^{-1}$$

is natural.

The transformation ST first applies T and then applies S :

$$v \xrightarrow{T} T(v) \xrightarrow{S} S(T(v)).$$

To return to v , we must undo these operations in the reverse order: first apply S^{-1} and then apply T^{-1} .

2.8 Matrix Transpose

We now introduce another basic operation on matrices. Unlike the inverse, which exists only for certain square matrices, the transpose is defined for every matrix.

Definition 2.27 — Transpose of a matrix

Let

$$A = (a_{ij}) \in \mathbb{K}^{m \times n}.$$

The **transpose** of A , denoted by

$$A^T,$$

is the $n \times m$ matrix obtained by interchanging the rows and columns of A .

Equivalently,

$$(A^T)_{ij} = a_{ji}.$$

Thus the rows of A become the columns of A^T , and the columns of A become the rows of A^T .

Example 2.28 — Transpose of a matrix

If

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix},$$

then

$$A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

Notice that A is a 2×3 matrix, whereas A^T is a 3×2 matrix.

Here are some basic properties of the matrix transpose.

Proposition 2.29 — Properties of the transpose

Let A and B be matrices of compatible dimensions and let $\lambda \in \mathbb{K}$. Then

$$(A^T)^T = A,$$

$$(A + B)^T = A^T + B^T,$$

$$(\lambda A)^T = \lambda A^T,$$

and

$$(AB)^T = B^T A^T.$$

Proof

The first three identities follow directly from the definition of the transpose. We verify the product identity.

Suppose

$$A \in \mathbb{K}^{m \times n}, \quad B \in \mathbb{K}^{n \times p}.$$

For every i and j ,

$$\begin{aligned} ((AB)^\top)_{ij} &= (AB)_{ji} \\ &= \sum_{k=1}^n a_{jk} b_{ki} \\ &= \sum_{k=1}^n (B^\top)_{ik} (A^\top)_{kj} \\ &= (B^\top A^\top)_{ij}. \end{aligned}$$

Hence

$$(AB)^\top = B^\top A^\top.$$

Remark 2.30 — The order reverses under transposition

The identity

$$(AB)^\top = B^\top A^\top$$

resembles the inverse-of-a-product formula

$$(AB)^{-1} = B^{-1}A^{-1}.$$

In both cases, applying the operation to a product reverses the order of the factors.

Proposition 2.31 — Transpose and inverse identities

Let A and B be square matrices of the same size. Whenever the corresponding inverses exist, the following identities hold:

$$(A^\top)^\top = A,$$

$$(AB)^\top = B^\top A^\top,$$

$$(AB)^{-1} = B^{-1}A^{-1},$$

and

$$(A^\top)^{-1} = (A^{-1})^\top.$$

In particular, A is invertible if and only if $A^\top$ is invertible.

Proof

The first identity follows immediately from the definition of the transpose, and the second was established above.

For the inverse of a product, observe that

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I,$$

and similarly,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I.$$

Hence

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Now suppose A is invertible. Since

$$AA^{-1} = I,$$

taking transposes gives

$$(AA^{-1})^T = I^T.$$

Using the transpose-of-a-product rule,

$$(A^{-1})^T A^T = I.$$

Similarly, transposing

$$A^{-1}A = I$$

gives

$$A^T(A^{-1})^T = I.$$

Thus $(A^{-1})^T$ is the inverse of A^T , and therefore

$$(A^T)^{-1} = (A^{-1})^T.$$

Hence A invertible implies A^T invertible. Applying the same argument to A^T , together with

$$(A^T)^T = A,$$

gives the converse.

2.9 Symmetric and Skew-Symmetric Matrices

Transpose allows us to distinguish two important classes of square matrices: those that remain unchanged under transposition, and those that change sign.

Definition 2.32 — Symmetric and skew-symmetric matrices

Let

$$A \in \mathbb{K}^{n \times n}.$$

The matrix A is called **symmetric** if

$$A^T = A.$$

The matrix A is called **skew-symmetric** (or **antisymmetric**) if

$$A^T = -A.$$

In terms of entries, A is symmetric if

$$a_{ij} = a_{ji}$$

for every i, j , whereas A is skew-symmetric if

$$a_{ij} = -a_{ji}.$$

Thus a symmetric matrix is reflected unchanged across the main diagonal, while a skew-symmetric matrix is reflected across the main diagonal and changes sign.

Example 2.33 — Symmetric and skew-symmetric matrices

A typical symmetric 3×3 matrix has the form

$$\begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix}.$$

A typical skew-symmetric 3×3 matrix has the form

$$\begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}.$$

Why are the diagonals zero for skew-symmetric matrices?

Proposition 2.34 — Diagonal entries of a skew-symmetric matrix

If A is skew-symmetric, then every diagonal entry of A is zero.

Proof

If A is skew-symmetric, then

$$a_{ii} = -a_{ii}$$

for every i . Hence

$$2a_{ii} = 0.$$

therefore

$$a_{ii} = 0.$$

Proposition 2.35 — Symmetric-skew-symmetric decomposition

Every square matrix

$$A \in \mathbb{K}^{n \times n}$$

can be written uniquely as

$$A = S + K,$$

where S is symmetric and K is skew-symmetric.

More precisely,

$$S = \frac{A + A^T}{2}, \quad K = \frac{A - A^T}{2}.$$

Proof

We first show that such a decomposition exists. Define

$$S = \frac{A + A^T}{2}, \quad K = \frac{A - A^T}{2}.$$

Then

$$\begin{aligned} S^T &= \left(\frac{A + A^T}{2} \right)^T \\ &= \frac{A^T + (A^T)^T}{2} \\ &= \frac{A^T + A}{2} \\ &= S, \end{aligned}$$

so S is symmetric.

Similarly,

$$\begin{aligned} K^\top &= \left(\frac{A - A^\top}{2} \right)^\top \\ &= \frac{A^\top - A}{2} \\ &= -\frac{A - A^\top}{2} \\ &= -K, \end{aligned}$$

so K is skew-symmetric.

Moreover,

$$S + K = \frac{A + A^\top}{2} + \frac{A - A^\top}{2} = A.$$

Thus the decomposition exists.

To prove uniqueness, suppose also that

$$A = S_1 + K_1,$$

where S_1 is symmetric and K_1 is skew-symmetric. Taking transposes,

$$A^\top = S_1^\top + K_1^\top = S_1 - K_1.$$

Adding the two equations gives

$$A + A^\top = 2S_1,$$

so

$$S_1 = \frac{A + A^\top}{2} = S.$$

Subtracting them gives

$$A - A^\top = 2K_1,$$

so

$$K_1 = \frac{A - A^\top}{2} = K.$$

Hence the decomposition is unique.

Remark 2.36 — A note on the scalar field

The decomposition

$$A = \frac{A + A^\top}{2} + \frac{A - A^\top}{2}$$

requires that the scalar 2 be invertible in the underlying field $\mathbb{K}$.

Thus, technically, the result does not hold over an arbitrary field: it can fail in fields of

characteristic 2. (If you don't know what characteristic is, don't worry. This is almost never used in economics. But if you do know an application of this concept somewhere in economics, do let me know since I am curious.)

For our purposes, however, we work primarily over

$$\mathbb{R} \quad \text{or} \quad \mathbb{C},$$

where $2 \neq 0$ and division by 2 is always valid. Hence the decomposition above applies throughout our discussion.

3 Rank of a Linear Map and a Matrix

The rank of a linear map measures how many independent directions remain after the map is applied. To understand this precisely, we first study two subspaces naturally associated with every linear map: its *kernel*, which records the directions that are collapsed to zero, and its *image*, which records the directions that can actually be reached.

3.1 Kernel and Image

Let

$$T : V \rightarrow W$$

be a linear map. There are two natural questions we can ask:

- ▶ Which vectors in V are sent to zero?
- ▶ Which vectors in W can actually be obtained as outputs of T ?

The answers give the kernel and the image of T .

Definition 3.1 — Kernel and image

Let

$$T : V \rightarrow W$$

be a linear map.

The **kernel** of T is

$$\ker T = \{v \in V : T(v) = 0_W\}.$$

The **image** of T is

$$\operatorname{Im} T = \{T(v) : v \in V\}.$$

Equivalently,

$$\operatorname{Im} T = \{w \in W : w = T(v) \text{ for some } v \in V\}.$$

Thus,

$$\ker T \subseteq V, \quad \operatorname{Im} T \subseteq W.$$

The kernel is therefore a subset of the *domain*, whereas the image is a subset of the *codomain*.

These two sets are vector subspaces.

Proposition 3.2 — Kernel and image are subspaces

Let

$$T : V \rightarrow W$$

be a linear map. Then

$$\ker T \leq V \quad \text{and} \quad \operatorname{Im} T \leq W.$$

Proof

We first consider the kernel.

Because T is linear,

$$T(0_V) = 0_W,$$

so

$$0_V \in \ker T.$$

Now let $u, v \in \ker T$ and let $\lambda, \mu \in \mathbb{K}$. Then

$$T(u) = T(v) = 0_W,$$

and hence

$$\begin{aligned} T(\lambda u + \mu v) &= \lambda T(u) + \mu T(v) \\ &= \lambda 0_W + \mu 0_W \\ &= 0_W. \end{aligned}$$

Thus

$$\lambda u + \mu v \in \ker T.$$

By the subspace test,

$$\ker T \leq V.$$

Now consider the image. Since

$$T(0_V) = 0_W,$$

we have

$$0_W \in \operatorname{Im} T.$$

Let $w_1, w_2 \in \operatorname{Im} T$. Then there exist $v_1, v_2 \in V$ such that

$$w_1 = T(v_1), \quad w_2 = T(v_2).$$

For any $\lambda, \mu \in \mathbb{K}$,

$$\begin{aligned} \lambda w_1 + \mu w_2 &= \lambda T(v_1) + \mu T(v_2) \\ &= T(\lambda v_1 + \mu v_2). \end{aligned}$$

Therefore

$$\lambda w_1 + \mu w_2 \in \operatorname{Im} T.$$

Again by the subspace test,

$$\operatorname{Im} T \leq W.$$

See the following remark for an interpretation of these two subspaces.

Remark 3.3 — What the kernel and image measure

The kernel measures what the linear map *loses*:

$$v \in \ker T \iff T(v) = 0_W.$$

Thus vectors in the kernel are directions that are completely collapsed by T .

The image measures what the linear map can *produce*:

$$w \in \operatorname{Im} T \iff w = T(v) \text{ for some } v \in V.$$

Thus $\operatorname{Im} T$ is the set of all attainable outputs of T .

To see the use of these two subspaces, let's see the following characterization of injectivity and surjectivity of linear maps.

Proposition 3.4 — Injectivity and the kernel

Let

$$T : V \rightarrow W$$

be a linear map. Then

$$T \text{ is injective} \iff \ker T = \{0_V\}.$$

Proof

Suppose first that T is injective. If

$$v \in \ker T,$$

then

$$T(v) = 0_W = T(0_V).$$

Since T is injective,

$$v = 0_V.$$

Therefore

$$\ker T = \{0_V\}.$$

Conversely, suppose

$$\ker T = \{0_V\}.$$

If

$$T(u) = T(v),$$

then by linearity,

$$T(u - v) = T(u) - T(v) = 0_W.$$

Hence

$$u - v \in \ker T.$$

Because $\ker T = \{0_V\}$,

$$u - v = 0_V,$$

and therefore

$$u = v.$$

Thus T is injective.

Proposition 3.5 — Surjectivity and the image

Let

$$T : V \rightarrow W$$

be a linear map. Then

$$T \text{ is surjective} \iff \operatorname{Im} T = W.$$

This is simply the definition of surjectivity: every vector in the codomain must be attained as an output of T .

One thing to note is that surjectivity depends on the choice of the codomain.

Remark 3.6 — The image and the codomain are not the same thing

For a linear map

$$T : V \rightarrow W,$$

we always have

$$\text{Im } T \subseteq W,$$

but equality need not hold.

The codomain W is part of the specification of the map, whereas $\text{Im } T$ consists only of those vectors in W that are actually reached.

Thus

$$T \text{ is surjective} \iff \text{Im } T = W.$$

Let's see an example.

Example 3.7 — Kernel and image of a linear map

Let

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

be defined by

$$T(x, y, z) = (x + y, y + z).$$

To find the kernel, solve

$$T(x, y, z) = (0, 0).$$

Thus

$$x + y = 0, \quad y + z = 0,$$

so

$$x = -y, \quad z = -y.$$

Writing $y = t$ gives

$$(x, y, z) = t(-1, 1, -1).$$

Hence

$$\ker T = \text{Span}\{(-1, 1, -1)\}.$$

For the image, observe that

$$\begin{aligned} T(x, y, z) &= (x + y, y + z) \\ &= x(1, 0) + y(1, 1) + z(0, 1). \end{aligned}$$

Therefore

$$\text{Im } T = \text{Span}\{(1, 0), (1, 1), (0, 1)\}.$$

Since

$$(1, 0) \quad \text{and} \quad (0, 1)$$

already form a basis of $\mathbb{R}^2$,

$$\text{Im } T = \mathbb{R}^2.$$

Thus this map has a one-dimensional kernel and a two-dimensional image.

How "big" are these two subspaces? Recall that we have discussed the dimension of a subspace, which leads to the following definitions.

Definition 3.8 — Rank and nullity of a linear map

Let

$$T : V \rightarrow W$$

be a linear map, with V finite-dimensional.

The **nullity** of T is the dimension of its kernel:

$$\text{nullity}(T) = \dim \ker T.$$

The **rank** of T is the dimension of its image:

$$\text{Rank}(T) = \dim \text{Im } T.$$

Remark 3.9 — Rank and nullity as dimensions

The nullity and rank measure two different aspects of the action of T :

$$\text{nullity}(T) = \dim \ker T$$

measures the number of independent directions that are collapsed to zero, whereas

$$\text{Rank}(T) = \dim \text{Im } T$$

measures the number of independent directions that remain visible in the output.

The rank–nullity theorem will show that these two quantities account for the entire dimension of the domain.

3.2 The Rank-Nullity Theorem

How are the rank and nullity of a linear map related? To study this, we need a proposition as a tool.

Proposition 3.10 — Basis extension

Let V be a finite-dimensional vector space. Every linearly independent list

$$v_1, \dots, v_k$$

in V can be extended to a basis of V .

Proof

Let

$$v_1, \dots, v_k$$

be linearly independent in V .

If these vectors already span V , then they form a basis and there is nothing to prove.

Otherwise, choose

$$v_{k+1} \notin \text{Span}(v_1, \dots, v_k).$$

Then

$$v_1, \dots, v_k, v_{k+1}$$

is linearly independent.

If these vectors still do not span V , choose another vector outside their span and repeat the argument.

Because V is finite-dimensional, this process must terminate after finitely many steps. The resulting linearly independent list spans V and therefore forms a basis of V .

Now we are ready to dive into the main result of this subsection and in my opinion, a fundamental theorem in linear algebra.

Theorem 3.11 — Rank-nullity theorem

Let

$$T : V \rightarrow W$$

be a linear map, where V is finite-dimensional. Then $\text{Im } T$ is finite-dimensional and

$$\dim V = \dim \ker T + \dim \text{Im } T.$$

Equivalently,

$$\dim V = \text{nullity}(T) + \text{Rank}(T).$$

Proof

Let

$$u_1, \dots, u_k$$

be a basis of $\ker T$. Thus

$$\dim \ker T = k.$$

By the basis extension theorem, extend this list to a basis of V :

$$u_1, \dots, u_k, v_1, \dots, v_r.$$

Hence

$$\dim V = k + r.$$

We will show that

$$T(v_1), \dots, T(v_r)$$

is a basis of $\text{Im } T$.

First, we show that these vectors span $\text{Im } T$.

Take any

$$w \in \text{Im } T.$$

By definition of the image, there exists $v \in V$ such that

$$w = T(v).$$

Because

$$u_1, \dots, u_k, v_1, \dots, v_r$$

is a basis of V , we can write

$$v = a_1 u_1 + \dots + a_k u_k + b_1 v_1 + \dots + b_r v_r.$$

Applying T gives

$$\begin{aligned} w &= T(v) \\ &= a_1 T(u_1) + \dots + a_k T(u_k) + b_1 T(v_1) + \dots + b_r T(v_r). \end{aligned}$$

Since

$$u_1, \dots, u_k \in \ker T,$$

we have

$$T(u_1) = \cdots = T(u_k) = 0.$$

Therefore

$$w = b_1 T(v_1) + \cdots + b_r T(v_r).$$

Thus

$$T(v_1), \dots, T(v_r)$$

spans $\text{Im } T$.

Next, we show that these vectors are linearly independent.

Suppose

$$c_1 T(v_1) + \cdots + c_r T(v_r) = 0.$$

By linearity,

$$T(c_1 v_1 + \cdots + c_r v_r) = 0.$$

Hence

$$c_1 v_1 + \cdots + c_r v_r \in \ker T.$$

Because

$$u_1, \dots, u_k$$

is a basis of $\ker T$, there exist scalars $d_1, \dots, d_k$ such that

$$c_1 v_1 + \cdots + c_r v_r = d_1 u_1 + \cdots + d_k u_k.$$

Therefore

$$d_1 u_1 + \cdots + d_k u_k - c_1 v_1 - \cdots - c_r v_r = 0.$$

But

$$u_1, \dots, u_k, v_1, \dots, v_r$$

is a basis of V , and hence is linearly independent. Therefore

$$c_1 = \cdots = c_r = 0.$$

Thus

$$T(v_1), \dots, T(v_r)$$

is linearly independent.

Consequently,

$$T(v_1), \dots, T(v_r)$$

is a basis of $\text{Im } T$. In particular, $\text{Im } T$ is finite-dimensional and

$$\dim \text{Im } T = r.$$

Therefore,

$$\dim V = k + r = \dim \ker T + \dim \operatorname{Im} T.$$

3.3 Applications of the Rank–Nullity Theorem

The rank–nullity theorem is especially useful when a problem can be expressed in terms of a linear map

$$T : V \rightarrow W.$$

Instead of solving the problem directly, we can often study the simpler homogeneous problem

$$T(v) = 0.$$

When the domain and codomain have the same finite dimension, this is particularly powerful: showing that the zero vector is the only vector mapped to zero is enough to obtain both existence and uniqueness.

Proposition 3.12 — Zero-kernel criterion for bijectivity

Let

$$T : V \rightarrow W$$

be a linear map between finite-dimensional vector spaces satisfying

$$\dim V = \dim W.$$

Then the following statements are equivalent:

$$\ker T = \{0\},$$

T is injective,

T is surjective,

and

T is bijective.

Proof

We already know that

$$T \text{ is injective} \iff \ker T = \{0\}.$$

Suppose now that

$$\ker T = \{0\}.$$

By the rank-nullity theorem,

$$\dim V = \dim \ker T + \dim \operatorname{Im} T = \dim \operatorname{Im} T.$$

Since

$$\dim V = \dim W,$$

we obtain

$$\dim \operatorname{Im} T = \dim W.$$

But $\operatorname{Im} T$ is a subspace of W , so

$$\operatorname{Im} T = W.$$

Thus T is surjective. Hence T is both injective and surjective, and therefore bijective.

Conversely, if T is bijective, then in particular it is injective, so

$$\ker T = \{0\}.$$

Remark 3.13 — The homogeneous-problem principle

Suppose

$$\dim V = r$$

and a problem with r pieces of data can be represented by a linear map

$$\Phi : V \rightarrow \mathbb{K}^r.$$

Because the domain and codomain have the same dimension, to prove that every possible data vector has a unique solution, it is enough to prove

$$\ker \Phi = \{0\}.$$

In other words,

zero data have only the zero solution

implies

arbitrary data have a unique solution.

This simple principle is the key to many interpolation problems.

Application 1: Linear systems

Consider a matrix

$$A \in \mathbb{K}^{m \times n}$$

and the associated linear map

$$T_A : \mathbb{K}^n \rightarrow \mathbb{K}^m, \quad T_A(x) = Ax.$$

The system of linear equations

$$Ax = b$$

is therefore the equation

$$T_A(x) = b.$$

The kernel of T_A consists precisely of the solutions of the corresponding homogeneous system

$$Ax = 0.$$

Proposition 3.14 — Solutions of a linear system

Suppose

$$Ax = b$$

has a particular solution x_p . Then x solves $Ax = b$ if and only if

$$x = x_p + z$$

for some

$$z \in \ker T_A.$$

Proof

Suppose first that x is another solution. Then

$$Ax = Ax_p = b,$$

and therefore

$$A(x - x_p) = 0.$$

Thus

$$x - x_p \in \ker T_A.$$

Writing

$$z = x - x_p$$

gives

$$x = x_p + z.$$

Conversely, if

$$x = x_p + z \quad \text{with} \quad z \in \ker T_A,$$

then

$$Ax = A(x_p + z) = Ax_p + Az = b + 0 = b.$$

Hence x is a solution.

If the system is consistent, that is, there exists a solution to the linear system, the freedom in its solutions is exactly the freedom contained in the kernel.

By rank-nullity,

$$n = \dim \ker T_A + \text{Rank}(T_A),$$

and therefore

$$\dim \ker T_A = n - \text{Rank}(T_A).$$

Thus the nullity measures the number of independent degrees of freedom remaining in a consistent linear system.

Corollary 3.15 — Variables, equations, and solutions

Let

$$A \in \mathbb{K}^{m \times n}.$$

(i) If $n > m$, then the homogeneous system

$$Ax = 0$$

has a nonzero solution.

(ii) If $m > n$, then there exist vectors

$$b \in \mathbb{K}^m$$

for which

$$Ax = b$$

has no solution.

(iii) If $m = n$, then

$$Ax = b$$

has a unique solution for every $b \in \mathbb{K}^n$ if and only if

$$Ax = 0$$

has only the zero solution.

Proof

If $n > m$, then

$$\dim \operatorname{Im} T_A \leq m < n.$$

Rank-nullity therefore implies

$$\dim \ker T_A = n - \dim \operatorname{Im} T_A > 0.$$

Hence the kernel contains a nonzero vector.

If $m > n$, then

$$\dim \operatorname{Im} T_A \leq n < m,$$

so

$$\operatorname{Im} T_A \neq \mathbb{K}^m.$$

Thus there exist right-hand sides b that are not in the image of T_A , and for such b the system $Ax = b$ has no solution.

Finally, if $m = n$, then T_A maps between vector spaces of the same finite dimension. The result therefore follows from [proposition 3.12](#).

Polynomial spaces

For a positive integer r , let

$$\mathbb{R}_{r-1}[x] = \operatorname{Span}(1, x, x^2, \dots, x^{r-1})$$

denote the vector space of real polynomials of degree at most $r - 1$, including the zero polynomial.

The list

$$1, x, x^2, \dots, x^{r-1}$$

is a basis, and hence

$$\dim \mathbb{R}_{r-1}[x] = r.$$

Before studying polynomial interpolation, we collect several elementary facts about roots of polynomials. These results will allow us to establish that the interpolation maps below have trivial kernels.

Proposition 3.16 — Factor theorem

Let

$$p \in \mathbb{R}[x]$$

and let $a \in \mathbb{R}$. Then

$$p(a) = 0 \iff (x - a) \text{ divides } p(x), \text{ written as } (x - a) \mid p(x).$$

Proof

Suppose first that

$$(x - a) \mid p(x).$$

Then

$$p(x) = (x - a)q(x)$$

for some polynomial q , and therefore

$$p(a) = 0.$$

Conversely, suppose

$$p(a) = 0.$$

Write

$$p(x) = c_0 + c_1x + \cdots + c_nx^n.$$

Then

$$p(x) - p(a) = \sum_{k=1}^n c_k(x^k - a^k).$$

For every $k \geq 1$,

$$x^k - a^k = (x - a)(x^{k-1} + x^{k-2}a + \cdots + xa^{k-2} + a^{k-1}).$$

Hence $x - a$ divides $p(x) - p(a)$. Since $p(a) = 0$,

$$p(x) = p(x) - p(a),$$

and therefore

$$(x - a) \mid p(x).$$

Definition 3.17 — Multiplicity of a root

Let

$$p \in \mathbb{R}[x],$$

and suppose $p \neq 0$.

A number $a \in \mathbb{R}$ is called a **root of multiplicity** m of p if

$$(x - a)^m \mid p(x)$$

but

$$(x - a)^{m+1} \nmid p(x).$$

More generally, we say that a is a root of multiplicity **at least** m if

$$(x - a)^m \mid p(x).$$

There is a nice link between the multiplicity of roots and derivatives.

Proposition 3.18 — Multiplicity and derivatives

Let

$$p \in \mathbb{R}[x], \quad a \in \mathbb{R},$$

and let $m \geq 1$. Then

$$(x - a)^m \mid p(x)$$

if and only if

$$p(a) = p'(a) = \cdots = p^{(m-1)}(a) = 0.$$

Proof

We prove the result by induction on m .

For $m = 1$, the statement is exactly the factor theorem:

$$(x - a) \mid p(x) \iff p(a) = 0.$$

Now suppose the result holds for $m - 1$.

We first record a useful identity. If

$$p(x) = (x - a)q(x),$$

then for every $j \geq 1$,

$$p^{(j)}(x) = (x - a)q^{(j)}(x) + jq^{(j-1)}(x).$$

Indeed, the case $j = 1$ follows from the product rule,

$$p'(x) = q(x) + (x - a)q'(x),$$

and the general formula follows by repeated differentiation.

In particular, evaluating at $x = a$ gives

$$p^{(j)}(a) = jq^{(j-1)}(a).$$

Suppose first that

$$(x - a)^m \mid p(x).$$

Then

$$p(x) = (x - a)q(x),$$

where

$$(x - a)^{m-1} \mid q(x).$$

By the induction hypothesis,

$$q(a) = q'(a) = \cdots = q^{(m-2)}(a) = 0.$$

Also $p(a) = 0$, and for $j = 1, \dots, m-1$,

$$p^{(j)}(a) = jq^{(j-1)}(a) = 0.$$

Thus

$$p(a) = p'(a) = \cdots = p^{(m-1)}(a) = 0.$$

Conversely, suppose

$$p(a) = p'(a) = \cdots = p^{(m-1)}(a) = 0.$$

Since $p(a) = 0$, the factor theorem gives

$$p(x) = (x - a)q(x)$$

for some polynomial q .

For $j = 1, \dots, m-1$,

$$0 = p^{(j)}(a) = jq^{(j-1)}(a).$$

Since $j \neq 0$ in $\mathbb{R}$, we obtain

$$q(a) = q'(a) = \cdots = q^{(m-2)}(a) = 0.$$

By the induction hypothesis,

$$(x - a)^{m-1} \mid q(x).$$

Therefore

$$(x - a)^m \mid p(x).$$

Proposition 3.19 — Roots counted with multiplicity

Let

$$p \in \mathbb{R}[x]$$

be a nonzero polynomial. Suppose that

$$a_1, \dots, a_s$$

are distinct roots of p with multiplicities at least

$$m_1, \dots, m_s,$$

respectively. Then

$$\deg p \geq m_1 + \dots + m_s.$$

Proof

Since a_i is a root of multiplicity at least m_i ,

$$(x - a_i)^{m_i} \mid p(x)$$

for every i .

Because the numbers

$$a_1, \dots, a_s$$

are distinct, these are distinct linear factors. Repeatedly factoring them out gives

$$p(x) = \left[\prod_{i=1}^s (x - a_i)^{m_i} \right] q(x)$$

for some polynomial q .

Therefore

$$\deg p = \sum_{i=1}^s m_i + \deg q \geq \sum_{i=1}^s m_i.$$

Corollary 3.20 — Number of roots of a polynomial

A nonzero polynomial of degree d has at most d distinct roots.

More generally, it has at most d roots when roots are counted according to their multiplicities.

Proof

Apply [proposition 3.19](#). If the distinct roots have multiplicities

$$m_1, \dots, m_s,$$

then

$$m_1 + \cdots + m_s \leq \deg p = d.$$

Since every multiplicity is at least one,

$$s \leq d.$$

Application 2: Interpolation at fixed points

Let

$$x_1, \dots, x_r$$

be r distinct real numbers. Suppose that we are given arbitrary values

$$y_1, \dots, y_r \in \mathbb{R}$$

and want to find a polynomial

$$p \in \mathbb{R}_{r-1}[x]$$

such that

$$p(x_i) = y_i, \quad i = 1, \dots, r.$$

Define the **evaluation map**

$$E : \mathbb{R}_{r-1}[x] \rightarrow \mathbb{R}^r$$

by

$$E(p) = \begin{pmatrix} p(x_1) \\ p(x_2) \\ \vdots \\ p(x_r) \end{pmatrix}.$$

The map E is linear, and the interpolation problem is precisely the equation

$$E(p) = \begin{pmatrix} y_1 \\ \vdots \\ y_r \end{pmatrix}.$$

Theorem 3.21 — Polynomial interpolation at distinct points

Let

$$x_1, \dots, x_r$$

be distinct real numbers. For every

$$y_1, \dots, y_r \in \mathbb{R},$$

there exists a unique polynomial

$$p \in \mathbb{R}_{r-1}[x]$$

such that

$$p(x_i) = y_i, \quad i = 1, \dots, r.$$

Proof

Because

$$\dim \mathbb{R}_{r-1}[x] = r = \dim \mathbb{R}^r,$$

it is enough to show that

$$\ker E = \{0\}.$$

Suppose

$$p \in \ker E.$$

Then

$$p(x_1) = p(x_2) = \dots = p(x_r) = 0.$$

Thus p has r distinct roots.

But a nonzero polynomial in $\mathbb{R}_{r-1}[x]$ has degree at most $r - 1$ and therefore cannot have r distinct roots. Indeed, by [corollary 3.20](#), a nonzero polynomial with r distinct roots must have degree at least r .

Hence

$$p = 0.$$

Thus

$$\ker E = \{0\}.$$

By [proposition 3.12](#), E is bijective. Therefore every vector

$$(y_1, \dots, y_r) \in \mathbb{R}^r$$

corresponds to exactly one polynomial

$$p \in \mathbb{R}_{r-1}[x].$$

Remark 3.22 — Existence and uniqueness without constructing the polynomial

The argument above proves existence and uniqueness without explicitly constructing the interpolating polynomial.

What is then this interpolating polynomial? Valid question. The *Lagrange interpolation formula*

gives an explicit expression for that polynomial. This is beyond the scope of this course. The key point is that: Rank-nullity explains why such a unique polynomial must exist in the first place.

Application 3: Hermite interpolation

Ordinary polynomial interpolation prescribes the value of a polynomial at several distinct points. A natural generalization is to prescribe not only the value of the polynomial, but also some of its derivatives.

Let

$$x_1, \dots, x_s$$

be distinct real numbers, and let

$$m_1, \dots, m_s$$

be positive integers satisfying

$$m_1 + \dots + m_s = r.$$

At the point x_i , we prescribe

$$p(x_i), p'(x_i), \dots, p^{(m_i-1)}(x_i).$$

Thus the point x_i contributes m_i pieces of data, and altogether there are

$$m_1 + \dots + m_s = r$$

pieces of data.

Theorem 3.23 — Hermite interpolation

Let

$$x_1, \dots, x_s$$

be distinct real numbers, and let

$$m_1 + \dots + m_s = r, \quad m_i \geq 1.$$

For arbitrary prescribed values

$$y_{i,j} \in \mathbb{R}, \quad i = 1, \dots, s, \quad j = 0, \dots, m_i - 1,$$

there exists a unique polynomial

$$p \in \mathbb{R}_{r-1}[x]$$

such that

$$p^{(j)}(x_i) = y_{i,j}$$

for every i and j .

Proof

Define the linear map

$$H : \mathbb{R}_{r-1}[x] \rightarrow \mathbb{R}^r$$

by

$$H(p) = (p(x_1), p'(x_1), \dots, p^{(m_1-1)}(x_1); \dots; p(x_s), p'(x_s), \dots, p^{(m_s-1)}(x_s)).$$

Because differentiation and evaluation at a fixed point are linear, H is a linear map.

Moreover,

$$\dim \mathbb{R}_{r-1}[x] = r = \dim \mathbb{R}^r.$$

By [proposition 3.12](#), it is therefore enough to show that

$$\ker H = \{0\}.$$

Suppose

$$p \in \ker H.$$

Then, for every i ,

$$p(x_i) = p'(x_i) = \dots = p^{(m_i-1)}(x_i) = 0.$$

By [proposition 3.18](#), this implies that x_i is a root of p of multiplicity at least m_i .

Thus p has roots whose total multiplicity is at least

$$m_1 + \dots + m_s = r.$$

By [proposition 3.19](#), if p were nonzero, then

$$\deg p \geq r.$$

But

$$p \in \mathbb{R}_{r-1}[x],$$

so

$$\deg p \leq r - 1.$$

This is impossible unless

$$p = 0.$$

Hence

$$\ker H = \{0\}.$$

Since the domain and codomain of H have the same finite dimension, [proposition 3.12](#) implies that H is bijective.

Therefore, for every collection of prescribed values

$$y_{i,j},$$

there exists a unique polynomial

$$p \in \mathbb{R}_{r-1}[x]$$

satisfying all the interpolation conditions.

Remark 3.24 — Lagrange and Taylor interpolation as special cases

Hermite interpolation contains several familiar interpolation problems as special cases.

If

$$m_1 = \cdots = m_r = 1,$$

then only function values are prescribed, and we recover ordinary interpolation at r distinct points.

At the other extreme, if

$$s = 1, \quad m_1 = r,$$

then we prescribe

$$p(x_1), p'(x_1), \dots, p^{(r-1)}(x_1),$$

which is the finite-dimensional algebraic structure underlying Taylor interpolation.

In each case the proof is the same: zero data force sufficiently many zeros, counted with multiplicity, to make a nonzero polynomial of degree at most $r - 1$ impossible.

Application 4: Interpolation from interval averages

The data used in an interpolation problem need not consist of values at individual points. We can instead prescribe averages of the polynomial over intervals.

Let

$$I_i = [a_i, b_i], \quad i = 1, \dots, r,$$

where

$$a_i < b_i,$$

and suppose the interiors of the intervals are pairwise disjoint:

$$(a_i, b_i) \cap (a_j, b_j) = \emptyset \quad \text{whenever } i \neq j.$$

Suppose we are given prescribed average values

$$\mu_1, \dots, \mu_r \in \mathbb{R}$$

and seek

$$p \in \mathbb{R}_{r-1}[x]$$

such that

$$\frac{1}{b_i - a_i} \int_{a_i}^{b_i} p(x) dx = \mu_i, \quad i = 1, \dots, r.$$

Define the linear map

$$\mathcal{A} : \mathbb{R}_{r-1}[x] \rightarrow \mathbb{R}^r$$

by

$$\mathcal{A}(p) = \begin{pmatrix} \frac{1}{b_1 - a_1} \int_{a_1}^{b_1} p(x) dx \\ \vdots \\ \frac{1}{b_r - a_r} \int_{a_r}^{b_r} p(x) dx \end{pmatrix}.$$

Theorem 3.25 — Interpolation from averages over disjoint intervals

Let

$$I_1, \dots, I_r$$

be nondegenerate intervals with pairwise disjoint interiors.

For every collection of prescribed averages

$$\mu_1, \dots, \mu_r \in \mathbb{R},$$

there exists a unique polynomial

$$p \in \mathbb{R}_{r-1}[x]$$

satisfying

$$\frac{1}{b_i - a_i} \int_{a_i}^{b_i} p(x) dx = \mu_i, \quad i = 1, \dots, r.$$

Proof

Because

$$\dim \mathbb{R}_{r-1}[x] = r = \dim \mathbb{R}^r,$$

it is enough to prove

$$\ker \mathcal{A} = \{0\}.$$

Suppose

$$p \in \ker \mathcal{A}.$$

Then

$$\int_{a_i}^{b_i} p(x) dx = 0 \quad \text{for every } i.$$

Let P be an antiderivative of p . Then

$$P(b_i) - P(a_i) = \int_{a_i}^{b_i} p(x) dx = 0.$$

Hence

$$P(a_i) = P(b_i).$$

By Rolle's theorem, there exists

$$\xi_i \in (a_i, b_i)$$

such that

$$P'(\xi_i) = 0.$$

Since

$$P' = p,$$

we obtain

$$p(\xi_i) = 0.$$

Because the interiors of the intervals are pairwise disjoint, the points

$$\xi_1, \dots, \xi_r$$

are distinct. Thus p has at least r distinct roots.

But

$$p \in \mathbb{R}_{r-1}[x],$$

so a nonzero p can have at most $r - 1$ distinct roots. Therefore

$$p = 0.$$

Hence

$$\ker \mathcal{A} = \{0\}.$$

By [proposition 3.12](#), $\mathcal{A}$ is bijective, proving existence and uniqueness.

Remark 3.26 — One principle behind all four applications

The preceding applications have the same mathematical structure.

We begin with a finite-dimensional space of unknown objects and construct a linear map that records the available data:

$$\text{unknown object} \longmapsto \text{observed data}.$$

For example,

$$x \longmapsto Ax$$

records linear equations,

$$p \longmapsto (p(x_1), \dots, p(x_r))$$

records point values,

$$p \longmapsto (p(x_i), p'(x_i), \dots)$$

records values and derivatives, and

$$p \longmapsto \left(\frac{1}{|I_i|} \int_{I_i} p \right)_{i=1}^r$$

records interval averages.

When the unknown space and the data space have the same finite dimension, rank–nullity reduces the entire existence-and-uniqueness question to the homogeneous problem:

$$\ker T = \{0\}.$$

Thus a recurring strategy in linear algebra is:

1. formulate the problem as a linear transformation;
2. study what zero data imply;
3. prove that the kernel is trivial;
4. use rank–nullity to conclude that the map is bijective.

3.4 Rank of a Matrix

We have defined the rank of a linear map

$$T : V \rightarrow W$$

intrinsically by

$$\text{Rank}(T) = \dim \text{Im } T.$$

We now connect this definition with the matrix representing T .

Proposition 3.27 — The image is generated by the images of a basis

Let

$$T : V \rightarrow W$$

be a linear map, and let

$$\mathcal{E} = (e_1, \dots, e_n)$$

be a basis of V . Then

$$\text{Im } T = \text{Span}(T(e_1), \dots, T(e_n)).$$

Proof

Every vector $v \in V$ can be written as

$$v = x_1 e_1 + \dots + x_n e_n.$$

By linearity,

$$T(v) = x_1 T(e_1) + \dots + x_n T(e_n).$$

Hence every vector in $\text{Im } T$ belongs to

$$\text{Span}(T(e_1), \dots, T(e_n)).$$

Thus

$$\text{Im } T \subseteq \text{Span}(T(e_1), \dots, T(e_n)).$$

Conversely, each $T(e_j)$ belongs to $\text{Im } T$, and $\text{Im } T$ is a subspace of W . Therefore every linear combination of

$$T(e_1), \dots, T(e_n)$$

also belongs to $\text{Im } T$. Hence

$$\text{Span}(T(e_1), \dots, T(e_n)) \subseteq \text{Im } T.$$

Therefore

$$\text{Im } T = \text{Span}(T(e_1), \dots, T(e_n)).$$

Now let

$$\mathcal{F} = (f_1, \dots, f_m)$$

be a basis of W , and let

$$A = [T]_{\mathcal{F} \leftarrow \mathcal{E}} \in \mathbb{K}^{m \times n}.$$

Recall that

$$T(e_j) = \sum_{i=1}^m a_{ij} f_i.$$

Therefore the j th column of A is exactly the coordinate vector:

If

$$A = \begin{pmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{pmatrix},$$

then

$$a_j = [T(e_j)]_{\mathcal{F}}.$$

Definition 3.28 — Column space and rank of a matrix

Let

$$A = \begin{pmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{pmatrix} \in \mathbb{K}^{m \times n},$$

where

$$a_1, \dots, a_n \in \mathbb{K}^m$$

are the columns of A .

The **column space** of A is

$$\text{Col}(A) = \text{Span}(a_1, \dots, a_n) \leq \mathbb{K}^m.$$

The **rank** of A is the dimension of its column space:

$$\text{Rank}(A) = \dim \text{Col}(A).$$

Remark 3.29 — What the column space represents

If

$$A = \begin{pmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{pmatrix}$$

and

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

then

$$Ax = x_1 a_1 + \cdots + x_n a_n.$$

Hence

$$\{Ax : x \in \mathbb{K}^n\} = \text{Col}(A).$$

Thus the column space is exactly the set of vectors that can be produced by multiplication by A .

Proposition 3.30 — Rank of a linear map and rank of its matrix

Let

$$T : V \rightarrow W$$

be a linear map between finite-dimensional vector spaces. Let

$$\mathcal{E} = (e_1, \dots, e_n)$$

and

$$\mathcal{F} = (f_1, \dots, f_m)$$

be bases of V and W , respectively, and let

$$A = [T]_{\mathcal{F} \leftarrow \mathcal{E}}.$$

Then

$$\text{Rank}(T) = \text{Rank}(A).$$

Proof

By [proposition 3.27](#),

$$\text{Im } T = \text{Span}(T(e_1), \dots, T(e_n)).$$

The columns of

$$A = [T]_{\mathcal{F} \leftarrow \mathcal{E}}$$

are

$$[T(e_1)]_{\mathcal{F}}, \dots, [T(e_n)]_{\mathcal{F}}.$$

Therefore

$$\text{Col}(A) = \text{Span}([T(e_1)]_{\mathcal{F}}, \dots, [T(e_n)]_{\mathcal{F}}).$$

Passing from a vector in W to its coordinate vector with respect to $\mathcal{F}$ is a linear bijection

$$W \longrightarrow \mathbb{K}^m.$$

Hence it preserves linear independence, spanning, and dimension. (Check this statement as an exercise. Search "Linear Isomorphism" online for hints.)

Therefore

$$\dim \text{Im } T = \dim \text{Col}(A).$$

By the definitions of rank,

$$\text{Rank}(T) = \text{Rank}(A).$$

Remark 3.31 — The column space is the image in coordinates

If

$$A = [T]_{\mathcal{F} \leftarrow \mathcal{E}},$$

then

$$\text{Col}(A)$$

is the coordinate representation, with respect to $\mathcal{F}$, of

$$\text{Im } T.$$

Thus

$$\text{Im } T \quad \text{and} \quad \text{Col}(A)$$

represent the same collection of outputs, one intrinsically in W and the other in coordinates in $\mathbb{K}^m$.

Now let's turn the rank-nullity theorem into the world of matrices.

Definition 3.32 — Kernel and nullity of a matrix

Let

$$A \in \mathbb{K}^{m \times n}.$$

The **kernel** or **null space** of A is

$$\ker A = \{x \in \mathbb{K}^n : Ax = 0\}.$$

Its dimension is called the **nullity** of A :

$$\text{nullity}(A) = \dim \ker A.$$

Corollary 3.33 — Rank-nullity for matrices

Let

$$A \in \mathbb{K}^{m \times n}.$$

Then

$$n = \text{Rank}(A) + \text{nullity}(A).$$

Equivalently,

$$n = \text{Rank}(A) + \dim \ker A.$$

Proof

Consider the linear map

$$T_A : \mathbb{K}^n \rightarrow \mathbb{K}^m, \quad T_A(x) = Ax.$$

Its kernel is

$$\ker T_A = \{x \in \mathbb{K}^n : Ax = 0\} = \ker A,$$

and its image is the column space of A :

$$\text{Im } T_A = \text{Col}(A).$$

Hence

$$\text{Rank}(T_A) = \text{Rank}(A).$$

Applying the rank-nullity theorem to T_A gives

$$\dim \mathbb{K}^n = \dim \ker T_A + \text{Rank}(T_A).$$

Therefore

$$n = \dim \ker A + \text{Rank}(A).$$

Remark 3.34 — Why the number of columns appears in rank–nullity

For

$$A \in \mathbb{K}^{m \times n},$$

the matrix represents a linear map

$$\mathbb{K}^n \rightarrow \mathbb{K}^m.$$

Thus the domain has dimension n , which is exactly the number of columns of A .

This is why the matrix rank–nullity formula is

$$\text{Rank}(A) + \text{nullity}(A) = n,$$

rather than m .

As a useful corollary, we obtain the following bound on the rank of a matrix.

Corollary 3.35 — Basic bounds on matrix rank

If

$$A \in \mathbb{K}^{m \times n},$$

then

$$0 \leq \text{Rank}(A) \leq \min\{m, n\}.$$

Proof

Because

$$\text{Col}(A) \leq \mathbb{K}^m,$$

we have

$$\text{Rank}(A) = \dim \text{Col}(A) \leq m.$$

Also,

$$\text{Col}(A)$$

is spanned by the n columns of A , so

$$\text{Rank}(A) \leq n.$$

Therefore

$$\text{Rank}(A) \leq \min\{m, n\}.$$

There are all sorts of other rank inequalities for matrices. We will not touch upon those in this course.

Remark 3.36 — Rank does not depend on the choice of bases

The matrix representing a linear map depends on the chosen bases, but its rank does not.

Indeed, if

$$A = [T]_{\mathcal{F} \leftarrow \mathcal{E}},$$

then

$$\text{Rank}(A) = \text{Rank}(T) = \dim \text{Im } T.$$

The quantity on the right is defined entirely in terms of the linear map T and does not involve any choice of coordinates.

Consequently, every matrix representation of the same linear map has the same rank.

Thus the entries of a matrix depend on the choice of coordinates, but its rank measures an intrinsic property of the underlying linear map.

3.5 Change of Bases for a Linear Map

We have already seen how the coordinate vector of a vector changes when the basis is changed. We now apply the same idea to the matrix representing a linear map.

Let

$$T : V \rightarrow W$$

be linear. Let

$$\mathcal{E} = (e_1, \dots, e_n), \quad \mathcal{E}' = (e'_1, \dots, e'_n)$$

be bases of V , and let

$$\mathcal{F} = (f_1, \dots, f_m), \quad \mathcal{F}' = (f'_1, \dots, f'_m)$$

be bases of W .

Suppose

$$(e'_1, \dots, e'_n) = (e_1, \dots, e_n)P$$

and

$$(f'_1, \dots, f'_m) = (f_1, \dots, f_m)Q,$$

where P and Q are invertible.

Recall that

$$[v]_{\mathcal{E}} = P[v]_{\mathcal{E}'}$$

for every $v \in V$, and

$$[w]_{\mathcal{F}} = Q[w]_{\mathcal{F}'}$$

for every $w \in W$.

Equivalently,

$$P = [I_V]_{\mathcal{E} \leftarrow \mathcal{E}'}, \quad Q = [I_W]_{\mathcal{F} \leftarrow \mathcal{F}'}$$

Proposition 3.37 — Change of bases for a linear map

Let

$$A = [T]_{\mathcal{F} \leftarrow \mathcal{E}}, \quad B = [T]_{\mathcal{F}' \leftarrow \mathcal{E}'}$$

Then

$$B = Q^{-1}AP.$$

Proof

For every $v \in V$,

$$[T(v)]_{\mathcal{F}} = A[v]_{\mathcal{E}}.$$

Since

$$[v]_{\mathcal{E}} = P[v]_{\mathcal{E}'},$$

we have

$$[T(v)]_{\mathcal{F}} = AP[v]_{\mathcal{E}'}$$

On the other hand,

$$[T(v)]_{\mathcal{F}'} = B[v]_{\mathcal{E}'},$$

and because

$$[T(v)]_{\mathcal{F}} = Q[T(v)]_{\mathcal{F}'},$$

we obtain

$$[T(v)]_{\mathcal{F}} = QB[v]_{\mathcal{E}'}$$

Hence

$$AP[v]_{\mathcal{E}'} = QB[v]_{\mathcal{E}'}$$

for every $v \in V$. Since the coordinate vectors $[v]_{\mathcal{E}'}$ range over all of $\mathbb{K}^n$,

$$AP = QB.$$

Therefore

$$B = Q^{-1}AP.$$

Remark 3.38 — How to read the change-of-bases formula

The formula

$$[T]_{\mathcal{F}' \leftarrow \mathcal{E}'} = Q^{-1}[T]_{\mathcal{F} \leftarrow \mathcal{E}}P$$

reflects the order of the coordinate changes:

$$[v]_{\mathcal{E}'} \xrightarrow{P} [v]_{\mathcal{E}} \xrightarrow{[T]_{\mathcal{F} \leftarrow \mathcal{E}}} [T(v)]_{\mathcal{F}} \xrightarrow{Q^{-1}} [T(v)]_{\mathcal{F}'}.$$

Thus changing the basis of the domain acts on the right of the matrix, whereas changing the basis of the codomain acts on the left.

Corollary 3.39 — Change of basis for a linear operator

Let

$$T : V \rightarrow V$$

be a linear transformation, and suppose

$$\mathcal{E}' = \mathcal{E}P.$$

Then

$$[T]_{\mathcal{E}' \leftarrow \mathcal{E}'} = P^{-1}[T]_{\mathcal{E} \leftarrow \mathcal{E}}P.$$

Definition 3.40 — Similar matrices

Two square matrices A and B are called **similar** if there exists an invertible matrix P such that

$$B = P^{-1}AP.$$

Remark 3.41 — One basis change versus two

For a linear transformation

$$T : V \rightarrow V,$$

using the same basis for the domain and codomain leads to similarity:

$$B = P^{-1}AP.$$

For a general linear map

$$T : V \rightarrow W,$$

the bases of V and W may be chosen independently, giving

$$B = Q^{-1}AP.$$

This greater freedom will allow us to choose bases adapted to the kernel and image of T and obtain the rank normal form

$$[T]_{\mathcal{F}' \leftarrow \mathcal{E}'} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}, \quad r = \text{Rank}(T).$$

Equivalently, every matrix A of rank r can be transformed by invertible matrices on the left and right into this form. Coming in the next subsection!

3.6 Row Rank and the Rank Normal Form

We have defined the rank of a matrix using its columns: if

$$A \in \mathbb{K}^{m \times n},$$

then

$$\text{Rank}(A) = \dim \text{Col}(A).$$

There is an analogous space generated by the rows of A . At first, there is no obvious reason why the dimensions of these two spaces should be the same. We will prove, however, that they are equal.

Definition 3.42 — Row space and row rank

Let

$$A \in \mathbb{K}^{m \times n},$$

and let

$$\rho_1, \dots, \rho_m \in \mathbb{K}_n$$

denote the rows of A .

The **row space** of A is

$$\text{Row}(A) = \text{Span}(\rho_1, \dots, \rho_m) \leq \mathbb{K}_n.$$

The **row rank** of A is

$$\dim \text{Row}(A).$$

At this point we therefore have two quantities:

$$\underbrace{\dim \text{Col}(A)}_{\text{column rank}} \quad \text{and} \quad \underbrace{\dim \text{Row}(A)}_{\text{row rank}}.$$

Our definition of $\text{Rank}(A)$ is the first of these. We will show below that the second always has exactly the same value.

A basis adapted to a linear map

The rank–nullity theorem gives more than a dimension formula. Its proof shows how to choose bases of the domain and codomain that reveal the structure of a linear map particularly clearly.

Proposition 3.43 — Adapted-basis form of a linear map

Let

$$T : V \rightarrow W$$

be a linear map between finite-dimensional vector spaces, and suppose

$$\dim V = n, \quad \dim W = m, \quad \text{Rank}(T) = r.$$

Then there exist bases

$$\mathcal{E}' = (v_1, \dots, v_r, u_1, \dots, u_{n-r})$$

of V and

$$\mathcal{F}' = (T(v_1), \dots, T(v_r), w_{r+1}, \dots, w_m)$$

of W such that

$$[T]_{\mathcal{F}' \leftarrow \mathcal{E}'} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Proof

Choose a basis

$$u_1, \dots, u_{n-r}$$

of $\ker T$. By rank–nullity,

$$\dim \ker T = n - r.$$

By the basis extension theorem, extend this basis of $\ker T$ to a basis of V . After reordering the basis vectors, write this basis as

$$\mathcal{E}' = (v_1, \dots, v_r, u_1, \dots, u_{n-r}).$$

As we proved in the rank–nullity theorem,

$$T(v_1), \dots, T(v_r)$$

is a basis of $\text{Im } T$.

Since $\text{Im } T$ is a subspace of W , extend this basis of $\text{Im } T$ to a basis of W :

$$\mathcal{F}' = (T(v_1), \dots, T(v_r), w_{r+1}, \dots, w_m).$$

Now consider the matrix of T with respect to these bases.

For $j = 1, \dots, r$,

$$T(v_j)$$

is the j th vector of the basis $\mathcal{F}'$. Hence

$$[T(v_j)]_{\mathcal{F}'} = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix},$$

with the 1 in position j .

Thus the first r columns of

$$[T]_{\mathcal{F}' \leftarrow \mathcal{E}'}$$

form the block

$$\begin{pmatrix} I_r \\ 0 \end{pmatrix}.$$

For the remaining basis vectors,

$$u_j \in \ker T,$$

and hence

$$T(u_j) = 0.$$

Therefore all the remaining columns are zero.

Consequently,

$$[T]_{\mathcal{F}' \leftarrow \mathcal{E}'} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Remark 3.44 — What the adapted-basis form means

The matrix

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

shows the essential structure of a rank- r linear map.

After suitable choices of coordinates,

$$T$$

keeps r independent directions and sends the remaining $n - r$ independent directions to zero. Thus, up to independent choices of bases in the domain and codomain, the integer

$$r = \text{Rank}(T)$$

is the only information needed to describe the basic structure of the linear map.

The rank normal form

We now translate the preceding statement about linear maps into a statement about matrices.

Recall that if

$$A = [T]_{\mathcal{F} \leftarrow \mathcal{E}}$$

and we change the bases in the domain and codomain, then

$$[T]_{\mathcal{F}' \leftarrow \mathcal{E}'} = Q^{-1}AP$$

for suitable invertible change-of-basis matrices P and Q .

Theorem 3.45 — Rank normal form

Let

$$A \in \mathbb{K}^{m \times n}$$

and suppose

$$\text{Rank}(A) = r.$$

Then there exist invertible matrices

$$P \in \mathbb{K}^{m \times m}, \quad Q \in \mathbb{K}^{n \times n}$$

such that

$$PAQ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Proof

Regard A as the matrix of the associated linear map

$$T_A : \mathbb{K}^n \rightarrow \mathbb{K}^m, \quad T_A(x) = Ax,$$

with respect to the standard bases.

We have already shown that

$$\text{Rank}(T_A) = \text{Rank}(A) = r.$$

By [proposition 3.43](#), there exist bases $\mathcal{E}'$ of $\mathbb{K}^n$ and $\mathcal{F}'$ of $\mathbb{K}^m$ such that

$$[T_A]_{\mathcal{F}' \leftarrow \mathcal{E}'} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

By the change-of-bases formula, there exist invertible matrices $Q_0 \in \mathbb{K}^{n \times n}$ and $P_0 \in \mathbb{K}^{m \times m}$ such that

$$[T_A]_{\mathcal{F}' \leftarrow \mathcal{E}'} = P_0^{-1} A Q_0.$$

Therefore

$$P_0^{-1} A Q_0 = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Setting

$$P = P_0^{-1}, \quad Q = Q_0,$$

gives

$$PAQ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Remark 3.46 — The rank normal form is a change-of-basis statement

The rank normal form should not be viewed merely as a matrix simplification procedure.

Left multiplication by an invertible matrix corresponds to changing coordinates in the codomain, while right multiplication by an invertible matrix corresponds to changing coordinates in the domain.

Thus

$$PAQ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

is the matrix version of the statement that every rank- r linear map takes the same simple form after suitable choices of bases.

Behavior of the row space under invertible multiplication

To compare row rank with column rank, we first record how the row space behaves when a matrix is multiplied by invertible matrices.

Proposition 3.47 — Row rank is preserved by invertible multiplication

Let

$$A \in \mathbb{K}^{m \times n},$$

and let

$$P \in \mathbb{K}^{m \times m}, \quad Q \in \mathbb{K}^{n \times n}$$

be invertible. Then

$$\dim \text{Row}(PAQ) = \dim \text{Row}(A).$$

Proof

Write

$$P = (p_{ij})_{i,j=1}^m,$$

and denote the rows of A by

$$A_{1,\cdot}, \dots, A_{m,\cdot} \in \mathbb{K}_n.$$

We consider left and right multiplication separately.

First consider left multiplication by P .

By the rule for matrix multiplication, the i th row of PA is

$$(PA)_{i,\cdot} = \sum_{j=1}^m p_{ij} A_{j,\cdot}.$$

Indeed, for each column k ,

$$(PA)_{ik} = \sum_{j=1}^m p_{ij} a_{jk},$$

which is precisely the k th entry of the row vector

$$\sum_{j=1}^m p_{ij} A_{j,\cdot}.$$

Thus every row of PA is a linear combination of the rows of A . Therefore

$$\text{Row}(PA) \subseteq \text{Row}(A).$$

To obtain the reverse inclusion, use the invertibility of P . Since

$$A = P^{-1}(PA),$$

write

$$P^{-1} = (c_{ij})_{i,j=1}^m.$$

Applying the same row formula gives

$$A_{i,\cdot} = \sum_{j=1}^m c_{ij}(PA)_{j,\cdot}.$$

Hence every row of A is a linear combination of the rows of PA . Therefore

$$\text{Row}(A) \subseteq \text{Row}(PA).$$

Combining the two inclusions,

$$\text{Row}(PA) = \text{Row}(A).$$

In particular,

$$\dim \text{Row}(PA) = \dim \text{Row}(A).$$

Now consider right multiplication by Q .

For each $i = 1, \dots, m$, the i th row of AQ is

$$(AQ)_{i,\cdot} = A_{i,\cdot}Q.$$

Thus right multiplication by Q applies the same linear transformation

$$R_Q : \mathbb{K}_n \rightarrow \mathbb{K}_n, \quad R_Q(x) = xQ,$$

to every row of A .

We now show that

$$\text{Row}(AQ) = R_Q(\text{Row}(A)).$$

Take any vector

$$y \in \text{Row}(AQ).$$

Then y is a linear combination of the rows of AQ , so for some scalars $\lambda_1, \dots, \lambda_m$,

$$y = \sum_{i=1}^m \lambda_i (AQ)_{i,\cdot}.$$

Using

$$(AQ)_{i,\cdot} = A_{i,\cdot}Q,$$

we obtain

$$\begin{aligned} y &= \sum_{i=1}^m \lambda_i(A_{i,\cdot}, Q) \\ &= \left(\sum_{i=1}^m \lambda_i A_{i,\cdot} \right) Q. \end{aligned}$$

Since

$$\sum_{i=1}^m \lambda_i A_{i,\cdot} \in \text{Row}(A),$$

we have

$$y \in R_Q(\text{Row}(A)).$$

Hence

$$\text{Row}(AQ) \subseteq R_Q(\text{Row}(A)).$$

Conversely, take

$$y \in R_Q(\text{Row}(A)).$$

Then

$$y = xQ$$

for some

$$x \in \text{Row}(A).$$

Write

$$x = \sum_{i=1}^m \lambda_i A_{i,\cdot}.$$

Then

$$\begin{aligned} y &= \left(\sum_{i=1}^m \lambda_i A_{i,\cdot} \right) Q \\ &= \sum_{i=1}^m \lambda_i(A_{i,\cdot}, Q) \\ &= \sum_{i=1}^m \lambda_i(AQ)_{i,\cdot}. \end{aligned}$$

Therefore

$$y \in \text{Row}(AQ).$$

Thus

$$\text{Row}(AQ) = R_Q(\text{Row}(A)).$$

Because Q is invertible, the map

$$R_Q : x \mapsto xQ$$

is a linear bijection, with inverse

$$R_{Q^{-1}} : x \mapsto xQ^{-1}.$$

A linear bijection preserves dimension, so

$$\dim \text{Row}(AQ) = \dim \text{Row}(A).$$

Finally, apply these two results successively:

$$\dim \text{Row}(PAQ) = \dim \text{Row}(AQ) = \dim \text{Row}(A).$$

Therefore

$$\dim \text{Row}(PAQ) = \dim \text{Row}(A).$$

Equality of row rank and column rank

We can now prove one of the fundamental facts about matrix rank.

Theorem 3.48 — Row rank equals column rank

For every matrix

$$A \in \mathbb{K}^{m \times n},$$

the dimension of the row space equals the dimension of the column space:

$$\dim \text{Row}(A) = \dim \text{Col}(A).$$

Hence

$$\dim \text{Row}(A) = \text{Rank}(A).$$

Proof

Let

$$r = \text{Rank}(A) = \dim \text{Col}(A).$$

By the rank normal form theorem, there exist invertible matrices P and Q such that

$$PAQ = R,$$

where

$$R = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

By [proposition 3.47](#),

$$\dim \text{Row}(A) = \dim \text{Row}(R).$$

The matrix R has exactly r nonzero rows, namely

$$(1, 0, \dots, 0), \quad (0, 1, \dots, 0), \quad \dots, \quad (0, \dots, 1, 0, \dots, 0),$$

and these rows are linearly independent. Therefore

$$\dim \text{Row}(R) = r.$$

Hence

$$\dim \text{Row}(A) = r.$$

But

$$r = \dim \text{Col}(A).$$

Therefore

$$\dim \text{Row}(A) = \dim \text{Col}(A).$$

Remark 3.49 — The two meanings of matrix rank

We originally defined

$$\text{Rank}(A) = \dim \text{Col}(A).$$

The preceding theorem shows that we may equally write

$$\text{Rank}(A) = \dim \text{Row}(A).$$

Thus

$$\text{Rank}(A) = \text{number of independent columns} = \text{number of independent rows}.$$

Corollary 3.50 — Rank of the transpose

For every matrix

$$A \in \mathbb{K}^{m \times n},$$

we have

$$\text{Rank}(A^\top) = \text{Rank}(A).$$

Proof

The columns of $A^\top$ are precisely the transposes of the rows of A . Hence

$$\dim \text{Col}(A^\top) = \dim \text{Row}(A).$$

By the equality of row rank and column rank,

$$\dim \text{Row}(A) = \text{Rank}(A).$$

Therefore

$$\text{Rank}(A^\top) = \dim \text{Col}(A^\top) = \text{Rank}(A).$$

Matrix equivalence

The rank normal form also gives a complete classification of matrices under independent changes of coordinates in the domain and codomain.

Definition 3.51 — Equivalent matrices

Two matrices

$$A, B \in \mathbb{K}^{m \times n}$$

are called **equivalent** if there exist invertible matrices

$$P \in \mathbb{K}^{m \times m}, \quad Q \in \mathbb{K}^{n \times n}$$

such that

$$B = PAQ.$$

Remark 3.52 — Equivalence versus similarity

Matrix equivalence should be distinguished from matrix similarity.

Similarity has the form

$$B = P^{-1}AP$$

and corresponds to changing the basis of a linear transformation

$$T : V \rightarrow V$$

in the same way in both the domain and codomain.

Matrix equivalence has the more general form

$$B = PAQ,$$

where the two invertible matrices are independent. It corresponds to changing bases independently in the domain and codomain of a linear map

$$T : V \rightarrow W.$$

Corollary 3.53 — Rank is invariant under matrix equivalence

If

$$B = PAQ,$$

where P and Q are invertible, then

$$\text{Rank}(B) = \text{Rank}(A).$$

Proof

By the equality of row rank and column rank,

$$\text{Rank}(B) = \dim \text{Row}(B).$$

Since

$$B = PAQ,$$

[proposition 3.47](#) gives

$$\dim \text{Row}(B) = \dim \text{Row}(A).$$

Hence

$$\text{Rank}(B) = \text{Rank}(A).$$

Theorem 3.54 — Rank classifies matrices up to equivalence

Let

$$A, B \in \mathbb{K}^{m \times n}.$$

Then A and B are equivalent if and only if

$$\text{Rank}(A) = \text{Rank}(B).$$

Proof

Suppose first that A and B are equivalent. Then

$$B = PAQ$$

for some invertible P and Q . By [corollary 3.53](#),

$$\text{Rank}(B) = \text{Rank}(A).$$

Conversely, suppose

$$\text{Rank}(A) = \text{Rank}(B) = r.$$

By the rank normal form theorem, there exist invertible matrices P_A, Q_A, P_B, Q_B such that

$$P_A A Q_A = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} = P_B B Q_B.$$

Therefore

$$B = P_B^{-1} P_A A Q_A Q_B^{-1}.$$

Both

$$P_B^{-1} P_A \quad \text{and} \quad Q_A Q_B^{-1}$$

are invertible. Hence A and B are equivalent.

Remark 3.55 — Rank as the complete invariant of matrix equivalence

For matrices of a fixed size,

$$\text{Rank}(A) = r$$

completely determines the equivalence class of A .

Indeed, every $m \times n$ matrix of rank r is equivalent to the same matrix

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus, after arbitrary choices of bases in both the domain and codomain, all matrices of the same rank represent the same basic linear-map structure.

Full-rank matrices

The equality of row rank and column rank also gives useful terminology for matrices whose rows or columns are as independent as their dimensions allow.

Definition 3.56 — Full row rank and full column rank

Let

$$A \in \mathbb{K}^{m \times n}.$$

We say that A has **full row rank** if

$$\text{Rank}(A) = m,$$

and that A has **full column rank** if

$$\text{Rank}(A) = n.$$

Thus full row rank means that the m rows of A are linearly independent, whereas full column rank means that the n columns of A are linearly independent.

Square matrices as a special class of matrices have the following corollary.

Corollary 3.57 — Rank and invertibility of a square matrix

Let

$$A \in \mathbb{K}^{n \times n}.$$

Then the following statements are equivalent:

- (i) A is invertible;
- (ii) $\text{Rank}(A) = n$;
- (iii) the columns of A form a basis of $\mathbb{K}^n$;
- (iv) the rows of A form a basis of $\mathbb{K}^n$;
- (v) $\ker A = \{0\}$.

Proof

We prove the equivalences by linking each statement to the rank condition

$$\text{Rank}(A) = n.$$

First, we show that

$$(i) \implies (v).$$

Suppose A is invertible and

$$x \in \ker A.$$

Then

$$Ax = 0.$$

Multiplying by A^{-1} gives

$$x = A^{-1}Ax = A^{-1}0 = 0.$$

Hence

$$\ker A = \{0\}.$$

Next, by the matrix form of the rank–nullity theorem, [corollary 3.33](#),

$$n = \text{Rank}(A) + \dim \ker A.$$

Therefore,

$$\ker A = \{0\} \iff \dim \ker A = 0 \iff \text{Rank}(A) = n.$$

Thus

$$(v) \iff (ii).$$

Now consider the columns of A .

By [definition 3.28](#),

$$\text{Rank}(A) = \dim \text{Col}(A),$$

where $\text{Col}(A)$ is the span of the n columns of A . Hence

$$\text{Rank}(A) = n$$

means that the column space has dimension n .

Since

$$\text{Col}(A) \leq \mathbb{K}^n \quad \text{and} \quad \dim \mathbb{K}^n = n,$$

we must have

$$\text{Col}(A) = \mathbb{K}^n.$$

Thus the n columns of A span $\mathbb{K}^n$.

By [proposition 1.21](#), n vectors that span an n -dimensional vector space form a basis. Therefore,

$$\text{Rank}(A) = n \iff \text{the columns of } A \text{ form a basis of } \mathbb{K}^n.$$

Hence

$$(ii) \iff (iii).$$

For the rows, [theorem 3.48](#) gives

$$\dim \text{Row}(A) = \text{Rank}(A).$$

Therefore

$$\text{Rank}(A) = n \iff \dim \text{Row}(A) = n.$$

Since the row space is a subspace of $\mathbb{K}_n$, which has dimension n , this is equivalent to

$$\text{Row}(A) = \mathbb{K}_n.$$

Thus the n rows of A span $\mathbb{K}_n$.

Again, by [proposition 1.21](#), the n rows therefore form a basis of $\mathbb{K}_n$. Hence

$$(ii) \iff (iv).$$

It remains to close the implication back to invertibility.

Suppose

$$\text{Rank}(A) = n.$$

By the rank normal form theorem, [theorem 3.45](#), there exist invertible matrices $P, Q \in \mathbb{K}^{n \times n}$ such that

$$PAQ = \begin{pmatrix} I_n \\ 0 \end{pmatrix} = I_n.$$

Hence

$$A = P^{-1}Q^{-1}.$$

Since P^{-1} and Q^{-1} are invertible, their product is invertible. Therefore A is invertible.

Thus

$$(ii) \implies (i).$$

Combining the implications,

$$(i) \iff (ii) \iff (iii) \iff (iv) \iff (v).$$

4 Sums, Direct Sums, and Invariant Subspaces

We have studied individual subspaces of a vector space. We now study how different subspaces are related to one another.

4.1 Intersections and Sums of Subspaces

Let U and W be subspaces of a vector space V .

Definition 4.1 — Intersection and sum of subspaces

The **intersection** of U and W is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

The **sum** of U and W is

$$U + W = \{u + w : u \in U, w \in W\}.$$

The intersection consists of those vectors belonging to both subspaces. The sum consists of all vectors that can be obtained by taking one vector from each subspace and adding them.

Proposition 4.2 — Intersection and sum are subspaces

Let U and W be subspaces of V . Then

$$U \cap W \quad \text{and} \quad U + W$$

are both subspaces of V .

Moreover:

- (1) $U \cap W$ is the largest subspace contained in both U and W ;
- (2) $U + W$ is the smallest subspace containing both U and W .

Proof

We first consider the intersection.

Because U and W are subspaces,

$$0 \in U \quad \text{and} \quad 0 \in W,$$

so

$$0 \in U \cap W.$$

If

$$x, y \in U \cap W,$$

then

$$x, y \in U \quad \text{and} \quad x, y \in W.$$

Since both U and W are subspaces,

$$x + y \in U \quad \text{and} \quad x + y \in W.$$

Therefore

$$x + y \in U \cap W.$$

Similarly, if

$$c \in \mathbb{K} \quad \text{and} \quad x \in U \cap W,$$

then

$$cx \in U \quad \text{and} \quad cx \in W,$$

so

$$cx \in U \cap W.$$

Hence $U \cap W$ is a subspace.

Now consider the sum. Since

$$0 = 0 + 0, \quad 0 \in U, \quad 0 \in W,$$

we have

$$0 \in U + W.$$

Suppose

$$x = u_1 + w_1, \quad y = u_2 + w_2$$

belong to $U + W$, where

$$u_1, u_2 \in U, \quad w_1, w_2 \in W.$$

Then

$$x + y = (u_1 + u_2) + (w_1 + w_2).$$

Because

$$u_1 + u_2 \in U, \quad w_1 + w_2 \in W,$$

we have

$$x + y \in U + W.$$

Similarly, for $c \in \mathbb{K}$,

$$cx = c(u_1 + w_1) = cu_1 + cw_1,$$

where

$$cu_1 \in U, \quad cw_1 \in W.$$

Thus

$$cx \in U + W.$$

Hence $U + W$ is also a subspace.

It remains to verify the two extremal properties.

Clearly,

$$U \cap W \subseteq U \quad \text{and} \quad U \cap W \subseteq W.$$

If S is any subspace satisfying

$$S \subseteq U \quad \text{and} \quad S \subseteq W,$$

then every vector of S belongs to both U and W . Hence

$$S \subseteq U \cap W.$$

Thus $U \cap W$ is the largest subspace contained in both.

For the sum, observe that

$$u = u + 0 \in U + W \quad \text{for every } u \in U,$$

and similarly

$$w = 0 + w \in U + W \quad \text{for every } w \in W.$$

Thus

$$U \subseteq U + W, \quad W \subseteq U + W.$$

Now suppose S is any subspace containing both U and W . If

$$x \in U + W,$$

then

$$x = u + w$$

for some

$$u \in U, \quad w \in W.$$

Since

$$U, W \subseteq S$$

and S is closed under addition,

$$u + w \in S.$$

Therefore

$$U + W \subseteq S.$$

Hence $U + W$ is the smallest subspace containing both U and W .

Remark 4.3 — Sum versus union

In general,

$$U \cup W$$

is not a subspace. If

$$u \in U \setminus W \quad \text{and} \quad w \in W \setminus U,$$

then typically

$$u + w \notin U \cup W.$$

The sum $U + W$ repairs this problem by including all linear combinations generated jointly by the two subspaces.

Indeed,

$$U + W = \text{Span}(U \cup W).$$

(Try to prove this, it should not be too hard.)

Example 4.4 — Intersection and sum in $\mathbb{R}^3$

Let

$$U = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} : x, y \in \mathbb{R} \right\}$$

be the xy -plane, and let

$$W = \left\{ \begin{pmatrix} 0 \\ y \\ z \end{pmatrix} : y, z \in \mathbb{R} \right\}$$

be the yz -plane.

A vector belongs to both U and W precisely when its first and third coordinates are zero. Hence

$$U \cap W = \left\{ \begin{pmatrix} 0 \\ y \\ 0 \end{pmatrix} : y \in \mathbb{R} \right\},$$

which is the y -axis.

On the other hand,

$$U + W = \mathbb{R}^3.$$

Indeed, for every

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3,$$

we may write

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix},$$

where the first vector belongs to U and the second belongs to W .

4.2 The Dimension Formula

The dimensions of the sum and intersection are related by an important counting formula.

Theorem 4.5 — Dimension formula for two subspaces

Let U and W be subspaces of a finite-dimensional vector space V . Then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Proof

Let

$$v_1, \dots, v_r$$

be a basis of

$$U \cap W.$$

Thus

$$\dim(U \cap W) = r.$$

Because

$$v_1, \dots, v_r$$

is linearly independent in U , by [proposition 3.10](#) we may extend it to a basis

$$v_1, \dots, v_r, u_1, \dots, u_p$$

of U .

Similarly, extend the same basis of $U \cap W$ to a basis

$$v_1, \dots, v_r, w_1, \dots, w_q$$

of W .

We claim that

$$v_1, \dots, v_r, u_1, \dots, u_p, w_1, \dots, w_q$$

is a basis of $U + W$.

First, this list spans $U + W$. Indeed, every

$$x \in U + W$$

can be written as

$$x = u + w, \quad u \in U, \quad w \in W.$$

Since the first extended list is a basis of U and the second is a basis of W , both u and w are linear combinations of vectors in the displayed list. Hence so is x .

It remains to prove linear independence.

Suppose

$$\sum_{i=1}^r a_i v_i + \sum_{j=1}^p b_j u_j + \sum_{k=1}^q c_k w_k = 0.$$

Rearranging,

$$\sum_{k=1}^q c_k w_k = -\sum_{i=1}^r a_i v_i - \sum_{j=1}^p b_j u_j.$$

The right-hand side belongs to U , whereas the left-hand side belongs to W . Therefore

$$\sum_{k=1}^q c_k w_k \in U \cap W.$$

Because

$$v_1, \dots, v_r$$

is a basis of $U \cap W$, there exist scalars $d_1, \dots, d_r$ such that

$$\sum_{k=1}^q c_k w_k = \sum_{i=1}^r d_i v_i.$$

Hence

$$\sum_{i=1}^r d_i v_i - \sum_{k=1}^q c_k w_k = 0.$$

But

$$v_1, \dots, v_r, w_1, \dots, w_q$$

is a basis of W , and therefore is linearly independent. Thus

$$c_1 = \dots = c_q = 0.$$

The original relation now reduces to

$$\sum_{i=1}^r a_i v_i + \sum_{j=1}^p b_j u_j = 0.$$

Since

$$v_1, \dots, v_r, u_1, \dots, u_p$$

is a basis of U , all the remaining coefficients are zero.

Thus the displayed list is linearly independent and hence is a basis of $U + W$.

Consequently,

$$\dim(U + W) = r + p + q.$$

At the same time,

$$\dim U = r + p, \quad \dim W = r + q, \quad \dim(U \cap W) = r.$$

Therefore

$$\begin{aligned} \dim(U + W) &= r + p + q \\ &= (r + p) + (r + q) - r \\ &= \dim U + \dim W - \dim(U \cap W). \end{aligned}$$

Remark 4.6 — Analogy with inclusion–exclusion

The formula

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$$

is the vector-space analogue of the counting identity

$$\#(A \cup B) = \#A + \#B - \#(A \cap B)$$

for finite sets.

The common part must be subtracted once because it has been counted in both subspaces.

Corollary 4.7 — Dimension bound for a sum

Under the assumptions of [theorem 4.5](#),

$$\dim(U + W) \leq \dim U + \dim W,$$

with equality if and only if

$$U \cap W = \{0\}.$$

Proof

By [theorem 4.5](#),

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Since

$$\dim(U \cap W) \geq 0,$$

the inequality follows.

Equality holds precisely when

$$\dim(U \cap W) = 0,$$

which is equivalent to

$$U \cap W = \{0\}.$$

4.3 Direct Sums

Every vector in $U + W$ can be written as

$$u + w, \quad u \in U, \quad w \in W.$$

In general, however, this representation need not be unique.

Direct sums are precisely the case in which it is unique.

Definition 4.8 — Direct sum

Let

$$V_1, \dots, V_m$$

be subspaces of V .

Their sum

$$V_1 + \dots + V_m$$

is called a **direct sum** if every vector in the sum has a unique representation

$$v = v_1 + \dots + v_m, \quad v_i \in V_i.$$

In this case we write

$$V_1 \oplus \dots \oplus V_m.$$

Looking at the representation of every vector in a sum to determine whether this sum is direct is simply not feasible. We therefore seek some characterizations of a direct sum.

Theorem 4.9 — Characterizations of a direct sum

Let

$$V_1, \dots, V_m$$

be finite-dimensional subspaces of V , and let

$$S = V_1 + \dots + V_m.$$

Then the following statements are equivalent:

(i)

$$S = V_1 \oplus \dots \oplus V_m.$$

(ii) If

$$v_1 + \cdots + v_m = 0, \quad v_i \in V_i,$$

then

$$v_1 = \cdots = v_m = 0.$$

(iii) For every

$$i = 2, \dots, m,$$

we have

$$V_i \cap (V_1 + \cdots + V_{i-1}) = \{0\}.$$

(iv)

$$\dim S = \dim V_1 + \cdots + \dim V_m.$$

(v) If $\mathcal{B}_i$ is a basis of V_i for each i , then concatenating the lists

$$\mathcal{B}_1, \dots, \mathcal{B}_m$$

gives a basis of S .

Proof

We show the equivalences in several steps.

(i) $\iff$ (ii).

Suppose first that the sum is direct. The zero vector already has the representation

$$0 = 0 + \cdots + 0.$$

By uniqueness, any representation

$$0 = v_1 + \cdots + v_m, \quad v_i \in V_i,$$

must therefore satisfy

$$v_1 = \cdots = v_m = 0.$$

Conversely, suppose (ii) holds. Let

$$v = v_1 + \cdots + v_m$$

and

$$v = u_1 + \cdots + u_m$$

be two representations of the same vector, where

$$v_i, u_i \in V_i.$$

Subtracting gives

$$0 = (v_1 - u_1) + \cdots + (v_m - u_m).$$

Because

$$v_i - u_i \in V_i,$$

condition (ii) implies

$$v_i - u_i = 0 \quad \text{for every } i.$$

Thus

$$v_i = u_i \quad \text{for every } i,$$

so the representation is unique.

(ii) $\iff$ (iii).

Assume (ii). Let

$$x \in V_i \cap (V_1 + \cdots + V_{i-1})$$

for some $i \geq 2$. Then

$$x = v_1 + \cdots + v_{i-1}$$

for some

$$v_j \in V_j.$$

Hence

$$v_1 + \cdots + v_{i-1} - x = 0,$$

where

$$-x \in V_i.$$

By (ii), all these vectors are zero. In particular,

$$x = 0.$$

Thus (iii) holds.

Conversely, assume (iii), and suppose

$$v_1 + \cdots + v_m = 0, \quad v_i \in V_i.$$

Then

$$v_m = -(v_1 + \cdots + v_{m-1}).$$

Thus

$$v_m \in V_m \cap (V_1 + \cdots + V_{m-1}).$$

By (iii),

$$v_m = 0.$$

Repeating the same argument with

$$v_1 + \cdots + v_{m-1} = 0$$

shows successively that

$$v_{m-1} = 0, \dots, v_1 = 0.$$

Hence (ii) holds.

(iii) $\iff$ (v).

For each i , let

$$\mathcal{B}_i$$

be a basis of V_i . The concatenated list always spans

$$S = V_1 + \cdots + V_m.$$

Assume (iii). Suppose a linear combination of the vectors in the concatenated list equals zero. Grouping the terms according to the subspaces gives

$$v_1 + \cdots + v_m = 0, \quad v_i \in V_i.$$

By (iii), and hence by (ii),

$$v_1 = \cdots = v_m = 0.$$

Because each $\mathcal{B}_i$ is linearly independent, every coefficient in the original linear combination must be zero. Thus the concatenated list is linearly independent and hence is a basis of S .

Conversely, if the concatenated list is a basis, suppose

$$v_1 + \cdots + v_m = 0, \quad v_i \in V_i.$$

Express each v_i in the basis $\mathcal{B}_i$. This gives a linear combination of the concatenated basis equal to zero. Linear independence then implies that all coefficients are zero, and hence

$$v_i = 0 \quad \text{for every } i.$$

Thus (ii), and therefore (iii), holds.

(iv) $\iff$ (v).

The concatenated list

$$\mathcal{B}_1, \dots, \mathcal{B}_m$$

spans S and contains exactly

$$\dim V_1 + \cdots + \dim V_m$$

vectors.

If (iv) holds, this number equals $\dim S$. By [proposition 1.21](#), a spanning list containing exactly $\dim S$ vectors is a basis of S . Thus (v) holds.

Conversely, if the concatenated list is a basis of S , the number of vectors in that basis equals $\dim S$. Hence

$$\dim S = \dim V_1 + \cdots + \dim V_m.$$

Therefore all five conditions are equivalent.

One common case is the (direct) sum of two subspaces, see the following corollary.

Corollary 4.10 — Direct sum of two subspaces

Let U and W be subspaces of V . Then

$$U + W$$

is a direct sum if and only if

$$U \cap W = \{0\}.$$

Thus

$$U + W = U \oplus W \iff U \cap W = \{0\}.$$

Proof

This is the case $m = 2$ of [theorem 4.9](#).

Example 4.11 — A direct-sum decomposition of $\mathbb{R}^3$

Let

$$U = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} : x, y \in \mathbb{R} \right\}, \quad W = \left\{ \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix} : z \in \mathbb{R} \right\}.$$

Then

$$U + W = \mathbb{R}^3.$$

Moreover,

$$U \cap W = \{0\}.$$

Hence, by [corollary 4.10](#),

$$\mathbb{R}^3 = U \oplus W.$$

Thus every vector has the unique decomposition

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix}.$$

Definition 4.12 — Complementary subspace

Let U be a subspace of V .

A subspace W of V is called a **complement** of U if

$$V = U \oplus W.$$

Proposition 4.13 — Every subspace has a complement

Let V be finite-dimensional and let U be a subspace of V . Then U has a complement in V .

Proof

Let

$$u_1, \dots, u_r$$

be a basis of U .

By [proposition 3.10](#), extend this basis to a basis

$$u_1, \dots, u_r, w_1, \dots, w_s$$

of V .

Define

$$W = \text{Span}(w_1, \dots, w_s).$$

Because the displayed list spans V ,

$$V = U + W.$$

Suppose

$$x \in U \cap W.$$

Then x can be written both as a linear combination of

$$u_1, \dots, u_r$$

and as a linear combination of

$$w_1, \dots, w_s.$$

Subtracting these two representations gives a linear relation among

$$u_1, \dots, u_r, w_1, \dots, w_s.$$

Since this list is a basis of V , it is linearly independent. Hence all coefficients are zero, so

$$x = 0.$$

Therefore

$$U \cap W = \{0\}.$$

By [corollary 4.10](#),

$$V = U \oplus W.$$

Thus W is a complement of U .

Remark 4.14 — Complements are generally not unique

A subspace may have many different complements. The proposition asserts existence, not uniqueness.

4.4 Invariant Subspaces

Direct sums allow us to decompose a vector space into smaller pieces. For a linear transformation

$$T : V \rightarrow V,$$

we would like these pieces to be compatible with T .

This leads to the notion of an invariant subspace.

Definition 4.15 — Invariant subspace

Let

$$T : V \rightarrow V$$

be a linear transformation.

A subspace U of V is called **invariant under T** , or **T -invariant**, if

$$T(u) \in U \quad \text{for every } u \in U.$$

Equivalently,

$$T(U) \subseteq U.$$

If U is invariant under T , then the restriction

$$T|_U : U \rightarrow U$$

is itself a linear transformation on the smaller vector space U . Thus we can study part of the behavior of T by studying the simpler operator $T|_U$.

Proposition 4.16 — Basic invariant subspaces

Let

$$T : V \rightarrow V$$

be a linear transformation. Then each of the following is invariant under T :

$$\{0\}, \quad V, \quad \ker T, \quad \operatorname{Im} T.$$

Proof

The subspaces $\{0\}$ and V are immediate.

If

$$u \in \ker T,$$

then

$$T(u) = 0.$$

Since

$$0 \in \ker T,$$

we have

$$T(u) \in \ker T.$$

Thus $\ker T$ is invariant.

Now let

$$u \in \operatorname{Im} T.$$

Then there exists some $v \in V$ such that

$$u = T(v).$$

Hence

$$T(u) = T(T(v)).$$

This is itself the image under T of the vector $T(v) \in V$, so

$$T(u) \in \operatorname{Im} T.$$

Therefore $\operatorname{Im} T$ is invariant.

Example 4.17 — Differentiation and polynomial subspaces

Let

$$T : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$$

be differentiation:

$$T(p) = p'.$$

For every nonnegative integer n , the subspace

$$\mathbb{R}_n[x]$$

is invariant under T , because

$$p \in \mathbb{R}_n[x] \implies p' \in \mathbb{R}_n[x].$$

Thus differentiation on the infinite-dimensional space $\mathbb{R}[x]$ can be restricted to a linear transformation

$$T|_{\mathbb{R}_n[x]} : \mathbb{R}_n[x] \rightarrow \mathbb{R}_n[x].$$

Proposition 4.18 — Sums and intersections of invariant subspaces

Let

$$T : V \rightarrow V$$

be a linear transformation, and let U and W be invariant under T . Then

$$U \cap W \quad \text{and} \quad U + W$$

are also invariant under T .

Proof

Let

$$x \in U \cap W.$$

Since U and W are both invariant,

$$T(x) \in U \quad \text{and} \quad T(x) \in W.$$

Therefore

$$T(x) \in U \cap W.$$

Thus $U \cap W$ is invariant.

Now let

$$x \in U + W.$$

Then

$$x = u + w$$

for some

$$u \in U, \quad w \in W.$$

By linearity,

$$T(x) = T(u) + T(w).$$

Because U and W are invariant,

$$T(u) \in U, \quad T(w) \in W.$$

Hence

$$T(x) \in U + W.$$

Thus $U + W$ is invariant.

4.5 Invariant Subspaces and the Matrix of a Linear Transformation

The usefulness of invariant subspaces becomes especially clear when we choose a basis adapted to them.

Proposition 4.19 — Invariant subspace and block upper-triangular form

Let

$$T : V \rightarrow V$$

be a linear transformation on a finite-dimensional vector space, and let U be invariant under T .

Let

$$u_1, \dots, u_r$$

be a basis of U , and extend it, using [proposition 3.10](#), to a basis

$$\mathcal{E} = (u_1, \dots, u_r, w_1, \dots, w_s)$$

of V .

Then the matrix of T with respect to $\mathcal{E}$ has the block form

$$[T]_{\mathcal{E} \leftarrow \mathcal{E}} = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix},$$

where

$$A = [T|_U]_{(u_1, \dots, u_r) \leftarrow (u_1, \dots, u_r)}.$$

Proof

Because U is invariant under T ,

$$T(u_j) \in U \quad \text{for } j = 1, \dots, r.$$

Hence each $T(u_j)$ is a linear combination only of

$$u_1, \dots, u_r.$$

Therefore the coordinate vectors

$$[T(u_j)]_{\mathcal{E}}, \quad j = 1, \dots, r,$$

have zero entries in their last s positions.

Recall that these coordinate vectors form the first r columns of

$$[T]_{\mathcal{E} \leftarrow \mathcal{E}}.$$

Hence the lower-left block is zero, and the matrix has the form

$$[T]_{\mathcal{E} \leftarrow \mathcal{E}} = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}.$$

The upper-left block records the action of T on the basis

$$u_1, \dots, u_r$$

of U , and hence it is precisely the matrix of the restricted transformation

$$T|_U : U \rightarrow U.$$

If the whole space decomposes into invariant subspaces, the matrix simplifies further.

Corollary 4.20 — Invariant direct sums and block-diagonal matrices

Suppose

$$V = U \oplus W$$

and both U and W are invariant under

$$T : V \rightarrow V.$$

Choose bases

$$\mathcal{B}_U = (u_1, \dots, u_r)$$

of U and

$$\mathcal{B}_W = (w_1, \dots, w_s)$$

of W .

Then

$$\mathcal{E} = (u_1, \dots, u_r, w_1, \dots, w_s)$$

is a basis of V , and

$$[T]_{\mathcal{E} \leftarrow \mathcal{E}} = \begin{pmatrix} [T|_U]_{\mathcal{B}_U \leftarrow \mathcal{B}_U} & 0 \\ 0 & [T|_W]_{\mathcal{B}_W \leftarrow \mathcal{B}_W} \end{pmatrix}.$$

Proof

By [theorem 4.9](#), concatenating bases of U and W gives a basis of V .

Since U is invariant, the lower-left block is zero by [proposition 4.19](#).

Since W is also invariant,

$$T(w_j) \in W \quad \text{for every } j,$$

so the coordinates of $T(w_j)$ along

$$u_1, \dots, u_r$$

are zero. Hence the upper-right block is also zero.

Therefore the matrix is block diagonal.

Remark 4.21 — Why invariant decompositions matter

If

$$V = U \oplus W$$

with both U and W invariant under T , then the study of T splits into two smaller problems:

$$T|_U : U \rightarrow U \quad \text{and} \quad T|_W : W \rightarrow W.$$

The block-diagonal matrix

$$\begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$$

is the coordinate expression of this decomposition.

More generally, if

$$V = V_1 \oplus \cdots \oplus V_m$$

and every V_i is invariant under T , then a basis obtained by concatenating bases of the V_i gives a block-diagonal matrix for T .

The simplest nonzero invariant subspaces are one-dimensional.

Proposition 4.22 — One-dimensional invariant subspaces

Let

$$T : V \rightarrow V$$

be a linear transformation and let

$$v \in V, \quad v \neq 0.$$

Then the one-dimensional subspace

$$\text{Span}(v)$$

is invariant under T if and only if there exists a scalar

$$\lambda \in \mathbb{K}$$

such that

$$T(v) = \lambda v.$$

Proof

Suppose first that

$$\text{Span}(v)$$

is invariant under T . Then

$$T(v) \in \text{Span}(v).$$

By the definition of span, there exists

$$\lambda \in \mathbb{K}$$

such that

$$T(v) = \lambda v.$$

Conversely, suppose

$$T(v) = \lambda v.$$

Every vector in $\text{Span}(v)$ has the form

$$cv$$

for some $c \in \mathbb{K}$. By linearity,

$$T(cv) = cT(v) = c\lambda v \in \text{Span}(v).$$

Hence

$$\text{Span}(v)$$

is invariant under T .

Remark 4.23 — From invariant subspaces to eigenvalues

The equation

$$T(v) = \lambda v, \quad v \neq 0,$$

says that T does not change the direction spanned by v : it only rescales vectors along that direction.

Thus one-dimensional invariant subspaces are exactly the directions on which T acts by scalar multiplication.

This observation leads directly to the concepts of **eigenvalues** and **eigenvectors**, which we study later in the course.

5 Determinants

Before going into eigenvalues and eigen vectors, let us turn back to square matrices for a moment. We have already developed several ways to understand the structure of a square matrix. In particular, by [corollary 3.57](#), for

$$A \in \mathbb{K}^{n \times n},$$

the following conditions are equivalent:

$$A \text{ is invertible}, \quad \text{Rank}(A) = n, \quad \ker A = \{0\}.$$

We now introduce another quantity associated with a square matrix: a single scalar

$$\det(A) \in \mathbb{K}$$

called its **determinant**.

The determinant will eventually give another criterion for invertibility:

$$A \text{ is invertible} \iff \det(A) \neq 0.$$

It will also play a central role in the study of eigenvalues, where we will encounter expressions of the form

$$\det(A - \lambda I).$$

Rather than beginning with a complicated formula, we first ask: *what properties should the determinant have?*

5.1 The Determinant and Its Defining Properties

Let

$$A = \begin{pmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{pmatrix} \in \mathbb{K}^{n \times n},$$

where

$$a_1, \dots, a_n \in \mathbb{K}^n$$

are the columns of A .

We will often write

$$\det(A) = \det(a_1, \dots, a_n)$$

to emphasize that the determinant is a function of the columns of the matrix.

The determinant is characterized by three fundamental properties: linearity in each column, alternation, and normalization.

Definition 5.1 — Multilinear and alternating functions

Let

$$D : (\mathbb{K}^n)^n \rightarrow \mathbb{K}$$

be a function of n vector arguments.

We say that D is **multilinear** if it is linear in each argument when all the other arguments are held fixed. Thus, for every $j = 1, \dots, n$,

$$\begin{aligned} D(a_1, \dots, a_{j-1}, \alpha u + \beta v, a_{j+1}, \dots, a_n) \\ = \alpha D(a_1, \dots, a_{j-1}, u, a_{j+1}, \dots, a_n) \\ + \beta D(a_1, \dots, a_{j-1}, v, a_{j+1}, \dots, a_n). \end{aligned}$$

We say that D is **alternating** if the value of D is zero whenever any two of its arguments are equal.

More precisely, if

$$a_i = a_j$$

for some distinct indices

$$i, j \in \{1, \dots, n\}, \quad i \neq j,$$

then

$$D(a_1, \dots, a_n) = 0.$$

The following proposition is a direct consequence of alternation.

Proposition 5.2 — Alternation implies sign change under interchange

Let

$$D : (\mathbb{K}^n)^n \rightarrow \mathbb{K}$$

be multilinear and alternating. Interchanging two arguments changes the sign of D .

More precisely, if $i \neq j$, then

$$D(a_1, \dots, a_i, \dots, a_j, \dots, a_n) = -D(a_1, \dots, a_j, \dots, a_i, \dots, a_n).$$

Proof

It is enough to focus on the two positions being interchanged and keep all other arguments fixed.

Because D is alternating, putting the same vector $a_i + a_j$ into both positions gives

$$D(\dots, a_i + a_j, \dots, a_i + a_j, \dots) = 0.$$

By multilinearity,

$$0 = D(\dots, a_i, \dots, a_i, \dots) + D(\dots, a_i, \dots, a_j, \dots) \\ + D(\dots, a_j, \dots, a_i, \dots) + D(\dots, a_j, \dots, a_j, \dots).$$

The first and last terms vanish because D is alternating. Therefore

$$D(\dots, a_i, \dots, a_j, \dots) + D(\dots, a_j, \dots, a_i, \dots) = 0.$$

Hence

$$D(\dots, a_i, \dots, a_j, \dots) = -D(\dots, a_j, \dots, a_i, \dots).$$

Thus alternation captures the familiar idea that interchanging two columns reverses the sign of the determinant.

To state an explicit formula for the determinant, we need a small amount of notation concerning permutations.

Definition 5.3 — Permutations and their signs

A **permutation** of

$$\{1, \dots, n\}$$

is a bijection

$$\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}.$$

The set of all such permutations is denoted by

$$S_n.$$

An **inversion** of σ is a pair

$$i < j$$

such that

$$\sigma(i) > \sigma(j).$$

If $\text{inv}(\sigma)$ denotes the number of inversions of σ , the **sign** of σ is

$$\text{sgn}(\sigma) = (-1)^{\text{inv}(\sigma)}.$$

Thus

$$\text{sgn}(\sigma) = \begin{cases} 1, & \sigma \text{ has an even number of inversions,} \\ -1, & \sigma \text{ has an odd number of inversions.} \end{cases}$$

For example, the identity permutation

$$\sigma(i) = i$$

has no inversions, so

$$\operatorname{sgn}(\sigma) = 1.$$

Interchanging two elements of a permutation changes its sign. Thus, if τ is obtained from σ by interchanging two positions, then

$$\operatorname{sgn}(\tau) = -\operatorname{sgn}(\sigma).$$

We can now state the fundamental theorem that defines the determinant.

Theorem 5.4 — Existence and uniqueness of the determinant

There exists a unique function

$$\det : (\mathbb{K}^n)^n \rightarrow \mathbb{K}$$

satisfying the following three properties:

(1) **Multilinearity.**

The determinant is linear in each column separately.

(2) **Alternation.**

If two columns are equal, then

$$\det(a_1, \dots, a_n) = 0.$$

(3) **Normalization.**

For the standard basis

$$e_1, \dots, e_n$$

of $\mathbb{K}^n$,

$$\det(e_1, \dots, e_n) = 1.$$

Equivalently,

$$\det(I_n) = 1.$$

For

$$A = (a_{ij}) \in \mathbb{K}^{n \times n},$$

this unique function is given explicitly by

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j), j}.$$

Proof

We prove existence and uniqueness separately.

Existence.

For

$$A = (a_{ij}) \in \mathbb{K}^{n \times n},$$

define

$$D(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j),j}.$$

We verify that D has the three stated properties.

First, D is multilinear in the columns.

Fix a column j . In every term

$$\text{sgn}(\sigma) \prod_{k=1}^n a_{\sigma(k),k},$$

exactly one factor,

$$a_{\sigma(j),j},$$

comes from column j .

Thus, if the j th column is replaced by

$$\alpha u + \beta v,$$

each term in the sum separates linearly into an α -term and a β -term. Therefore

$$\begin{aligned} D(a_1, \dots, \alpha u + \beta v, \dots, a_n) \\ = \alpha D(a_1, \dots, u, \dots, a_n) + \beta D(a_1, \dots, v, \dots, a_n). \end{aligned}$$

Hence D is multilinear.

Second, D is alternating.

Suppose columns r and s , with $r \neq s$, are equal.

For every permutation $\sigma \in S_n$, let τ be the permutation obtained by interchanging the values of σ at positions r and s . Then

$$\text{sgn}(\tau) = -\text{sgn}(\sigma).$$

Because columns r and s are equal, the products associated with σ and τ are identical:

$$\prod_{j=1}^n a_{\sigma(j),j} = \prod_{j=1}^n a_{\tau(j),j}.$$

Their contributions to $D(A)$ therefore cancel:

$$\operatorname{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j),j} + \operatorname{sgn}(\tau) \prod_{j=1}^n a_{\tau(j),j} = 0.$$

The permutations can be paired in this way, so

$$D(A) = 0.$$

Thus D is alternating.

Third, consider the identity matrix

$$I_n.$$

In the sum

$$D(I_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n (I_n)_{\sigma(j),j},$$

the product is nonzero only when

$$\sigma(j) = j \quad \text{for every } j.$$

Thus only the identity permutation contributes.

Since the identity permutation has sign 1,

$$D(I_n) = 1.$$

Therefore a function satisfying the three required properties exists.

Uniqueness.

Now suppose

$$D : (\mathbb{K}^n)^n \rightarrow \mathbb{K}$$

is any multilinear, alternating function satisfying

$$D(e_1, \dots, e_n) = 1.$$

Let

$$A = \begin{pmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{pmatrix},$$

where

$$a_j = \sum_{i=1}^n a_{ij} e_i.$$

Using multilinearity in every column,

$$\begin{aligned} D(a_1, \dots, a_n) &= D\left(\sum_{i_1=1}^n a_{i_1 1} e_{i_1}, \dots, \sum_{i_n=1}^n a_{i_n n} e_{i_n}\right) \\ &= \sum_{i_1=1}^n \cdots \sum_{i_n=1}^n a_{i_1 1} \cdots a_{i_n n} D(e_{i_1}, \dots, e_{i_n}). \end{aligned}$$

If two of the indices

$$i_1, \dots, i_n$$

are equal, then two arguments of D are equal. By alternation,

$$D(e_{i_1}, \dots, e_{i_n}) = 0.$$

Therefore the only nonzero terms are those for which

$$i_1, \dots, i_n$$

are all distinct.

Because there are n distinct indices chosen from

$$\{1, \dots, n\},$$

such a choice must be a permutation of

$$1, \dots, n.$$

Hence there exists

$$\sigma \in S_n$$

such that

$$i_j = \sigma(j) \quad \text{for } j = 1, \dots, n.$$

Thus

$$D(a_1, \dots, a_n) = \sum_{\sigma \in S_n} \left(\prod_{j=1}^n a_{\sigma(j), j} \right) D(e_{\sigma(1)}, \dots, e_{\sigma(n)}).$$

By [proposition 5.2](#), every interchange of two arguments changes the sign of D . Therefore, rearranging

$$e_{\sigma(1)}, \dots, e_{\sigma(n)}$$

into

$$e_1, \dots, e_n$$

gives

$$D(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = \text{sgn}(\sigma)D(e_1, \dots, e_n).$$

Since

$$D(e_1, \dots, e_n) = 1,$$

we obtain

$$D(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = \text{sgn}(\sigma).$$

Consequently,

$$D(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j),j}.$$

Thus every function satisfying multilinearity, alternation, and normalization must equal the function constructed above. Hence the determinant is unique.

Definition 5.5 — Determinant

Let

$$A = \begin{pmatrix} | & & | \\ a_1 & \cdots & a_n \\ | & & | \end{pmatrix} \in \mathbb{K}^{n \times n}.$$

The **determinant** of A , denoted by

$$\det(A) \quad \text{or simply} \quad \det A,$$

is the unique scalar-valued function of the columns of A that is

- (1) multilinear;
- (2) alternating;
- (3) normalized by

$$\det(I_n) = 1.$$

Equivalently,

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j),j}.$$

Although the explicit permutation formula gives a complete formula for the determinant, in practice we will rarely compute determinants directly from this formula. Its main role is to establish the existence of the determinant.

The three defining properties—multilinearity, alternation, and normalization—are much more useful for understanding and manipulating determinants.

Example 5.6 — Determinants in dimensions one and two

For a 1×1 matrix

$$A = (a),$$

there is only one permutation, so

$$\det(A) = a.$$

For a 2×2 matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

there are two permutations of $\{1, 2\}$.

The identity permutation contributes

$$ad,$$

whereas the permutation that interchanges 1 and 2 contributes

$$-bc.$$

Therefore

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

5.2 Basic Properties of Determinants

The defining properties of the determinant immediately imply a number of useful computational rules.

Let

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = \begin{pmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{pmatrix} \in \mathbb{K}^{n \times n}.$$

Thus

$$\det(A) = \det(a_1, \dots, a_n).$$

We begin with the consequences of multilinearity.

Proposition 5.7 — Linearity in each column

Fix all columns of a determinant except the j th column. Then the determinant is linear in the j th column. In particular:

(1) If the j th column is multiplied by a scalar $c \in \mathbb{K}$, then the determinant is multiplied by c :

$$\det(a_1, \dots, ca_j, \dots, a_n) = c \det(a_1, \dots, a_j, \dots, a_n).$$

In matrix form,

$$\det \begin{pmatrix} a_{11} & \cdots & ca_{1j} & \cdots & a_{1n} \\ a_{21} & \cdots & ca_{2j} & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & ca_{nj} & \cdots & a_{nn} \end{pmatrix} = c \det \begin{pmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & a_{nj} & \cdots & a_{nn} \end{pmatrix}.$$

(2) If the j th column is the sum of two vectors,

$$a_j = u + v,$$

then the determinant splits as

$$\det(a_1, \dots, u + v, \dots, a_n) = \det(a_1, \dots, u, \dots, a_n) + \det(a_1, \dots, v, \dots, a_n).$$

Writing

$$u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}, \quad v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix},$$

this becomes

$$\begin{aligned} & \det \begin{pmatrix} a_{11} & \cdots & u_1 + v_1 & \cdots & a_{1n} \\ a_{21} & \cdots & u_2 + v_2 & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & u_n + v_n & \cdots & a_{nn} \end{pmatrix} \\ &= \det \begin{pmatrix} a_{11} & \cdots & u_1 & \cdots & a_{1n} \\ a_{21} & \cdots & u_2 & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & u_n & \cdots & a_{nn} \end{pmatrix} + \det \begin{pmatrix} a_{11} & \cdots & v_1 & \cdots & a_{1n} \\ a_{21} & \cdots & v_2 & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & v_n & \cdots & a_{nn} \end{pmatrix}. \end{aligned}$$

(3) If the j th column is the zero vector, then the determinant is zero:

$$\det(a_1, \dots, 0, \dots, a_n) = 0.$$

In matrix form,

$$\det \begin{pmatrix} a_{11} & \cdots & 0 & \cdots & a_{1n} \\ a_{21} & \cdots & 0 & \cdots & a_{2n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & 0 & \cdots & a_{nn} \end{pmatrix} = 0.$$

Proof

The first two statements are precisely the multilinearity property in [theorem 5.4](#).

For the third statement, write the zero column as

$$0 = 0_{\mathbb{K}} a_j.$$

By linearity in the j th column,

$$\det(a_1, \dots, 0, \dots, a_n) = 0_{\mathbb{K}} \det(a_1, \dots, a_j, \dots, a_n) = 0.$$

Remark 5.8 — The determinant is not linear in the whole matrix

Linearity in each column separately does *not* mean that

$$\det(A + B) = \det(A) + \det(B).$$

For example, take

$$A = B = I_2.$$

Then

$$A + B = 2I_2 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix},$$

so

$$\det(A + B) = \det \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 4,$$

whereas

$$\det(A) + \det(B) = 1 + 1 = 2.$$

Thus

$$\det(A + B) \neq \det(A) + \det(B)$$

in general.

Multilinearity means that the determinant is linear when *one column at a time* is varied while all other columns are held fixed.

We now turn to the consequences of alternation.

Proposition 5.9 — Interchanging two columns

If two columns of a matrix are interchanged, then the determinant changes sign.

More precisely, if the i th and j th columns are interchanged, where $i \neq j$, then

$$\det(a_1, \dots, a_i, \dots, a_j, \dots, a_n) = -\det(a_1, \dots, a_j, \dots, a_i, \dots, a_n).$$

For example,

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix}$$

becomes, after interchanging the first two columns,

$$\det \begin{pmatrix} a_{12} & a_{11} & a_{13} & \cdots & a_{1n} \\ a_{22} & a_{21} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n2} & a_{n1} & a_{n3} & \cdots & a_{nn} \end{pmatrix},$$

and

$$\det \begin{pmatrix} a_{12} & a_{11} & a_{13} & \cdots & a_{1n} \\ a_{22} & a_{21} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n2} & a_{n1} & a_{n3} & \cdots & a_{nn} \end{pmatrix} = -\det \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix}.$$

Proof

This is exactly [proposition 5.2](#), applied to the determinant, which is alternating and multilinear by [theorem 5.4](#).

Corollary 5.10 — Equal or proportional columns

If two columns of a square matrix are equal, then its determinant is zero.

More generally, if two columns are proportional, then its determinant is zero.

Proof

Suppose first that

$$a_i = a_j$$

for some $i \neq j$. Then the alternating property in [theorem 5.4](#) gives immediately

$$\det(A) = 0.$$

Now suppose that the two columns are proportional:

$$a_j = ca_i$$

for some $c \in \mathbb{K}$.

By [proposition 5.7](#),

$$\begin{aligned}\det(a_1, \dots, a_i, \dots, ca_i, \dots, a_n) &= c \det(a_1, \dots, a_i, \dots, a_i, \dots, a_n) \\ &= 0,\end{aligned}$$

because the determinant on the right has two equal columns.

For example,

$$\det \begin{pmatrix} a_{11} & ca_{11} & a_{13} & \cdots & a_{1n} \\ a_{21} & ca_{21} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & ca_{n1} & a_{n3} & \cdots & a_{nn} \end{pmatrix} = 0.$$

The combination of multilinearity and alternation gives another especially important rule.

Proposition 5.11 — Adding a multiple of one column to another

Adding a scalar multiple of one column to a different column does not change the determinant.

More precisely, if

$$i \neq j$$

and $c \in \mathbb{K}$, then

$$\det(a_1, \dots, a_j + ca_i, \dots, a_n) = \det(a_1, \dots, a_j, \dots, a_n).$$

In matrix form, for example, adding c times the first column to the second column gives

$$\det \begin{pmatrix} a_{11} & a_{12} + ca_{11} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} + ca_{21} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} + ca_{n1} & a_{n3} & \cdots & a_{nn} \end{pmatrix} = \det \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix}.$$

Proof

By multilinearity in the j th column,

$$\begin{aligned}\det(a_1, \dots, a_j + ca_i, \dots, a_n) \\ = \det(a_1, \dots, a_j, \dots, a_n) + c \det(a_1, \dots, a_i, \dots, a_i, \dots, a_n).\end{aligned}$$

The second determinant contains the same column a_i in two different positions. By the alternating property in [theorem 5.4](#), it is zero. Hence

$$\det(a_1, \dots, a_j + ca_i, \dots, a_n) = \det(a_1, \dots, a_j, \dots, a_n).$$

The preceding result has an important strengthening. It is not only proportional columns that force the determinant to vanish; any linear dependence among the columns does so.

Proposition 5.12 — Linearly dependent columns imply zero determinant

If the columns

$$a_1, \dots, a_n$$

of a square matrix A are linearly dependent, then

$$\det(A) = 0.$$

Proof

Suppose

$$a_1, \dots, a_n$$

are linearly dependent. By [proposition 1.12](#), one of the columns can be written as a linear combination of the others.

Without loss of generality, suppose

$$a_n = c_1 a_1 + \dots + c_{n-1} a_{n-1}.$$

Then

$$A = \begin{pmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{pmatrix} = \begin{pmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & c_1 a_1 + \cdots + c_{n-1} a_{n-1} \\ | & | & & | \end{pmatrix}.$$

By multilinearity in the last column,

$$\begin{aligned}\det(A) &= c_1 \det(a_1, a_2, \dots, a_{n-1}, a_1) \\ &\quad + c_2 \det(a_1, a_2, \dots, a_{n-1}, a_2) \\ &\quad + \dots + c_{n-1} \det(a_1, a_2, \dots, a_{n-1}, a_{n-1}).\end{aligned}$$

Every determinant on the right contains two equal columns. Hence, by the alternating property,

$$\det(a_1, a_2, \dots, a_{n-1}, a_j) = 0 \quad \text{for every } j = 1, \dots, n-1.$$

Therefore

$$\det(A) = 0.$$

Remark 5.13 — What has been proved so far

We have proved the implication

$$a_1, \dots, a_n \text{ linearly dependent} \implies \det(A) = 0.$$

Equivalently,

$$\det(A) \neq 0 \implies a_1, \dots, a_n \text{ are linearly independent.}$$

We will later prove the converse and obtain the fundamental equivalence

$$\det(A) \neq 0 \iff \text{Rank}(A) = n \iff A \text{ is invertible.}$$

So far the defining properties have been stated in terms of columns. The next result shows that the determinant treats rows and columns symmetrically.

Theorem 5.14 — Determinant of the transpose

For every square matrix

$$A \in \mathbb{K}^{n \times n},$$

we have

$$\det(A^\top) = \det(A).$$

Proof

Let

$$A = (a_{ij}).$$

By the explicit determinant formula in [theorem 5.4](#),

$$\det(A^\top) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n (A^\top)_{\sigma(j),j}.$$

Since

$$(A^\top)_{\sigma(j),j} = a_{j,\sigma(j)},$$

we obtain

$$\det(A^\top) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{j,\sigma(j)}.$$

Now set

$$\tau = \sigma^{-1}.$$

As σ ranges over S_n , so does τ .

Moreover,

$$\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma).$$

Indeed, the inversions of σ are in one-to-one correspondence with the inversions of σ^{-1} : if

$$i < j \quad \text{and} \quad \sigma(i) > \sigma(j),$$

then

$$\sigma(j) < \sigma(i)$$

and

$$\sigma^{-1}(\sigma(j)) = j > i = \sigma^{-1}(\sigma(i)).$$

Hence σ and σ^{-1} have the same number of inversions and therefore the same sign.

Finally, using

$$\tau = \sigma^{-1},$$

we can reindex the product:

$$\prod_{j=1}^n a_{j,\sigma(j)} = \prod_{k=1}^n a_{\tau(k),k}.$$

Therefore

$$\begin{aligned} \det(A^\top) &= \sum_{\tau \in S_n} \operatorname{sgn}(\tau) \prod_{k=1}^n a_{\tau(k),k} \\ &= \det(A), \end{aligned}$$

where the final equality again follows from [theorem 5.4](#).

The transpose theorem allows every property proved for columns to be translated immediately into the corresponding property for rows.

Corollary 5.15 — The corresponding row properties

Let

$$A \in \mathbb{K}^{n \times n}.$$

Then:

- (1) If one row is zero, then

$$\det(A) = 0.$$

- (2) Multiplying one row by $c \in \mathbb{K}$ multiplies the determinant by c .

Thus, for example,

$$\det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ ca_{i1} & ca_{i2} & \cdots & ca_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = c \det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

- (3) Interchanging two rows changes the sign of the determinant.

For example,

$$\det \begin{pmatrix} a_{21} & a_{22} & \cdots & a_{2n} \\ a_{11} & a_{12} & \cdots & a_{1n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = -\det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

- (4) If two rows are equal or proportional, then

$$\det(A) = 0.$$

- (5) Adding a scalar multiple of one row to another row does not change the determinant.

For example,

$$\det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} + ca_{11} & a_{22} + ca_{12} & \cdots & a_{2n} + ca_{1n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = \det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

Proof

By [theorem 5.14](#),

$$\det(A^\top) = \det(A).$$

The rows of A are the columns of $A^\top$. Therefore each statement follows by applying the corresponding column result to $A^\top$ and then transposing back.

Specifically, the assertions follow from [propositions 5.7](#), [5.9](#) and [5.11](#) and [corollary 5.10](#).

We conclude with an important class of matrices whose determinant can be read directly from their entries.

Theorem 5.16 — Determinant of a triangular matrix

Let

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{pmatrix}$$

be an upper-triangular matrix. Then

$$\det(A) = a_{11}a_{22} \cdots a_{nn}.$$

The same conclusion holds for a lower-triangular matrix.

Proof

Suppose first that A is upper triangular. Thus

$$a_{ij} = 0 \quad \text{whenever } i > j.$$

By the explicit determinant formula from [theorem 5.4](#),

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j),j}.$$

Consider a permutation

$$\sigma \neq \operatorname{id}.$$

We claim that there must exist some j such that

$$\sigma(j) > j.$$

Indeed, if

$$\sigma(j) \leq j \quad \text{for every } j,$$

then

$$\sum_{j=1}^n \sigma(j) \leq \sum_{j=1}^n j.$$

But because σ is a permutation,

$$\sum_{j=1}^n \sigma(j) = \sum_{j=1}^n j.$$

Thus equality must hold at every position:

$$\sigma(j) = j \quad \text{for every } j,$$

contradicting

$$\sigma \neq \text{id}.$$

Hence every nonidentity permutation has some j satisfying

$$\sigma(j) > j.$$

Because A is upper triangular,

$$a_{\sigma(j),j} = 0.$$

Therefore

$$\prod_{j=1}^n a_{\sigma(j),j} = 0$$

for every nonidentity permutation.

Thus the only nonzero contribution to the determinant formula comes from the identity permutation. Since

$$\text{sgn}(\text{id}) = 1,$$

we obtain

$$\det(A) = a_{11}a_{22} \cdots a_{nn}.$$

Now suppose A is lower triangular. Then $A^\top$ is upper triangular. By [theorem 5.14](#),

$$\det(A) = \det(A^\top).$$

Applying the result just proved to $A^\top$ gives

$$\det(A^\top) = a_{11}a_{22} \cdots a_{nn}.$$

Therefore

$$\det(A) = a_{11}a_{22} \cdots a_{nn}.$$

Corollary 5.17 — Determinant of a diagonal matrix

If

$$A = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix},$$

then

$$\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n.$$

Proof

A diagonal matrix is both upper triangular and lower triangular, so the result follows immediately from [theorem 5.16](#).

Remark 5.18 — Three elementary operations and the determinant

The preceding results can be summarized in terms of the three elementary row operations:

row operation	effect on the determinant
$R_i \leftrightarrow R_j$	$\det \mapsto -\det$
$R_i \mapsto cR_i$	$\det \mapsto c \det$
$R_j \mapsto R_j + cR_i, \quad i \neq j$	$\det \mapsto \det$

Exactly the same rules hold for elementary column operations. Just look at the transpose.

5.3 Multiplicativity and Invertibility

The properties established in the previous subsection describe how the determinant changes when individual rows or columns of a matrix are modified. We now study how the determinant interacts with matrix multiplication.

The central result is

$$\det(AB) = \det(A) \det(B).$$

Because matrix multiplication represents composition of linear transformations, this property connects the determinant directly to the algebra of linear maps. It will also give a simple characterization

of invertible matrices in terms of a single scalar.

We first record a consequence of the uniqueness argument used in [theorem 5.4](#).

Proposition 5.19 — Alternating multilinear functions are multiples of the determinant

Let

$$D : (\mathbb{K}^n)^n \rightarrow \mathbb{K}$$

be an alternating multilinear function. Then, for all

$$a_1, \dots, a_n \in \mathbb{K}^n,$$

we have

$$D(a_1, \dots, a_n) = D(e_1, \dots, e_n) \det(a_1, \dots, a_n),$$

where

$$e_1, \dots, e_n$$

denotes the **standard basis** of $\mathbb{K}^n$, defined by

$$e_j = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

where the 1 appears in the j th position and all other entries are zero.

Thus every alternating multilinear function on n vectors in $\mathbb{K}^n$ is a scalar multiple of the determinant.

Proof

Write each vector a_j in the standard basis:

$$a_j = \sum_{i=1}^n a_{ij} e_i, \quad j = 1, \dots, n.$$

By multilinearity,

$$\begin{aligned} D(a_1, \dots, a_n) &= D\left(\sum_{i_1=1}^n a_{i_1 1} e_{i_1}, \dots, \sum_{i_n=1}^n a_{i_n n} e_{i_n}\right) \\ &= \sum_{i_1=1}^n \cdots \sum_{i_n=1}^n a_{i_1 1} \cdots a_{i_n n} D(e_{i_1}, \dots, e_{i_n}). \end{aligned}$$

If two of the indices

$$i_1, \dots, i_n$$

are equal, then two arguments of D are equal. Since D is alternating,

$$D(e_{i_1}, \dots, e_{i_n}) = 0.$$

Hence only those terms for which

$$i_1, \dots, i_n$$

are all distinct can remain. Such a choice of indices is a permutation of

$$1, \dots, n.$$

Therefore

$$D(a_1, \dots, a_n) = \sum_{\sigma \in S_n} \left(\prod_{j=1}^n a_{\sigma(j), j} \right) D(e_{\sigma(1)}, \dots, e_{\sigma(n)}).$$

By [proposition 5.2](#), interchanging two arguments changes the sign of D . Consequently,

$$D(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = \text{sgn}(\sigma) D(e_1, \dots, e_n).$$

Substituting this into the preceding expression gives

$$D(a_1, \dots, a_n) = D(e_1, \dots, e_n) \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j), j}.$$

By the determinant formula in [theorem 5.4](#),

$$\sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n a_{\sigma(j), j} = \det(a_1, \dots, a_n).$$

Therefore

$$D(a_1, \dots, a_n) = D(e_1, \dots, e_n) \det(a_1, \dots, a_n).$$

This characterization makes the proof of multiplicativity particularly transparent.

Theorem 5.20 — Multiplicativity of the determinant

Let

$$A, B \in \mathbb{K}^{n \times n}.$$

Then

$$\det(AB) = \det(A) \det(B).$$

Proof

Fix the matrix

$$A \in \mathbb{K}^{n \times n}.$$

Write the columns of B as

$$B = \begin{pmatrix} | & | & \cdots & | \\ b_1 & b_2 & \cdots & b_n \\ | & | & \cdots & | \end{pmatrix}.$$

Recall that multiplication by A acts separately on the columns of B . Thus

$$AB = \begin{pmatrix} | & | & \cdots & | \\ Ab_1 & Ab_2 & \cdots & Ab_n \\ | & | & \cdots & | \end{pmatrix}.$$

Define

$$D_A : (\mathbb{K}^n)^n \rightarrow \mathbb{K}$$

by

$$D_A(b_1, \dots, b_n) = \det(Ab_1, \dots, Ab_n).$$

We first show that D_A is multilinear.

Fix every argument except the j th one. For

$$u, v \in \mathbb{K}^n \quad \text{and} \quad \alpha, \beta \in \mathbb{K},$$

linearity of the map

$$x \mapsto Ax$$

gives

$$A(\alpha u + \beta v) = \alpha Au + \beta Av.$$

Hence, by the multilinearity of the determinant,

$$\begin{aligned}
D_A(b_1, \dots, \alpha u + \beta v, \dots, b_n) &= \det(Ab_1, \dots, A(\alpha u + \beta v), \dots, Ab_n) \\
&= \det(Ab_1, \dots, \alpha Au + \beta Av, \dots, Ab_n) \\
&= \alpha \det(Ab_1, \dots, Au, \dots, Ab_n) \\
&\quad + \beta \det(Ab_1, \dots, Av, \dots, Ab_n) \\
&= \alpha D_A(b_1, \dots, u, \dots, b_n) + \beta D_A(b_1, \dots, v, \dots, b_n).
\end{aligned}$$

Thus D_A is multilinear.

Next, suppose two arguments are equal:

$$b_i = b_j, \quad i \neq j.$$

Then

$$Ab_i = Ab_j.$$

Therefore the matrix

$$\begin{pmatrix} | & & | \\ Ab_1 & \cdots & Ab_n \\ | & & | \end{pmatrix}$$

has two equal columns, and by the alternating property of the determinant,

$$D_A(b_1, \dots, b_n) = 0.$$

Thus D_A is alternating.

We may therefore apply [proposition 5.19](#). It gives

$$D_A(b_1, \dots, b_n) = D_A(e_1, \dots, e_n) \det(b_1, \dots, b_n).$$

Now

$$Ae_1, \dots, Ae_n$$

are precisely the columns of A . Hence

$$D_A(e_1, \dots, e_n) = \det(Ae_1, \dots, Ae_n) = \det(A).$$

Also,

$$\det(b_1, \dots, b_n) = \det(B).$$

Therefore

$$D_A(b_1, \dots, b_n) = \det(A) \det(B).$$

But by definition,

$$D_A(b_1, \dots, b_n) = \det(AB).$$

Consequently,

$$\det(AB) = \det(A) \det(B).$$

Multiplicativity immediately gives several useful consequences.

Corollary 5.21 — Determinant of powers

Let

$$A \in \mathbb{K}^{n \times n}.$$

For every positive integer k ,

$$\det(A^k) = (\det(A))^k.$$

Proof

By repeated application of [theorem 5.20](#),

$$\begin{aligned} \det(A^k) &= \det(\underbrace{A \cdots A}_{k \text{ factors}}) \\ &= \underbrace{\det(A) \cdots \det(A)}_{k \text{ factors}} \\ &= (\det(A))^k. \end{aligned}$$

Corollary 5.22 — Determinant of an inverse

Let

$$A \in \mathbb{K}^{n \times n}$$

be invertible. Then

$$\det(A) \neq 0$$

and

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Proof

Because A is invertible,

$$AA^{-1} = I_n.$$

Taking determinants and using [theorem 5.20](#),

$$\det(A) \det(A^{-1}) = \det(I_n).$$

By the normalization property in [theorem 5.4](#),

$$\det(I_n) = 1.$$

Hence

$$\det(A) \det(A^{-1}) = 1.$$

Therefore

$$\det(A) \neq 0,$$

and solving for $\det(A^{-1})$ gives

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

We can now connect the determinant to the rank and invertibility theory developed in the previous section.

Theorem 5.23 — Determinant criterion for invertibility

Let

$$A \in \mathbb{K}^{n \times n}.$$

Then

$$A \text{ is invertible} \iff \det(A) \neq 0.$$

Proof

Suppose first that A is invertible.

By [corollary 5.22](#),

$$\det(A) \det(A^{-1}) = 1.$$

Therefore

$$\det(A) \neq 0.$$

Conversely, suppose

$$\det(A) \neq 0.$$

Write the columns of A as

$$A = \begin{pmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{pmatrix}.$$

By [proposition 5.12](#), if

$$a_1, \dots, a_n$$

were linearly dependent, then

$$\det(A) = 0.$$

Because

$$\det(A) \neq 0,$$

the columns

$$a_1, \dots, a_n$$

must therefore be linearly independent.

By the definition of matrix rank in [definition 3.28](#),

$$\text{Rank}(A) = \dim \text{Span}(a_1, \dots, a_n).$$

Since $a_1, \dots, a_n$ are n linearly independent vectors,

$$\text{Rank}(A) = n.$$

Finally, by [corollary 3.57](#),

$$\text{Rank}(A) = n \implies A \text{ is invertible.}$$

Thus

$$A \text{ is invertible} \iff \det(A) \neq 0.$$

Definition 5.24 — Singular and nonsingular matrices

Let

$$A \in \mathbb{K}^{n \times n}.$$

The matrix A is called **nonsingular** if it is invertible.

It is called **singular** if it is not invertible.

Corollary 5.25 — Equivalent characterizations of nonsingularity

Let

$$A \in \mathbb{K}^{n \times n}.$$

Then the following statements are equivalent:

- (i) A is invertible;
- (ii) A is nonsingular;
- (iii)

$$\det(A) \neq 0;$$

(iv)

$$\text{Rank}(A) = n;$$

(v)

$$\ker A = \{0\};$$

- (vi) the columns of A form a basis of $\mathbb{K}^n$;
- (vii) the rows of A form a basis of $\mathbb{K}_n$.

Equivalently,

$$A \text{ is singular} \iff \det(A) = 0 \iff \text{Rank}(A) < n.$$

Proof

By [theorem 5.23](#),

$$A \text{ is invertible} \iff \det(A) \neq 0.$$

By [corollary 3.57](#), invertibility is equivalent to

$$\text{Rank}(A) = n, \quad \ker A = \{0\},$$

and to the columns and rows of A forming bases of their respective coordinate spaces.

Finally, “nonsingular” is simply another name for “invertible” by [definition 5.24](#).

Thus all the stated conditions are equivalent.

Remark 5.26 — What the determinant detects

For a square matrix, the determinant detects whether the associated linear transformation loses a dimension.

If

$$\det(A) \neq 0,$$

then

$$\text{Rank}(A) = n,$$

so the transformation

$$x \mapsto Ax$$

preserves n independent directions.

If

$$\det(A) = 0,$$

then

$$\text{Rank}(A) < n,$$

so the image lies in a proper subspace of $\mathbb{K}^n$.

Thus

$$\det(A) = 0$$

is precisely the algebraic signal that the transformation collapses at least one independent direction.

The determinant criterion also gives a short proof of an important fact about products of square matrices.

Corollary 5.27 — Invertibility of a product

Let

$$A, B \in \mathbb{K}^{n \times n}.$$

Then

$$AB \text{ is invertible} \iff A \text{ and } B \text{ are both invertible.}$$

Proof

By [theorem 5.23](#),

$$AB \text{ is invertible} \iff \det(AB) \neq 0.$$

By [theorem 5.20](#),

$$\det(AB) = \det(A) \det(B).$$

Since $\mathbb{K}$ is a field,

$$\det(A) \det(B) \neq 0$$

if and only if

$$\det(A) \neq 0 \quad \text{and} \quad \det(B) \neq 0.$$

Applying [theorem 5.23](#) once more,

$$\det(A) \neq 0 \iff A \text{ is invertible,}$$

and similarly for B .

Therefore

$$AB \text{ is invertible} \iff A \text{ and } B \text{ are both invertible.}$$

A useful consequence is that, for square matrices, a one-sided inverse is automatically a two-sided inverse.

Corollary 5.28 — A one-sided inverse is an inverse

Let

$$A, B \in \mathbb{K}^{n \times n}.$$

If

$$AB = I_n,$$

then

$$BA = I_n$$

and

$$B = A^{-1}.$$

Similarly, if

$$BA = I_n,$$

then

$$AB = I_n$$

and again

$$B = A^{-1}.$$

Proof

Suppose

$$AB = I_n.$$

Taking determinants and using [theorem 5.20](#),

$$\det(A) \det(B) = \det(I_n) = 1.$$

Hence

$$\det(A) \neq 0.$$

By [theorem 5.23](#), A is invertible. Multiplying

$$AB = I_n$$

on the left by A^{-1} gives

$$A^{-1}AB = A^{-1}I_n,$$

and therefore

$$B = A^{-1}.$$

Consequently,

$$BA = A^{-1}A = I_n.$$

The case

$$BA = I_n$$

is proved analogously.

We now connect the determinant with the change-of-basis theory developed earlier.

Corollary 5.29 — Determinant is invariant under similarity

Let

$$A, B \in \mathbb{K}^{n \times n}.$$

If A and B are similar, so that

$$B = P^{-1}AP$$

for some invertible matrix

$$P \in \mathbb{K}^{n \times n},$$

then

$$\det(B) = \det(A).$$

Proof

By [theorem 5.20](#),

$$\begin{aligned}\det(B) &= \det(P^{-1}AP) \\ &= \det(P^{-1}) \det(A) \det(P).\end{aligned}$$

By [corollary 5.22](#),

$$\det(P^{-1}) = \frac{1}{\det(P)}.$$

Therefore

$$\begin{aligned}\det(B) &= \frac{1}{\det(P)} \det(A) \det(P) \\ &= \det(A).\end{aligned}$$

Remark 5.30 — Similarity versus matrix equivalence

Recall that similarity has the special form

$$B = P^{-1}AP,$$

and therefore preserves the value of the determinant:

$$\det(B) = \det(A).$$

For the more general matrix equivalence

$$B = PAQ,$$

where P and Q are independently chosen invertible matrices, multiplicativity gives

$$\det(B) = \det(P) \det(A) \det(Q).$$

Thus matrix equivalence does not in general preserve the numerical value of the determinant. It does, however, preserve whether the determinant is zero, because

$$\det(P) \neq 0, \quad \det(Q) \neq 0.$$

Hence

$$\det(B) = 0 \iff \det(A) = 0.$$

The invariance of the determinant under similarity allows us to define the determinant of an abstract linear transformation without referring to any particular coordinate system.

Definition 5.31 — Determinant of a linear transformation

Let

$$T : V \rightarrow V$$

be a linear transformation on a finite-dimensional vector space V , and let

$$\mathcal{E}$$

be any basis of V .

The **determinant** of T is defined by

$$\det(T) := \det([T]_{\mathcal{E} \leftarrow \mathcal{E}}).$$

We must verify that this definition does not depend on the chosen basis.

Proposition 5.32 — The determinant of a linear transformation is basis-independent

Let

$$T : V \rightarrow V$$

be a linear transformation on a finite-dimensional vector space.

If $\mathcal{E}$ and $\mathcal{E}'$ are two bases of V , then

$$\det([T]_{\mathcal{E} \leftarrow \mathcal{E}}) = \det([T]_{\mathcal{E}' \leftarrow \mathcal{E}'}).$$

Therefore [definition 5.31](#) is well-defined.

Proof

Suppose the transition matrix from $\mathcal{E}$ to $\mathcal{E}'$ is P , so that

$$\mathcal{E}' = \mathcal{E}P.$$

By the change-of-basis formula for linear transformations, [corollary 3.39](#),

$$[T]_{\mathcal{E}' \leftarrow \mathcal{E}'} = P^{-1}[T]_{\mathcal{E} \leftarrow \mathcal{E}}P.$$

Thus the two matrices representing T are similar. By [corollary 5.29](#),

$$\det([T]_{\mathcal{E}' \leftarrow \mathcal{E}'}) = \det([T]_{\mathcal{E} \leftarrow \mathcal{E}}).$$

Hence the value of the determinant is independent of the chosen basis.

Proposition 5.33 — Properties of the determinant of a linear transformation

Let

$$S, T : V \rightarrow V$$

be linear transformations on a finite-dimensional vector space V . Then:

(1)

$$\det(I_V) = 1.$$

(2)

$$\det(ST) = \det(S) \det(T).$$

(3)

$$T \text{ is invertible} \iff \det(T) \neq 0.$$

(4) If T is invertible, then

$$\det(T^{-1}) = \frac{1}{\det(T)}.$$

Proof

Choose any basis

$$\mathcal{E}$$

of V .

For (1), the matrix of the identity transformation is

$$[I_V]_{\mathcal{E} \leftarrow \mathcal{E}} = I_n.$$

Hence, by the normalization property in [theorem 5.4](#),

$$\det(I_V) = \det(I_n) = 1.$$

For (2), composition of linear transformations corresponds to matrix multiplication:

$$[ST]_{\mathcal{E} \leftarrow \mathcal{E}} = [S]_{\mathcal{E} \leftarrow \mathcal{E}}[T]_{\mathcal{E} \leftarrow \mathcal{E}}.$$

Therefore, by [theorem 5.20](#),

$$\begin{aligned} \det(ST) &= \det([ST]_{\mathcal{E} \leftarrow \mathcal{E}}) \\ &= \det([S]_{\mathcal{E} \leftarrow \mathcal{E}}[T]_{\mathcal{E} \leftarrow \mathcal{E}}) \\ &= \det([S]_{\mathcal{E} \leftarrow \mathcal{E}}) \det([T]_{\mathcal{E} \leftarrow \mathcal{E}}) \\ &= \det(S) \det(T). \end{aligned}$$

For (3), by the correspondence between invertibility of a linear transformation and invertibility of its matrix, together with [theorem 5.23](#),

$$\begin{aligned} T \text{ is invertible} &\iff [T]_{\mathcal{E} \leftarrow \mathcal{E}} \text{ is invertible} \\ &\iff \det([T]_{\mathcal{E} \leftarrow \mathcal{E}}) \neq 0 \\ &\iff \det(T) \neq 0. \end{aligned}$$

Finally, if T is invertible, then

$$T^{-1}T = I_V.$$

Using (1) and (2),

$$1 = \det(I_V) = \det(T^{-1}T) = \det(T^{-1}) \det(T).$$

Therefore

$$\det(T^{-1}) = \frac{1}{\det(T)}.$$

Remark 5.34 — The determinant is intrinsic

A matrix representing a linear transformation depends on the choice of basis:

$$[T]_{\mathcal{E} \leftarrow \mathcal{E}} \neq [T]_{\mathcal{E}' \leftarrow \mathcal{E}'}$$

in general.

The determinant does not:

$$\det([T]_{\mathcal{E} \leftarrow \mathcal{E}}) = \det([T]_{\mathcal{E}' \leftarrow \mathcal{E}'}).$$

Thus

$$\det(T)$$

is an intrinsic property of the linear transformation T .

5.4 Minors, Cofactors, and Cofactor Expansion

The explicit permutation formula for the determinant is useful theoretically, but it quickly becomes cumbersome for computation. A more practical recursive formula expresses the determinant of an $n \times n$ matrix in terms of determinants of $(n - 1) \times (n - 1)$ matrices.

The basic objects in this formula are minors and cofactors.

Definition 5.35 — Minors and cofactors

Let

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \in \mathbb{K}^{n \times n}.$$

For

$$1 \leq i, j \leq n,$$

delete the i th row and the j th column of A . The resulting $(n - 1) \times (n - 1)$ matrix is denoted by

$$A(i|j).$$

The **minor** of the entry a_{ij} is

$$M_{ij} := \det A(i|j).$$

The **cofactor** of a_{ij} is

$$C_{ij} := (-1)^{i+j} M_{ij} = (-1)^{i+j} \det A(i|j).$$

For example, if

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

then deleting the second row and third column gives

$$A(2|3) = \begin{pmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{pmatrix}.$$

Hence

$$M_{23} = \det \begin{pmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{pmatrix},$$

and

$$C_{23} = (-1)^{2+3} M_{23} = -\det \begin{pmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{pmatrix}.$$

Thus the signs of the cofactors follow the checkerboard pattern

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

We first consider a determinant whose one column has only one nonzero entry.

Proposition 5.36 — A column with a single nonzero entry

Suppose the j th column of

$$A \in \mathbb{K}^{n \times n}$$

has only one possibly nonzero entry, namely a_{ij} :

$$A = \begin{pmatrix} a_{11} & \cdots & 0 & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & 0 & \cdots & a_{nn} \end{pmatrix}.$$

Then

$$\det(A) = a_{ij}C_{ij} = (-1)^{i+j}a_{ij}M_{ij}.$$

Proof

We move the entry a_{ij} to the $(1, 1)$ position.

First, move the j th column to the first column. This requires

$$j - 1$$

successive column interchanges. By [proposition 5.9](#), each interchange changes the sign of the determinant. Hence this contributes the factor

$$(-1)^{j-1}.$$

Next, move the i th row to the first row. This requires

$$i - 1$$

successive row interchanges.

To see the effect of a row interchange, note that interchanging two rows of a matrix A is equivalent, after transposition, to interchanging the corresponding two columns of A^T . By [theorem 5.14](#) and [proposition 5.9](#), each row interchange also changes the sign of the determinant. Hence these row interchanges contribute the factor

$$(-1)^{i-1}.$$

After these operations, we obtain a matrix of the form

$$B = \begin{pmatrix} a_{ij} & * & \cdots & * \\ 0 & & & \\ \vdots & & A(i|j) & \\ 0 & & & \end{pmatrix},$$

where the lower-right $(n-1) \times (n-1)$ block is exactly the matrix $A(i|j)$ obtained by deleting row i and column j from A .

The total number of interchanges is

$$(i-1) + (j-1) = i+j-2.$$

Therefore

$$\det(A) = (-1)^{i+j-2} \det(B) = (-1)^{i+j} \det(B).$$

It remains to compute $\det(B)$. In the first column of B , the only nonzero entry is a_{ij} . Using the explicit determinant formula from [theorem 5.4](#), every nonzero term in $\det(B)$ must select the entry a_{ij} from the first column. The remaining factors then form exactly one of the terms in

$$\det A(i|j).$$

Hence

$$\det(B) = a_{ij} \det A(i|j) = a_{ij} M_{ij}.$$

Consequently,

$$\det(A) = (-1)^{i+j} a_{ij} M_{ij}.$$

By the definition of the cofactor,

$$C_{ij} = (-1)^{i+j} M_{ij},$$

so

$$\det(A) = a_{ij} C_{ij}.$$

We can now expand a determinant along any column.

Theorem 5.37 — Cofactor expansion along a column

Let

$$A = (a_{ij}) \in \mathbb{K}^{n \times n}.$$

Fix a column

$$j \in \{1, \dots, n\}.$$

Then

$$\det(A) = \sum_{i=1}^n a_{ij} C_{ij}.$$

In full,

$$\det(A) = a_{1j} C_{1j} + a_{2j} C_{2j} + \dots + a_{nj} C_{nj}.$$

Equivalently,

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{ij} M_{ij}.$$

Proof

Write the j th column as

$$\begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj} \end{pmatrix} = a_{1j}e_1 + a_{2j}e_2 + \cdots + a_{nj}e_n,$$

where

$$e_1, \dots, e_n$$

is the standard basis of $\mathbb{K}^n$.

By linearity of the determinant in the j th column, [proposition 5.7](#),

$$\det(A) = \sum_{i=1}^n a_{ij} \det(A_i),$$

where A_i is obtained from A by replacing its j th column with e_i .

The j th column of A_i has exactly one nonzero entry, equal to 1 in row i . Therefore, by [proposition 5.36](#),

$$\det(A_i) = C_{ij}.$$

Consequently,

$$\det(A) = \sum_{i=1}^n a_{ij} C_{ij}.$$

For example, expansion along the first column gives

$$\det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} = a_{11}C_{11} + a_{21}C_{21} + \cdots + a_{n1}C_{n1}.$$

For a 3×3 matrix,

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix},$$

expansion along the first column gives

$$\begin{aligned}\det(A) &= a_{11} \det \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix} \\ &\quad - a_{21} \det \begin{pmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{pmatrix} \\ &\quad + a_{31} \det \begin{pmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{pmatrix}.\end{aligned}$$

Because the determinant is unchanged by transposition, the corresponding row formula follows immediately.

Corollary 5.38 — Cofactor expansion along a row

Let

$$A = (a_{ij}) \in \mathbb{K}^{n \times n}.$$

Fix a row

$$i \in \{1, \dots, n\}.$$

Then

$$\det(A) = \sum_{j=1}^n a_{ij} C_{ij}.$$

Thus

$$\det(A) = a_{i1} C_{i1} + a_{i2} C_{i2} + \dots + a_{in} C_{in}.$$

Proof

By [theorem 5.14](#),

$$\det(A) = \det(A^\top).$$

The i th row of A is the i th column of $A^\top$.

Applying [theorem 5.37](#) to that column of $A^\top$ gives

$$\det(A^\top) = \sum_{j=1}^n a_{ij} C_{ij}.$$

Therefore

$$\det(A) = \sum_{j=1}^n a_{ij} C_{ij}.$$

Remark 5.39 — Which row or column should be used?

A determinant may be expanded along *any* row or any column.

For computation, one should usually choose a row or column containing as many zeros as possible, because every zero entry contributes

$$0 \cdot C_{ij} = 0$$

to the expansion.

Example 5.40 — Expansion along a sparse column

Consider

$$A = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix}.$$

The second column contains only one nonzero entry, so expanding along that column gives

$$\det(A) = 3C_{22}.$$

Since

$$C_{22} = (-1)^{2+2} \det \begin{pmatrix} 2 & 1 \\ 4 & 5 \end{pmatrix},$$

we obtain

$$\begin{aligned} \det(A) &= 3 \det \begin{pmatrix} 2 & 1 \\ 4 & 5 \end{pmatrix} \\ &= 3(10 - 4) \\ &= 18. \end{aligned}$$

Cofactor expansion also gives an identity that will be useful when we construct the inverse of a matrix.

Proposition 5.41 — Cofactor orthogonality

Let

$$A = (a_{ij}) \in \mathbb{K}^{n \times n}.$$

For a fixed column r ,

$$\sum_{i=1}^n a_{ir} C_{ir} = \det(A).$$

If

$$r \neq s,$$

then

$$\sum_{i=1}^n a_{ir} C_{is} = 0.$$

Similarly, for rows,

$$\sum_{j=1}^n a_{rj} C_{rj} = \det(A),$$

whereas for

$$r \neq s,$$

$$\sum_{j=1}^n a_{rj} C_{sj} = 0.$$

Proof

The first and third identities are precisely the column and row cofactor expansions from [theorem 5.37](#) and [corollary 5.38](#).

Now let

$$r \neq s.$$

Form a new matrix B by replacing column s of A with column r of A .

Then columns r and s of B are equal. Hence, by the alternating property of the determinant,

$$\det(B) = 0.$$

Expand $\det(B)$ along column s . The entries of that column are

$$a_{1r}, \dots, a_{nr}.$$

Moreover, deleting column s in forming the corresponding cofactors removes the replaced column, so those cofactors are exactly

$$C_{1s}, \dots, C_{ns}$$

from the original matrix A .

Therefore

$$0 = \det(B) = \sum_{i=1}^n a_{ir} C_{is}.$$

The corresponding row identity follows either by the same argument or by applying the

column identity to $A^\top$ and using [theorem 5.14](#).

5.5 The Adjugate Matrix and Cramer's Rule

The cofactor identities from the previous subsection can be assembled into a matrix identity. This leads to an explicit formula for the inverse of a nonsingular matrix and, as a consequence, to Cramer's rule for solving square linear systems.

Definition 5.42 — Cofactor matrix and adjugate

Let

$$A = (a_{ij}) \in \mathbb{K}^{n \times n},$$

and let

$$C_{ij} = (-1)^{i+j} M_{ij}$$

be the cofactor of the entry a_{ij} , as defined in [definition 5.35](#).

The **cofactor matrix** of A is

$$C(A) = \begin{pmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{pmatrix}.$$

The **adjugate matrix** of A is the transpose of the cofactor matrix:

$$\text{adj}(A) = C(A)^\top.$$

Thus

$$\text{adj}(A) = \begin{pmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{pmatrix},$$

so that

$$(\text{adj}(A))_{ij} = C_{ji}.$$

The transpose in the definition is important. It places the cofactors in precisely the positions needed for ordinary matrix multiplication.

Theorem 5.43 — Adjugate identity

For every

$$A \in \mathbb{K}^{n \times n},$$

we have

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = \det(A)I_n.$$

Proof

We first compute

$$A \operatorname{adj}(A).$$

Because

$$(\operatorname{adj}(A))_{kj} = C_{jk},$$

the (i, j) entry of the product is

$$\begin{aligned} (A \operatorname{adj}(A))_{ij} &= \sum_{k=1}^n a_{ik} (\operatorname{adj}(A))_{kj} \\ &= \sum_{k=1}^n a_{ik} C_{jk}. \end{aligned}$$

Consider first the diagonal case

$$i = j.$$

Then

$$(A \operatorname{adj}(A))_{ii} = \sum_{k=1}^n a_{ik} C_{ik}.$$

By the cofactor expansion along row i in [corollary 5.38](#),

$$\sum_{k=1}^n a_{ik} C_{ik} = \det(A).$$

Hence

$$(A \operatorname{adj}(A))_{ii} = \det(A).$$

Now suppose

$$i \neq j.$$

By the row version of the cofactor orthogonality identity in [proposition 5.41](#),

$$\sum_{k=1}^n a_{ik} C_{jk} = 0.$$

Therefore

$$(A \operatorname{adj}(A))_{ij} = 0 \quad (i \neq j).$$

Thus

$$A \operatorname{adj}(A) = \begin{pmatrix} \det(A) & 0 & \cdots & 0 \\ 0 & \det(A) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \det(A) \end{pmatrix} = \det(A)I_n.$$

The identity in the opposite order is proved similarly. Since

$$(\operatorname{adj}(A))_{ik} = C_{ki},$$

we have

$$(\operatorname{adj}(A)A)_{ij} = \sum_{k=1}^n C_{ki}a_{kj}.$$

If

$$i = j,$$

the cofactor expansion along column i gives

$$\sum_{k=1}^n a_{ki}C_{ki} = \det(A).$$

If

$$i \neq j,$$

the column version of [proposition 5.41](#) gives

$$\sum_{k=1}^n a_{kj}C_{ki} = 0.$$

Hence

$$\operatorname{adj}(A)A = \det(A)I_n.$$

Therefore

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = \det(A)I_n.$$

The adjugate identity immediately gives an explicit inverse formula.

Corollary 5.44 — Inverse formula using the adjugate

Let

$$A \in \mathbb{K}^{n \times n}$$

satisfy

$$\det(A) \neq 0.$$

Then

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Proof

By [theorem 5.43](#),

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = \det(A)I_n.$$

Because

$$\det(A) \neq 0,$$

we may divide by $\det(A)$. Hence

$$A \left(\frac{1}{\det(A)} \operatorname{adj}(A) \right) = I_n$$

and

$$\left(\frac{1}{\det(A)} \operatorname{adj}(A) \right) A = I_n.$$

Therefore

$$\frac{1}{\det(A)} \operatorname{adj}(A)$$

is the inverse of A , and thus

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

Example 5.45 — Inverse of a 2×2 matrix

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Its cofactors are

$$C_{11} = d, \quad C_{12} = -c, \quad C_{21} = -b, \quad C_{22} = a.$$

Hence the cofactor matrix is

$$C(A) = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix},$$

and therefore

$$\text{adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Since

$$\det(A) = ad - bc,$$

if

$$ad - bc \neq 0,$$

then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Remark 5.46 — The adjugate formula is mainly theoretical

The formula

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

gives an explicit expression for the inverse in terms of determinants.

For a large matrix, however, computing all n^2 cofactors is generally inefficient.

We now apply the inverse formula to a square system of linear equations.

Consider

$$Ax = b,$$

where

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}.$$

For each

$$j = 1, \dots, n,$$

let $A_j(b)$ denote the matrix obtained from A by replacing its j th column by b :

$$A_j(b) = \begin{pmatrix} a_{11} & \cdots & a_{1,j-1} & b_1 & a_{1,j+1} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2,j-1} & b_2 & a_{2,j+1} & \cdots & a_{2n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & \cdots & a_{n,j-1} & b_n & a_{n,j+1} & \cdots & a_{nn} \end{pmatrix}.$$

Theorem 5.47 — Cramer's rule

Let

$$A \in \mathbb{K}^{n \times n}$$

and suppose

$$\det(A) \neq 0.$$

Then, for every

$$b \in \mathbb{K}^n,$$

the system

$$Ax = b$$

has a unique solution. Its components are

$$x_j = \frac{\det A_j(b)}{\det A}, \quad j = 1, \dots, n.$$

Proof

Because

$$\det(A) \neq 0,$$

[theorem 5.23](#) implies that A is invertible. Hence

$$Ax = b$$

has the unique solution

$$x = A^{-1}b.$$

By [corollary 5.44](#),

$$x = \frac{1}{\det(A)} \operatorname{adj}(A)b.$$

The j th row of $\operatorname{adj}(A)$ is

$$(C_{1j}, C_{2j}, \dots, C_{nj}).$$

Therefore the j th component of x is

$$x_j = \frac{1}{\det(A)} \sum_{i=1}^n C_{ij} b_i.$$

Now consider the matrix $A_j(b)$ obtained by replacing the j th column of A by b .

Expand its determinant along the j th column. By [theorem 5.37](#),

$$\det A_j(b) = \sum_{i=1}^n b_i \tilde{C}_{ij},$$

where $\tilde{C}_{ij}$ denotes the cofactor of the (i, j) entry of $A_j(b)$.

Deleting row i and column j removes the replaced column entirely. Thus the matrix used to compute $\tilde{C}_{ij}$ is exactly the same as the matrix used to compute C_{ij} in A . Hence

$$\tilde{C}_{ij} = C_{ij}.$$

Therefore

$$\det A_j(b) = \sum_{i=1}^n b_i C_{ij}.$$

Substituting this identity into the formula for x_j gives

$$x_j = \frac{\det A_j(b)}{\det A}.$$

Since this holds for every

$$j = 1, \dots, n,$$

we obtain

$$x_j = \frac{\det A_j(b)}{\det A}, \quad j = 1, \dots, n.$$

Example 5.48 — Cramer's rule for a 2×2 system

Consider

$$ax + by = r,$$

$$cx + dy = s,$$

or equivalently,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} r \\ s \end{pmatrix}.$$

Suppose

$$ad - bc \neq 0.$$

Then

$$A_1(b) = \begin{pmatrix} r & b \\ s & d \end{pmatrix}, \quad A_2(b) = \begin{pmatrix} a & r \\ c & s \end{pmatrix}.$$

By [theorem 5.47](#),

$$x = \frac{\det \begin{pmatrix} r & b \\ s & d \end{pmatrix}}{\det \begin{pmatrix} a & b \\ c & d \end{pmatrix}} = \frac{rd - bs}{ad - bc},$$

and

$$y = \frac{\det \begin{pmatrix} a & r \\ c & s \end{pmatrix}}{\det \begin{pmatrix} a & b \\ c & d \end{pmatrix}} = \frac{as - cr}{ad - bc}.$$

Remark 5.49 — What Cramer's rule tells us

Cramer's rule gives an explicit formula for each component of the solution:

$$x_j = \frac{\det A_j(b)}{\det A}.$$

Its main importance is theoretical rather than computational. For large systems, Gaussian elimination is far more efficient. (You should have seen this in your undergraduate linear algebra course).

5.6 Geometric Meaning of the Determinant

So far, we have studied the determinant algebraically: through its multilinearity, its alternating property, its multiplicativity, and its relation to invertibility. We now explain its geometric meaning.

The key idea is that the determinant measures how a linear transformation changes size. In dimension 2, it measures the scaling of area; in dimension 3, it measures the scaling of volume. Its sign records whether orientation is preserved or reversed.

The Determinant in $\mathbb{R}^2$ Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} | & | \\ u & v \\ | & | \end{pmatrix}, \quad u = \begin{pmatrix} a \\ c \end{pmatrix}, \quad v = \begin{pmatrix} b \\ d \end{pmatrix}.$$

Then the two columns u and v span a parallelogram in $\mathbb{R}^2$.

The determinant

$$\det(A) = ad - bc$$

is the *signed area* of this parallelogram. In particular,

$$|\det(A)| = \text{the area of the parallelogram spanned by } u \text{ and } v.$$

Thus the determinant gives more than just area: its sign also carries orientation information.

Orientation in $\mathbb{R}^2$. The standard basis

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

determines the standard counterclockwise orientation of the plane.

► If

$$\det(A) > 0,$$

then the ordered pair (u, v) has the same orientation as (e_1, e_2) .

► If

$$\det(A) < 0,$$

then the orientation is reversed.

► If

$$\det(A) = 0,$$

then u and v are linearly dependent, so the parallelogram collapses to a line segment and its area is zero.

A useful way to think about this is to start from the unit square. The matrix A sends the unit square to the parallelogram spanned by its two columns, and the area changes by the factor $|\det(A)|$.

Therefore a linear map

$$A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

scales all areas by the factor

$$|\det(A)|.$$

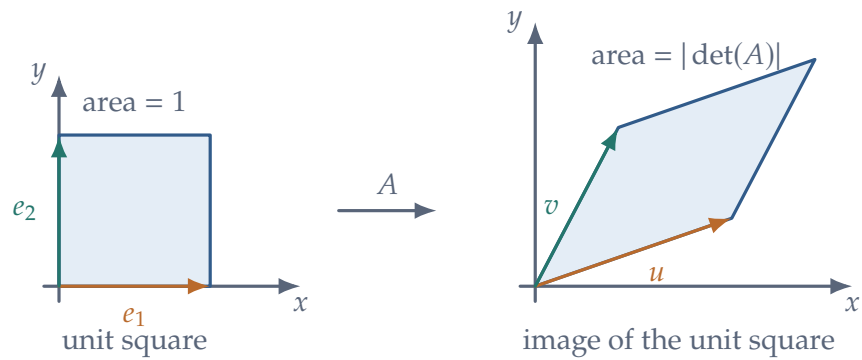


Figure 1: In $\mathbb{R}^2$, the determinant measures signed area.

Examples.

1. If

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix},$$

then

$$\det(A) = 6.$$

So the unit square is stretched to a rectangle of area 6.

2. If

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

then

$$\det(A) = 1.$$

This is a shear: it changes the shape, but not the area.

3. If

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

then

$$\det(A) = -1.$$

Thus area is preserved, but orientation is reversed.

The Determinant in $\mathbb{R}^3$ Now let

$$A = \begin{pmatrix} | & | & | \\ u & v & w \\ | & | & | \end{pmatrix} \in \mathbb{R}^{3 \times 3},$$

where

$$u, v, w \in \mathbb{R}^3.$$

These three vectors span a parallelepiped in $\mathbb{R}^3$.

The determinant

$$\det(A)$$

is the *signed volume* of this parallelepiped. Hence

$$|\det(A)| = \text{the volume of the parallelepiped spanned by } u, v, w.$$

As in dimension 2, the sign records orientation.

► If

$$\det(A) > 0,$$

then the ordered triple (u, v, w) has the same orientation as the standard basis of $\mathbb{R}^3$.

► If

$$\det(A) < 0,$$

then the orientation is reversed.

► If

$$\det(A) = 0,$$

then u, v, w are linearly dependent, so the parallelepiped collapses into a lower-dimensional object and has zero volume.

There is also a familiar vector formula:

$$\det(u, v, w) = u \cdot (v \times w).$$

Thus the determinant in $\mathbb{R}^3$ is exactly the scalar triple product.

Geometrically, one may think of this as

$$\text{volume} = \text{base area} \times \text{height}.$$

Indeed, the magnitude of

$$v \times w$$

is the area of the base parallelogram spanned by v and w , and then taking the dot product with u extracts the signed height in the normal direction.

Therefore a linear map

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

scales all volumes by the factor

$$|\det(A)|.$$

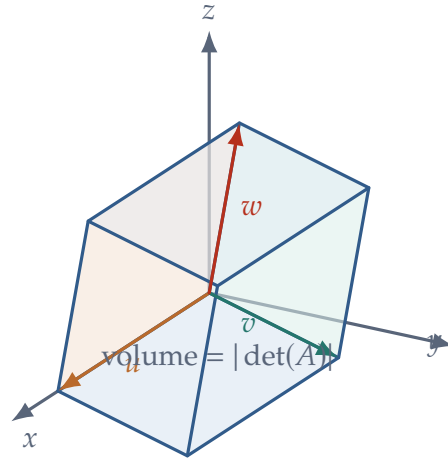


Figure 2: In $\mathbb{R}^3$, the determinant measures signed volume.

Examples.

1. If

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix},$$

then

$$\det(A) = 24.$$

So volumes are multiplied by 24.

2. If two columns of A are equal, then

$$\det(A) = 0,$$

and the parallelepiped collapses.

3. If

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

then

$$\det(A) = -1.$$

Thus volume is preserved, but orientation is reversed.

General Rule The two- and three-dimensional pictures suggest the general rule:

The determinant of a linear map measures the scaling of oriented volume.

In dimension n , if

$$A \in \mathbb{R}^{n \times n},$$

then

$$|\det(A)|$$

is the factor by which A scales n -dimensional volume, and the sign of $\det(A)$ records whether orientation is preserved or reversed.

This geometric interpretation explains many of the algebraic properties proved earlier:

- If

$$\det(A) = 0,$$

then volume collapses, so A cannot be invertible.

- If

$$\det(AB) = \det(A) \det(B),$$

then the total scaling under a composition is the product of the two individual scalings.

- If two columns are swapped, orientation reverses, so the determinant changes sign.
- If one column is replaced by itself plus a multiple of another, the spanned area or volume does not change (you can verify this with a plot), which explains why the determinant is unchanged under such an operation.

6 Eigenvalues, Eigenvectors, and Diagonalization

6.1 Eigenvalues and Eigenvectors

In [proposition 4.22](#), we saw that a one-dimensional subspace

$$\text{Span}(v), \quad v \neq 0,$$

is invariant under a linear transformation

$$T : V \rightarrow V$$

if and only if there exists a scalar $\lambda \in \mathbb{K}$ such that

$$T(v) = \lambda v.$$

Thus, on the one-dimensional subspace $\text{Span}(v)$, the transformation T acts simply by multiplication by the scalar λ .

This observation leads to the fundamental notions of eigenvalues and eigenvectors.

Definition 6.1 — Eigenvalues and eigenvectors

Let

$$T : V \rightarrow V$$

be a linear transformation.

A scalar

$$\lambda \in \mathbb{K}$$

is called an **eigenvalue** of T if there exists a nonzero vector

$$v \in V$$

such that

$$T(v) = \lambda v.$$

Any nonzero vector v satisfying

$$T(v) = \lambda v$$

is called an **eigenvector** of T corresponding to the eigenvalue λ .

The condition

$$v \neq 0$$

is essential. Indeed, the zero vector satisfies

$$T(0) = 0 = \lambda 0$$

for every scalar $\lambda \in \mathbb{K}$. Thus allowing $v = 0$ would make every scalar an eigenvalue of every linear transformation.

Remark 6.2 — Geometric meaning of an eigenvector

An eigenvector is a nonzero vector whose direction is preserved by the linear transformation.

If

$$T(v) = \lambda v,$$

then $T(v)$ lies on the same line

$$\text{Span}(v)$$

as v .

The scalar λ describes what T does along that direction:

► if

$$\lambda > 1,$$

the vector is stretched;

► if

$$0 < \lambda < 1,$$

the vector is contracted;

► if

$$\lambda < 0,$$

the direction is reversed as well as rescaled;

► if

$$\lambda = 0,$$

then

$$T(v) = 0.$$

Thus eigenvectors identify directions on which a linear transformation acts in the simplest possible way: by scalar multiplication.

Example 6.3 — A simple example

Consider

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

defined by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 3y \end{pmatrix}.$$

For

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

we have

$$T(e_1) = \begin{pmatrix} 2 \\ 0 \end{pmatrix} = 2e_1.$$

Hence

$$2$$

is an eigenvalue of T , and e_1 is a corresponding eigenvector.

Similarly,

$$T(e_2) = \begin{pmatrix} 0 \\ 3 \end{pmatrix} = 3e_2,$$

so

$$3$$

is an eigenvalue and e_2 is a corresponding eigenvector.

More generally, every nonzero vector on the x -axis is an eigenvector corresponding to 2, and every nonzero vector on the y -axis is an eigenvector corresponding to 3.

The eigenvalue equation can be rewritten in a form that connects it directly to the kernel and invertibility theory developed earlier.

Starting from

$$T(v) = \lambda v,$$

we have

$$T(v) - \lambda v = 0.$$

Since

$$\lambda v = (\lambda I_V)(v),$$

this is equivalent to

$$(T - \lambda I_V)(v) = 0.$$

Thus finding eigenvectors corresponding to λ amounts to finding nonzero vectors in the kernel of

$$T - \lambda I_V.$$

Theorem 6.4 — Equivalent conditions for an eigenvalue

Let V be finite-dimensional, let

$$T : V \rightarrow V$$

be a linear transformation, and let

$$\lambda \in \mathbb{K}.$$

Then the following statements are equivalent:

- (i) λ is an eigenvalue of T ;
- (ii) there exists a nonzero vector $v \in V$ such that

$$(T - \lambda I_V)(v) = 0;$$

(iii)

$$\ker(T - \lambda I_V) \neq \{0\};$$

- (iv) $T - \lambda I_V$ is not injective;
- (v) $T - \lambda I_V$ is not surjective;
- (vi) $T - \lambda I_V$ is not invertible.

Proof

By [definition 6.1](#), λ is an eigenvalue of T if and only if there exists

$$v \neq 0$$

such that

$$T(v) = \lambda v.$$

But

$$T(v) = \lambda v$$

if and only if

$$T(v) - \lambda v = 0,$$

which is equivalent to

$$(T - \lambda I_V)(v) = 0.$$

Hence

$$(i) \iff (ii).$$

Statement (ii) says precisely that the kernel of $T - \lambda I_V$ contains a nonzero vector. Thus

$$(ii) \iff (iii).$$

By the criterion for injectivity established earlier,

$$T - \lambda I_V \text{ is injective} \iff \ker(T - \lambda I_V) = \{0\}.$$

Therefore

$$(iii) \iff (iv).$$

Finally,

$$T - \lambda I_V : V \rightarrow V$$

is a linear transformation from a finite-dimensional vector space to itself. Because the domain and codomain have the same dimension, the equivalence between injectivity, surjectivity, and invertibility proved earlier gives

$$(iv) \iff (v) \iff (vi).$$

Hence all six statements are equivalent.

We now translate the preceding discussion into matrix language.

Let

$$\mathcal{E} = (e_1, \dots, e_n)$$

be a basis of V , and let

$$A = [T]_{\mathcal{E} \leftarrow \mathcal{E}}.$$

If

$$v \in V$$

has coordinate vector

$$x = [v]_{\mathcal{E}},$$

then

$$[T(v)]_{\mathcal{E}} = Ax.$$

Hence the eigenvalue equation

$$T(v) = \lambda v$$

is equivalent to

$$Ax = \lambda x.$$

Equivalently,

$$(A - \lambda I_n)x = 0.$$

Definition 6.5 — Eigenvalues and eigenvectors of a matrix

Let

$$A \in \mathbb{K}^{n \times n}.$$

A scalar

$$\lambda \in \mathbb{K}$$

is an **eigenvalue of A** if there exists a nonzero vector

$$x \in \mathbb{K}^n$$

such that

$$Ax = \lambda x.$$

Such a nonzero vector x is called an **eigenvector of A** corresponding to λ .

Thus

$$Ax = \lambda x$$

if and only if

$$(A - \lambda I_n)x = 0.$$

Consequently, λ is an eigenvalue of A precisely when the homogeneous system

$$(A - \lambda I_n)x = 0$$

has a nonzero solution.

Proposition 6.6 — Matrix criterion for an eigenvalue

Let

$$A \in \mathbb{K}^{n \times n}$$

and let

$$\lambda \in \mathbb{K}.$$

Then the following statements are equivalent:

(i) λ is an eigenvalue of A ;

(ii)

$$\ker(A - \lambda I_n) \neq \{0\};$$

(iii)

$$\text{Rank}(A - \lambda I_n) < n;$$

(iv) $A - \lambda I_n$ is not invertible;

(v)

$$\det(A - \lambda I_n) = 0.$$

Proof

By [definition 6.5](#), λ is an eigenvalue of A if and only if there exists

$$x \neq 0$$

such that

$$Ax = \lambda x.$$

Equivalently,

$$(A - \lambda I_n)x = 0.$$

Thus

$$(i) \iff (ii).$$

By the matrix form of the rank–nullity theorem,

$$n = \text{Rank}(A - \lambda I_n) + \dim \ker(A - \lambda I_n).$$

Therefore

$$\ker(A - \lambda I_n) \neq \{0\}$$

if and only if

$$\text{Rank}(A - \lambda I_n) < n.$$

Hence

$$(ii) \iff (iii).$$

By [corollary 3.57](#),

$$\text{Rank}(A - \lambda I_n) = n \iff A - \lambda I_n \text{ is invertible.}$$

Thus

$$(iii) \iff (iv).$$

Finally, by the determinant criterion for invertibility, [theorem 5.23](#),

$$A - \lambda I_n \text{ is invertible} \iff \det(A - \lambda I_n) \neq 0.$$

Therefore

$$(iv) \iff (v).$$

Hence all five statements are equivalent.

The final condition,

$$\det(A - \lambda I_n) = 0,$$

is particularly important. It converts the search for eigenvalues into the problem of solving a scalar polynomial equation. We will formalize this observation when we introduce the characteristic polynomial.

Example 6.7 — Finding eigenvalues of a 2×2 matrix

Consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

A scalar λ is an eigenvalue precisely when

$$\det(A - \lambda I_2) = 0.$$

Now

$$A - \lambda I_2 = \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix},$$

so

$$\begin{aligned}\det(A - \lambda I_2) &= \det \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix} \\ &= (2 - \lambda)^2 - 1 \\ &= \lambda^2 - 4\lambda + 3 \\ &= (\lambda - 1)(\lambda - 3).\end{aligned}$$

Thus the eigenvalues are

$$\lambda = 1 \quad \text{and} \quad \lambda = 3.$$

For $\lambda = 3$, we solve

$$\begin{aligned}(A - 3I_2)x &= 0 : \\ \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix}.\end{aligned}$$

Hence

$$x_1 = x_2.$$

Thus every nonzero vector of the form

$$\begin{pmatrix} t \\ t \end{pmatrix}, \quad t \neq 0,$$

is an eigenvector corresponding to $\lambda = 3$.

For $\lambda = 1$,

$$(A - I_2)x = 0$$

becomes

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

so

$$x_1 = -x_2.$$

Hence every nonzero vector of the form

$$\begin{pmatrix} t \\ -t \end{pmatrix}, \quad t \neq 0,$$

is an eigenvector corresponding to $\lambda = 1$.

6.2 Eigenspaces

For a fixed eigenvalue λ , there are usually many corresponding eigenvectors. Because scalar multiples and sums of eigenvectors corresponding to the same eigenvalue remain associated with that eigenvalue, it is natural to collect them into a subspace.

Recall from [theorem 6.4](#) that

$$T(v) = \lambda v$$

is equivalent to

$$(T - \lambda I_V)(v) = 0.$$

Thus the relevant subspace is the kernel of $T - \lambda I_V$.

Definition 6.8 — Eigenspace

Let

$$T : V \rightarrow V$$

be a linear transformation and let

$$\lambda \in \mathbb{K}.$$

The **eigenspace of T corresponding to λ** is

$$E_\lambda(T) := \ker(T - \lambda I_V).$$

Equivalently,

$$E_\lambda(T) = \{v \in V : T(v) = \lambda v\}.$$

Because $E_\lambda(T)$ is a kernel, it is automatically a subspace of V by the earlier result that kernels of linear maps are subspaces.

Notice an important distinction:

$$E_\lambda(T)$$

contains the zero vector, whereas an eigenvector is required to be nonzero.

Thus, if λ is an eigenvalue,

$$E_\lambda(T) \setminus \{0\}$$

is precisely the set of eigenvectors corresponding to λ .

Proposition 6.9 — Eigenvalues and nonzero eigenspaces

Let

$$T : V \rightarrow V$$

be linear and let

$$\lambda \in \mathbb{K}.$$

Then

$$\lambda \text{ is an eigenvalue of } T \iff E_\lambda(T) \neq \{0\}.$$

Proof

By [definition 6.1](#), λ is an eigenvalue of T if and only if there exists a nonzero vector $v \in V$ such that

$$T(v) = \lambda v.$$

By [definition 6.8](#), this is equivalent to the existence of a nonzero vector

$$v \in E_\lambda(T).$$

Hence

$$\lambda \text{ is an eigenvalue} \iff E_\lambda(T) \neq \{0\}.$$

Proposition 6.10 — An eigenspace is invariant

Let

$$T : V \rightarrow V$$

be linear and let $\lambda \in \mathbb{K}$. Then

$$E_\lambda(T)$$

is invariant under T .

Moreover, the restriction of T to $E_\lambda(T)$ is simply scalar multiplication by λ :

$$T|_{E_\lambda(T)} = \lambda I_{E_\lambda(T)}.$$

Proof

Let

$$v \in E_\lambda(T).$$

By [definition 6.8](#),

$$T(v) = \lambda v.$$

Because $E_\lambda(T)$ is a subspace and $v \in E_\lambda(T)$,

$$\lambda v \in E_\lambda(T).$$

Hence

$$T(v) \in E_\lambda(T).$$

Therefore $E_\lambda(T)$ is invariant under T .

Furthermore, for every

$$v \in E_\lambda(T),$$

we have

$$T(v) = \lambda v.$$

Thus, on this subspace,

$$T|_{E_\lambda(T)} = \lambda I_{E_\lambda(T)}.$$

Thus an eigenspace is a subspace on which the transformation has the simplest possible form: it acts as multiplication by a single scalar.

Example 6.11 — Eigenspaces of a matrix

Consider again

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

In [example 6.7](#), we found the eigenvalues

$$\lambda = 3 \quad \text{and} \quad \lambda = 1.$$

For $\lambda = 3$,

$$E_3(A) = \ker(A - 3I_2).$$

Since

$$A - 3I_2 = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix},$$

the equation

$$(A - 3I_2)x = 0$$

implies

$$x_1 = x_2.$$

Therefore

$$E_3(A) = \text{Span} \left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} \right).$$

Similarly,

$$E_1(A) = \ker(A - I_2),$$

and

$$A - I_2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Thus

$$x_1 = -x_2,$$

and hence

$$E_1(A) = \text{Span} \left(\begin{pmatrix} 1 \\ -1 \end{pmatrix} \right).$$

In this example,

$$\mathbb{R}^2 = E_3(A) \oplus E_1(A).$$

The last equation in the example illustrates a general phenomenon: eigenspaces corresponding to distinct eigenvalues do not overlap except at the zero vector.

Proposition 6.12 — Distinct eigenspaces intersect trivially

Let

$$T : V \rightarrow V$$

be linear, and let

$$\lambda, \mu \in \mathbb{K}, \quad \lambda \neq \mu.$$

Then

$$E_\lambda(T) \cap E_\mu(T) = \{0\}.$$

Proof

Let

$$v \in E_\lambda(T) \cap E_\mu(T).$$

Then

$$T(v) = \lambda v$$

and

$$T(v) = \mu v.$$

Therefore

$$\lambda v = \mu v,$$

so

$$(\lambda - \mu)v = 0.$$

Because

$$\lambda \neq \mu,$$

we have

$$\lambda - \mu \neq 0.$$

Since $\mathbb{K}$ is a field, it follows that

$$v = 0.$$

Hence

$$E_\lambda(T) \cap E_\mu(T) = \{0\}.$$

By [corollary 4.10](#), we immediately obtain

$$E_\lambda(T) + E_\mu(T) = E_\lambda(T) \oplus E_\mu(T)$$

whenever

$$\lambda \neq \mu.$$

For more than two eigenspaces, pairwise trivial intersections alone are not sufficient to establish that the whole sum is direct. Nevertheless, distinct eigenspaces satisfy the stronger condition required for a direct sum.

Theorem 6.13 — Eigenspaces corresponding to distinct eigenvalues form a direct sum

Let

$$T : V \rightarrow V$$

be a linear transformation, and let

$$\lambda_1, \dots, \lambda_m$$

be distinct eigenvalues of T .

Then

$$E_{\lambda_1}(T) + \dots + E_{\lambda_m}(T)$$

is a direct sum. Thus

$$E_{\lambda_1}(T) \oplus \dots \oplus E_{\lambda_m}(T).$$

Proof

By [theorem 4.9](#), it is enough to prove that the only way to write

$$0 = v_1 + \dots + v_m, \quad v_i \in E_{\lambda_i}(T),$$

is with

$$v_1 = \cdots = v_m = 0.$$

We prove this by induction on m .

For

$$m = 1,$$

the result is immediate.

Now suppose the result holds for $m - 1$ distinct eigenvalues, and suppose

$$v_1 + \cdots + v_m = 0, \quad v_i \in E_{\lambda_i}(T).$$

Apply the linear transformation

$$T - \lambda_m I_V$$

to both sides. We obtain

$$(T - \lambda_m I_V)(v_1) + \cdots + (T - \lambda_m I_V)(v_m) = 0.$$

Because

$$v_i \in E_{\lambda_i}(T),$$

we have

$$T(v_i) = \lambda_i v_i.$$

Hence

$$(T - \lambda_m I_V)(v_i) = (\lambda_i - \lambda_m)v_i.$$

For $i = m$,

$$(\lambda_m - \lambda_m)v_m = 0.$$

Therefore

$$(\lambda_1 - \lambda_m)v_1 + \cdots + (\lambda_{m-1} - \lambda_m)v_{m-1} = 0.$$

For each

$$i = 1, \dots, m-1,$$

the vector

$$(\lambda_i - \lambda_m)v_i$$

belongs to $E_{\lambda_i}(T)$ because $E_{\lambda_i}(T)$ is a subspace.

The eigenvalues

$$\lambda_1, \dots, \lambda_{m-1}$$

are distinct. By the induction hypothesis,

$$(\lambda_i - \lambda_m)v_i = 0 \quad \text{for every } i = 1, \dots, m-1.$$

Since the eigenvalues are all distinct,

$$\lambda_i - \lambda_m \neq 0.$$

Thus

$$v_i = 0, \quad i = 1, \dots, m-1.$$

Returning to

$$v_1 + \dots + v_m = 0,$$

we obtain

$$v_m = 0.$$

Hence

$$v_1 = \dots = v_m = 0.$$

By [theorem 4.9](#), the sum is direct.

This theorem gives two important consequences.

Corollary 6.14 — Eigenvectors corresponding to distinct eigenvalues are linearly independent

Let

$$v_1, \dots, v_m$$

be eigenvectors of T corresponding to distinct eigenvalues

$$\lambda_1, \dots, \lambda_m.$$

Then

$$v_1, \dots, v_m$$

are linearly independent.

Proof

Suppose

$$c_1v_1 + \dots + c_mv_m = 0.$$

For each i ,

$$c_iv_i \in E_{\lambda_i}(T).$$

By [theorem 6.13](#), the representation of the zero vector as a sum of vectors from the distinct eigenspaces is unique. Hence

$$c_i v_i = 0 \quad \text{for every } i.$$

Because each v_i is nonzero,

$$c_i = 0 \quad \text{for every } i.$$

Therefore

$$v_1, \dots, v_m$$

are linearly independent.

Corollary 6.15 — Number and dimensions of distinct eigenspaces

Suppose V is finite-dimensional and

$$\lambda_1, \dots, \lambda_m$$

are distinct eigenvalues of T . Then

$$\dim E_{\lambda_1}(T) + \dots + \dim E_{\lambda_m}(T) \leq \dim V.$$

In particular,

$$m \leq \dim V.$$

Thus a linear transformation on an n -dimensional vector space has at most n distinct eigenvalues.

Proof

By [theorem 6.13](#),

$$E_{\lambda_1}(T) + \dots + E_{\lambda_m}(T) = E_{\lambda_1}(T) \oplus \dots \oplus E_{\lambda_m}(T).$$

Therefore, by the dimension characterization of a direct sum in [theorem 4.9](#),

$$\dim (E_{\lambda_1}(T) + \dots + E_{\lambda_m}(T)) = \sum_{i=1}^m \dim E_{\lambda_i}(T).$$

The sum is a subspace of V , so

$$\dim (E_{\lambda_1}(T) + \dots + E_{\lambda_m}(T)) \leq \dim V.$$

Hence

$$\sum_{i=1}^m \dim E_{\lambda_i}(T) \leq \dim V.$$

Because each λ_i is an eigenvalue,

$$E_{\lambda_i}(T) \neq \{0\}$$

by [proposition 6.9](#). Thus

$$\dim E_{\lambda_i}(T) \geq 1.$$

Therefore

$$m \leq \sum_{i=1}^m \dim E_{\lambda_i}(T) \leq \dim V.$$

For a matrix, the same construction takes a particularly concrete form.

Definition 6.16 — Eigenspace of a matrix

Let

$$A \in \mathbb{K}^{n \times n}$$

and let $\lambda \in \mathbb{K}$.

The eigenspace of A corresponding to λ is

$$E_{\lambda}(A) = \ker(A - \lambda I_n).$$

Thus

$$E_{\lambda}(A) = \{x \in \mathbb{K}^n : Ax = \lambda x\}.$$

If λ is an eigenvalue, then

$$\dim E_{\lambda}(A) = \dim \ker(A - \lambda I_n).$$

By the matrix rank–nullity theorem,

$$\dim E_{\lambda}(A) = n - \text{Rank}(A - \lambda I_n).$$

Thus the dimension of an eigenspace can be computed directly from the rank of

$$A - \lambda I_n.$$

Remark 6.17 — The role of eigenspaces

An eigenspace is more than a collection of eigenvectors plus the zero vector. It is an invariant subspace on which T acts as scalar multiplication:

$$T|_{E_{\lambda}(T)} = \lambda I.$$

Moreover, eigenspaces corresponding to distinct eigenvalues combine without overlap:

$$E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T).$$

The central question for diagonalization will therefore be whether these eigenspaces are large enough, collectively, to fill the entire vector space:

$$V \stackrel{?}{=} \bigoplus_{\lambda} E_{\lambda}(T).$$

6.3 Eigenvalues and the Characteristic Polynomial

In [proposition 6.6](#), we showed that a scalar $\lambda \in \mathbb{K}$ is an eigenvalue of

$$A \in \mathbb{K}^{n \times n}$$

if and only if

$$A - \lambda I_n$$

is singular. By the determinant criterion for invertibility, [theorem 5.23](#), this is equivalent to

$$\det(A - \lambda I_n) = 0.$$

Thus the eigenvalues of A can be found by studying the determinant

$$\det(A - \lambda I_n)$$

as a polynomial in λ .

Definition 6.18 — Characteristic polynomial and characteristic equation

Let

$$A \in \mathbb{K}^{n \times n}.$$

The **characteristic polynomial** of A is

$$\chi_A(\lambda) := \det(A - \lambda I_n).$$

The equation

$$\chi_A(\lambda) = 0,$$

that is,

$$\det(A - \lambda I_n) = 0,$$

is called the **characteristic equation** of A .

Proposition 6.19 — Degree and leading term of the characteristic polynomial

Let

$$A \in \mathbb{K}^{n \times n}.$$

Then $\chi_A(\lambda)$ is a polynomial of degree n , with leading coefficient $(-1)^n$.

Thus

$$\chi_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_1 \lambda + c_0$$

for some

$$c_0, \dots, c_{n-1} \in \mathbb{K}.$$

Moreover,

$$c_0 = \det(A).$$

Proof

By definition,

$$\chi_A(\lambda) = \det(A - \lambda I_n).$$

Using the permutation formula from [theorem 5.4](#),

$$\chi_A(\lambda) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^n (A - \lambda I_n)_{\sigma(j), j}.$$

Consider first the term corresponding to the identity permutation. It is

$$(a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda).$$

Its highest-degree term is

$$(-\lambda)^n = (-1)^n \lambda^n.$$

Now consider a nonidentity permutation σ . Then for at least one j ,

$$\sigma(j) \neq j.$$

The corresponding factor

$$(A - \lambda I_n)_{\sigma(j), j} = a_{\sigma(j), j}$$

is an off-diagonal entry and therefore contains no λ . Consequently, every term arising from a nonidentity permutation has degree at most $n - 1$.

Hence

$$\deg \chi_A = n$$

and its leading coefficient is

$$(-1)^n.$$

Finally, evaluating at $\lambda = 0$ gives

$$\chi_A(0) = \det(A - 0I_n) = \det(A).$$

Thus the constant term is

$$c_0 = \det(A).$$

The characteristic polynomial is important because its roots are exactly the eigenvalues.

Theorem 6.20 — Eigenvalues are roots of the characteristic polynomial

Let

$$A \in \mathbb{K}^{n \times n}$$

and let

$$\lambda \in \mathbb{K}.$$

Then

$$\lambda \text{ is an eigenvalue of } A \iff \chi_A(\lambda) = 0.$$

Proof

See the beginning discussion of this subsection.

Thus finding the eigenvalues of a matrix can be separated into two steps:

$$A \longrightarrow \chi_A(\lambda) = \det(A - \lambda I_n) \longrightarrow \chi_A(\lambda) = 0.$$

Once an eigenvalue λ has been found, the corresponding eigenspace is obtained by solving

$$(A - \lambda I_n)x = 0 :$$

$$E_\lambda(A) = \ker(A - \lambda I_n).$$

Example 6.21 — Characteristic polynomial of a 2×2 matrix

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then

$$A - \lambda I_2 = \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix},$$

and therefore

$$\begin{aligned} \chi_A(\lambda) &= \det \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} \\ &= (a - \lambda)(d - \lambda) - bc \\ &= \lambda^2 - (a + d)\lambda + (ad - bc). \end{aligned}$$

Since

$$\operatorname{tr}(A) = a + d$$

and

$$\det(A) = ad - bc,$$

we obtain

$$\chi_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A).$$

Thus the eigenvalues of A are precisely the roots of

$$\lambda^2 - \operatorname{tr}(A)\lambda + \det(A) = 0.$$

Example 6.22 — A 3×3 example

Consider

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

Then

$$A - \lambda I_3 = \begin{pmatrix} 2 - \lambda & 1 & 0 \\ 0 & 3 - \lambda & 0 \\ 0 & 0 & 4 - \lambda \end{pmatrix}.$$

Because this matrix is upper triangular, [theorem 5.16](#) gives

$$\begin{aligned} \chi_A(\lambda) &= \det(A - \lambda I_3) \\ &= (2 - \lambda)(3 - \lambda)(4 - \lambda). \end{aligned}$$

Thus the eigenvalues are

$$2, \quad 3, \quad 4.$$

Notice that the leading term is

$$-\lambda^3,$$

as predicted by [proposition 6.19](#).

The characteristic polynomial should depend only on the linear transformation represented by a matrix, not on the particular basis used to represent it. The next result verifies this fact.

Proposition 6.23 — Similar matrices have the same characteristic polynomial

Let

$$A, B \in \mathbb{K}^{n \times n}.$$

If A and B are similar, then

$$\chi_A(\lambda) = \chi_B(\lambda).$$

Proof

Suppose

$$B = P^{-1}AP$$

for some invertible matrix

$$P \in \mathbb{K}^{n \times n}.$$

Then

$$\begin{aligned} B - \lambda I_n &= P^{-1}AP - \lambda I_n \\ &= P^{-1}AP - \lambda P^{-1}P \\ &= P^{-1}(A - \lambda I_n)P. \end{aligned}$$

Hence

$$B - \lambda I_n$$

is similar to

$$A - \lambda I_n.$$

By [corollary 5.29](#),

$$\det(B - \lambda I_n) = \det(A - \lambda I_n).$$

Therefore

$$\chi_B(\lambda) = \chi_A(\lambda).$$

This similarity invariance allows us to define the characteristic polynomial of an abstract linear transformation.

Definition 6.24 — Characteristic polynomial of a linear transformation

Let

$$T : V \rightarrow V$$

be a linear transformation on an n -dimensional vector space V .

Choose any basis

$$\mathcal{E}$$

of V , and let

$$A = [T]_{\mathcal{E} \leftarrow \mathcal{E}}.$$

The **characteristic polynomial** of T is defined by

$$\chi_T(\lambda) := \chi_A(\lambda) = \det(A - \lambda I_n).$$

Proposition 6.25 — The characteristic polynomial is basis-independent

The definition in [definition 6.24](#) does not depend on the chosen basis of V .

Proof

Let

$$\mathcal{E} \quad \text{and} \quad \mathcal{E}'$$

be two bases of V .

By the change-of-basis formula,

$$[T]_{\mathcal{E}' \leftarrow \mathcal{E}'} = P^{-1}[T]_{\mathcal{E} \leftarrow \mathcal{E}}P$$

for some invertible matrix P .

Thus the two matrices representing T are similar. By [proposition 6.23](#), they have the same characteristic polynomial.

Corollary 6.26 — Eigenvalues of a linear transformation

Let V be finite-dimensional and let

$$T : V \rightarrow V$$

be linear. Then

$$\lambda \in \mathbb{K}$$

is an eigenvalue of T if and only if

$$\chi_T(\lambda) = 0.$$

Proof

Choose a basis, and then invoke [theorem 6.20](#) and [definition 6.24](#).

Remark 6.27 — The characteristic polynomial and the scalar field

The characteristic polynomial may fail to have a root in the scalar field over which the vector space is defined.

For example,

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

has

$$\begin{aligned} \chi_A(\lambda) &= \det \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix} \\ &= \lambda^2 + 1. \end{aligned}$$

Over $\mathbb{R}$, the equation

$$\lambda^2 + 1 = 0$$

has no solution. Thus A has no real eigenvalues.

Over $\mathbb{C}$, however,

$$\lambda^2 + 1 = (\lambda - i)(\lambda + i),$$

so the eigenvalues are

$$i \quad \text{and} \quad -i.$$

Thus the existence of eigenvalues depends on the scalar field.

Corollary 6.28 — Existence of an eigenvalue over $\mathbb{C}$

Let V be a nonzero finite-dimensional complex vector space and let

$$T : V \rightarrow V$$

be linear. Then T has at least one eigenvalue.

Proof

Let

$$n = \dim V \geq 1.$$

By [proposition 6.19](#),

$$\chi_T(\lambda)$$

is a polynomial of degree n , and hence is nonconstant.

By the Fundamental Theorem of Algebra, every nonconstant polynomial with complex coefficients has a complex root. Thus there exists

$$\lambda \in \mathbb{C}$$

such that

$$\chi_T(\lambda) = 0.$$

By [corollary 6.26](#), λ is an eigenvalue of T .

We also need to distinguish how many times an eigenvalue occurs as a root of the characteristic polynomial from the dimension of its eigenspace.

Definition 6.29 — Algebraic and geometric multiplicity

Let

$$T : V \rightarrow V$$

be a linear transformation on a finite-dimensional vector space, and let λ_0 be an eigenvalue of T .

The **algebraic multiplicity** of λ_0 is its multiplicity as a root of the characteristic polynomial

$$\chi_T(\lambda).$$

More precisely, the algebraic multiplicity is the largest positive integer m such that

$$(\lambda - \lambda_0)^m$$

divides $\chi_T(\lambda)$.

The **geometric multiplicity** of λ_0 is the dimension of the corresponding eigenspace:

$$\dim E_{\lambda_0}(T) = \dim \ker(T - \lambda_0 I_V).$$

Example 6.30 — Algebraic multiplicity need not determine the eigenspace

Consider the two matrices

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

For both matrices,

$$\chi(\lambda) = \det \begin{pmatrix} 2 - \lambda & * \\ 0 & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2.$$

Thus 2 has algebraic multiplicity 2 for both A and B .

For A ,

$$A - 2I_2 = 0,$$

so

$$E_2(A) = \mathbb{R}^2$$

and

$$\dim E_2(A) = 2.$$

For B ,

$$B - 2I_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

so

$$E_2(B) = \ker(B - 2I_2) = \text{Span} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right),$$

and therefore

$$\dim E_2(B) = 1.$$

Thus the same characteristic polynomial can correspond to different eigenspace dimensions.

Remark 6.31 — Algebraic versus geometric multiplicity

For every eigenvalue λ_0 , its geometric multiplicity is at least one:

$$\dim E_{\lambda_0}(T) \geq 1.$$

We will shortly prove the stronger inequality

$$1 \leq \dim E_{\lambda_0}(T) \leq \text{algebraic multiplicity of } \lambda_0.$$

The relation between these two multiplicities is central to diagonalization.

6.4 Diagonalization

The natural question in this section is whether the entire vector space can be decomposed into such directions.

Equivalently:

Does V have a basis consisting entirely of eigenvectors of T ?

If the answer is yes, then the matrix of T in that basis is diagonal. Recall the form of a diagonal matrix from [corollary 5.17](#).

Definition 6.32 — Diagonalizable linear transformation

Let

$$T : V \rightarrow V$$

be a linear transformation on a finite-dimensional vector space.

We say that T is **diagonalizable** if there exists a basis $\mathcal{E}$ of V such that

$$[T]_{\mathcal{E} \leftarrow \mathcal{E}}$$

is diagonal.

The definition is equivalent to the existence of a basis of eigenvectors.

Theorem 6.33 — Diagonalization and eigenvector bases

Let

$$T : V \rightarrow V$$

be a linear transformation on an n -dimensional vector space.

Then T is diagonalizable if and only if V has a basis

$$v_1, \dots, v_n$$

consisting of eigenvectors of T .

More precisely, if

$$T(v_i) = \lambda_i v_i, \quad i = 1, \dots, n,$$

then with respect to the basis

$$\mathcal{E} = (v_1, \dots, v_n),$$

the matrix

$$[T]_{\mathcal{E} \leftarrow \mathcal{E}}$$

is diagonal, with diagonal entries

$$\lambda_1, \dots, \lambda_n.$$

Proof

Suppose first that

$$\mathcal{E} = (v_1, \dots, v_n)$$

is a basis consisting of eigenvectors, with

$$T(v_i) = \lambda_i v_i.$$

Then

$$[T(v_i)]_{\mathcal{E}} = \lambda_i [v_i]_{\mathcal{E}}.$$

Thus the i th column of

$$[T]_{\mathcal{E} \leftarrow \mathcal{E}}$$

has λ_i in the i th position and zeros elsewhere. Hence the matrix is diagonal.

Conversely, suppose

$$[T]_{\mathcal{E} \leftarrow \mathcal{E}}$$

is diagonal for some basis

$$\mathcal{E} = (v_1, \dots, v_n).$$

If its i th diagonal entry is λ_i , then its i th column gives

$$[T(v_i)]_{\mathcal{E}} = \lambda_i [v_i]_{\mathcal{E}}.$$

Therefore

$$T(v_i) = \lambda_i v_i.$$

Thus every vector in $\mathcal{E}$ is an eigenvector of T .

Hence diagonalization has a simple geometric interpretation:

diagonalization = choosing a basis along eigenvector directions.

We can now express this in terms of the eigenspaces introduced earlier.

Theorem 6.34 — Equivalent characterizations of diagonalizability

Let

$$T : V \rightarrow V$$

be a linear transformation on a finite-dimensional vector space, and let

$$\lambda_1, \dots, \lambda_m$$

be the distinct eigenvalues of T .

Then the following statements are equivalent:

- (i) T is diagonalizable;
- (ii) V has a basis consisting of eigenvectors of T ;
- (iii)

$$V = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T);$$

- (iv)

$$\dim V = \dim E_{\lambda_1}(T) + \cdots + \dim E_{\lambda_m}(T).$$

Proof

The equivalence

$$(i) \iff (ii)$$

is [theorem 6.33](#).

Suppose (ii) holds. Every vector in an eigenvector basis belongs to one of the eigenspaces

$$E_{\lambda_1}(T), \dots, E_{\lambda_m}(T).$$

Hence

$$V = E_{\lambda_1}(T) + \cdots + E_{\lambda_m}(T).$$

By [theorem 6.13](#), this sum is direct. Therefore

$$(ii) \implies (iii).$$

If (iii) holds, then the dimension formula for a direct sum gives

$$\dim V = \sum_{i=1}^m \dim E_{\lambda_i}(T),$$

so

$$(iii) \implies (iv).$$

Finally, suppose (iv) holds. For each i , choose a basis $\mathcal{B}_i$ of $E_{\lambda_i}(T)$. Since distinct eigenspaces form a direct sum, the concatenation

$$\mathcal{B}_1, \dots, \mathcal{B}_m$$

is linearly independent by [theorem 4.9](#).

By (iv), this list contains exactly $\dim V$ vectors. Hence, by [proposition 1.21](#), it is a basis of V . Every vector in the list is an eigenvector, and therefore

$$(iv) \implies (ii).$$

Thus all four statements are equivalent.

The matrix formulation follows immediately from the linear transformation associated with a square matrix.

Corollary 6.35 — Diagonalization of a matrix

Let

$$A \in \mathbb{K}^{n \times n},$$

and define

$$T_A : \mathbb{K}^n \rightarrow \mathbb{K}^n, \quad T_A(x) = Ax.$$

Then A is similar to a diagonal matrix if and only if $\mathbb{K}^n$ has a basis consisting of eigenvectors of A .

In particular, suppose

$$v_1, \dots, v_n$$

is such a basis, with

$$Av_i = \lambda_i v_i.$$

Let

$$P = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{pmatrix},$$

and let D be the diagonal matrix whose diagonal entries are

$$\lambda_1, \dots, \lambda_n$$

in the corresponding order.

Then

$$P^{-1}AP = D.$$

Proof

The matrix of T_A in the standard basis is A .

By [theorem 6.33](#), the basis

$$(v_1, \dots, v_n)$$

gives a diagonal matrix D for the same linear transformation.

The change-of-basis formula in [corollary 3.39](#) therefore gives

$$D = P^{-1}AP.$$

Corollary 6.36 — Distinct eigenvalues imply diagonalizability

Let

$$T : V \rightarrow V$$

be linear and suppose

$$\dim V = n.$$

If T has n distinct eigenvalues, then T is diagonalizable.

Proof

Choose one eigenvector corresponding to each of the n distinct eigenvalues.

By [corollary 6.14](#), these n eigenvectors are linearly independent. Since

$$\dim V = n,$$

they form a basis by [proposition 1.21](#).

Hence T is diagonalizable by [theorem 6.33](#).

The converse is false. A diagonalizable transformation need not have $\dim V$ distinct eigenvalues. What matters is the number of *linearly independent eigenvectors*, or equivalently, the dimensions of the eigenspaces.

This brings us back to the two notions of multiplicity introduced in [definition 6.29](#).

Theorem 6.37 — Geometric multiplicity is at most algebraic multiplicity

Let

$$T : V \rightarrow V$$

be linear on a finite-dimensional vector space, and let λ be an eigenvalue of T .

Then

$$1 \leq \dim E_\lambda(T) \leq m_a(\lambda),$$

where

$$m_a(\lambda)$$

denotes the algebraic multiplicity of λ .

Thus

$$1 \leq m_g(\lambda) \leq m_a(\lambda).$$

Proof

Because λ is an eigenvalue,

$$E_\lambda(T) \neq \{0\},$$

so

$$\dim E_\lambda(T) \geq 1.$$

Let

$$r = \dim E_\lambda(T),$$

and choose a basis

$$v_1, \dots, v_r$$

of $E_\lambda(T)$. They are linearly independent in V . By the basis extension result, [proposition 3.10](#), we may extend this list to a basis

$$\mathcal{E} = (v_1, \dots, v_r, v_{r+1}, \dots, v_n)$$

of V .

Write

$$A = [T]_{\mathcal{E} \leftarrow \mathcal{E}}.$$

For

$$i = 1, \dots, r,$$

we have

$$T(v_i) = \lambda v_i.$$

Hence the first r columns of A have λ in their corresponding diagonal positions and zero elsewhere.

Now consider

$$A - \mu I_n.$$

For each

$$i = 1, \dots, r,$$

its i th column has the single nonzero entry

$$\lambda - \mu$$

in position i .

Because

$$v_1, \dots, v_r$$

belong to $E_\lambda(T)$, we have

$$T(v_i) = \lambda v_i, \quad i = 1, \dots, r.$$

Therefore, with respect to the basis

$$\mathcal{E} = (v_1, \dots, v_r, v_{r+1}, \dots, v_n),$$

the first r columns of

$$A = [T]_{\mathcal{E} \leftarrow \mathcal{E}}$$

are

$$\lambda e_1, \dots, \lambda e_r.$$

Thus $A - \mu I_n$ has the form

$$A - \mu I_n = \left(\begin{array}{cccc|cccc} \lambda - \mu & 0 & \cdots & 0 & * & \cdots & * & \\ 0 & \lambda - \mu & \cdots & 0 & * & \cdots & * & \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots & \\ 0 & 0 & \cdots & \lambda - \mu & * & \cdots & * & \\ \hline 0 & 0 & \cdots & 0 & & & & \\ \vdots & \vdots & & \vdots & & & B - \mu I_{n-r} & \\ 0 & 0 & \cdots & 0 & & & & \end{array} \right).$$

The vertical line separates the first r columns, corresponding to the basis vectors

$$v_1, \dots, v_r,$$

from the remaining columns.

Now expand the determinant along the first column. Since its only nonzero entry is

$$\lambda - \mu,$$

we obtain one factor

$$\lambda - \mu.$$

Continuing in this way through the first r columns gives

$$\det(A - \mu I_n) = (\lambda - \mu)^r \det(B - \mu I_{n-r}).$$

Hence

$$\det(A - \mu I_n) = (\lambda - \mu)^r q(\mu),$$

where

$$q(\mu) = \det(B - \mu I_{n-r}).$$

Thus λ is a root of the characteristic polynomial with multiplicity at least r . Therefore

$$m_a(\lambda) \geq r = \dim E_\lambda(T).$$

We immediately obtain the main multiplicity criterion for diagonalizability.

Theorem 6.38 — Diagonalizability and multiplicities

Let

$$T : V \rightarrow V$$

be linear on an n -dimensional vector space. Suppose its characteristic polynomial splits completely over $\mathbb{K}$.

Let

$$\lambda_1, \dots, \lambda_m$$

be its distinct eigenvalues.

Then T is diagonalizable if and only if

$$m_g(\lambda_i) = m_a(\lambda_i) \quad \text{for every } i = 1, \dots, m.$$

Proof

Because the characteristic polynomial has degree n and splits over $\mathbb{K}$,

$$\sum_{i=1}^m m_a(\lambda_i) = n.$$

By [theorem 6.37](#),

$$m_g(\lambda_i) \leq m_a(\lambda_i) \quad \text{for every } i.$$

Suppose first that

$$m_g(\lambda_i) = m_a(\lambda_i) \quad \text{for every } i.$$

Then

$$\sum_{i=1}^m \dim E_{\lambda_i}(T) = \sum_{i=1}^m m_g(\lambda_i) = \sum_{i=1}^m m_a(\lambda_i) = n.$$

By [theorem 6.34](#), T is diagonalizable.

Conversely, suppose T is diagonalizable. Then

$$\sum_{i=1}^m m_g(\lambda_i) = \sum_{i=1}^m \dim E_{\lambda_i}(T) = n.$$

But also

$$\sum_{i=1}^m m_a(\lambda_i) = n$$

and

$$m_g(\lambda_i) \leq m_a(\lambda_i)$$

for every i .

Therefore equality must hold term by term:

$$m_g(\lambda_i) = m_a(\lambda_i) \quad \text{for every } i.$$

Remark 6.39 — What prevents diagonalization

When the characteristic polynomial splits, failure of diagonalizability has a precise meaning. For some eigenvalue λ ,

$$m_g(\lambda) < m_a(\lambda).$$

Thus the characteristic polynomial says that λ occurs several times algebraically, but its eigenspace does not contain enough independent eigenvectors to account for all of those occurrences.

There is another, very compact way to characterize this failure. It is based on the minimal polynomial.

Definition 6.40 — Polynomial of a linear transformation

Let

$$p(t) = a_0 + a_1t + \cdots + a_k t^k \in \mathbb{K}[t].$$

For

$$T : V \rightarrow V,$$

define

$$p(T) = a_0 I_V + a_1 T + \cdots + a_k T^k.$$

Definition 6.41 — Minimal polynomial

Let V be finite-dimensional and let

$$T : V \rightarrow V$$

be linear.

The **minimal polynomial** of T , denoted

$$m_T(t),$$

is the unique monic polynomial of smallest degree such that

$$m_T(T) = 0.$$

Such a polynomial exists. Indeed, if

$$n = \dim V,$$

then

$$I_V, T, T^2, \dots, T^{n^2}$$

are $n^2 + 1$ elements of the n^2 -dimensional vector space $\mathcal{L}(V)$, and hence are linearly dependent. Thus some nonzero polynomial p satisfies

$$p(T) = 0.$$

After dividing by its leading coefficient, we obtain a monic annihilating polynomial, and among all such polynomials we choose one of smallest degree.

Uniqueness follows because the difference of two distinct monic annihilating polynomials of the same smallest degree would be a nonzero annihilating polynomial of smaller degree.

Proposition 6.42 — The minimal polynomial divides every annihilating polynomial

Let

$$p \in \mathbb{K}[t].$$

If

$$p(T) = 0,$$

then

$$m_T(t) \mid p(t).$$

Proof

By polynomial division, write

$$p(t) = q(t)m_T(t) + r(t),$$

where either $r = 0$ or

$$\deg r < \deg m_T.$$

Applying this identity to T gives

$$0 = p(T) = q(T)m_T(T) + r(T) = r(T).$$

If $r \neq 0$, then after dividing r by its leading coefficient we would obtain a monic annihilating polynomial of degree smaller than m_T , contradicting the definition of the minimal polynomial.

Hence

$$r = 0,$$

so

$$m_T \mid p.$$

The minimal polynomial contains exactly the obstruction to diagonalizability.

Theorem 6.43 — Minimal-polynomial criterion for diagonalizability

Let

$$T : V \rightarrow V$$

be linear on a finite-dimensional vector space.

Then T is diagonalizable over $\mathbb{K}$ if and only if its minimal polynomial splits over $\mathbb{K}$ into distinct linear factors:

$$m_T(t) = (t - \lambda_1) \cdots (t - \lambda_m),$$

where

$$\lambda_1, \dots, \lambda_m$$

are distinct.

Proof

Suppose first that T is diagonalizable, and let

$$\lambda_1, \dots, \lambda_m$$

be its distinct eigenvalues.

Define

$$p(t) = (t - \lambda_1) \cdots (t - \lambda_m).$$

Because V has a basis consisting of eigenvectors, it is enough to evaluate $p(T)$ on an eigenvector v corresponding to some λ_i . We obtain

$$p(T)v = p(\lambda_i)v = 0.$$

Hence

$$p(T) = 0.$$

By [proposition 6.42](#),

$$m_T \mid p.$$

On the other hand, every eigenvalue λ_i must be a root of m_T . Indeed, if

$$T(v) = \lambda_i v, \quad v \neq 0,$$

then

$$0 = m_T(T)v = m_T(\lambda_i)v,$$

which implies

$$m_T(\lambda_i) = 0.$$

Thus each factor

$$t - \lambda_i$$

divides m_T . Since the λ_i are distinct,

$$p \mid m_T.$$

Both polynomials are monic, so

$$m_T(t) = (t - \lambda_1) \cdots (t - \lambda_m).$$

Conversely, suppose

$$m_T(t) = (t - \lambda_1) \cdots (t - \lambda_m)$$

with the λ_i distinct.

For each i , define the following polynomial (also called the Lagrange polynomial)

$$\ell_i(t) = \prod_{j \neq i} \frac{t - \lambda_j}{\lambda_i - \lambda_j}.$$

Then

$$\ell_i(\lambda_j) = \begin{cases} 1, & j = i, \\ 0, & j \neq i. \end{cases}$$

Hence, by polynomial interpolation,

$$\ell_1(t) + \cdots + \ell_m(t) = 1.$$

Applying this identity to T gives

$$I_V = \ell_1(T) + \cdots + \ell_m(T).$$

Therefore every

$$v \in V$$

can be written as

$$v = \ell_1(T)v + \cdots + \ell_m(T)v.$$

Moreover,

$$(t - \lambda_i)\ell_i(t)$$

is a scalar multiple of $m_T(t)$. Hence

$$(T - \lambda_i I_V)\ell_i(T) = 0.$$

Therefore

$$\ell_i(T)v \in E_{\lambda_i}(T).$$

It follows that

$$V = E_{\lambda_1}(T) + \cdots + E_{\lambda_m}(T).$$

By [theorem 6.13](#), this sum is direct:

$$V = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_m}(T).$$

Hence T is diagonalizable by [theorem 6.34](#).

Remark 6.44 — Three views of diagonalizability

When the characteristic polynomial splits over $\mathbb{K}$, we now have three ways to recognize diagonalizability:

$$\begin{aligned}
T \text{ is diagonalizable} &\iff V \text{ has a basis of eigenvectors,} \\
&\iff m_g(\lambda) = m_a(\lambda) \text{ for every eigenvalue } \lambda, \\
&\iff m_T(t) \text{ has no repeated linear factor.}
\end{aligned}$$

The first condition is geometric, the second compares the eigenspaces with the characteristic polynomial, and the third records the same information compactly in the minimal polynomial.

Example 6.45 — Repeated eigenvalues: two different possibilities

Consider

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Both matrices have characteristic polynomial

$$(2 - \lambda)^2.$$

Thus 2 has algebraic multiplicity 2 in both cases.

For A ,

$$E_2(A) = \mathbb{K}^2,$$

so

$$m_g(2) = 2 = m_a(2).$$

Moreover,

$$m_A(t) = t - 2.$$

Hence A is diagonalizable.

For B ,

$$E_2(B) = \text{Span} \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \right),$$

so

$$m_g(2) = 1 < 2 = m_a(2).$$

Its minimal polynomial is

$$m_B(t) = (t - 2)^2.$$

Hence B is not diagonalizable.

Thus a repeated root of the characteristic polynomial does not by itself prevent diagonalization. The issue is whether there are enough independent eigenvectors, or equivalently, whether the corresponding factor is repeated in the minimal polynomial.

Remark 6.46 — Why diagonalization is useful

If

$$A = PDP^{-1}$$

with D diagonal, then for every positive integer k ,

$$A^k = PD^kP^{-1}.$$

Powers of a diagonal matrix are immediate to compute: each diagonal entry is simply raised to the k th power.

Thus diagonalization converts repeated application of a complicated linear transformation into repeated scalar multiplication along its eigenvector directions.

7 Inner Product Spaces and Quadratic Forms

So far, our study of vector spaces has been primarily algebraic. We can add vectors, multiply them by scalars, choose bases, and study linear transformations between them. What is still missing is the geometric structure familiar from $\mathbb{R}^2$ and $\mathbb{R}^3$: length, angle, and perpendicularity.

An *inner product* equips a vector space with this geometry.

7.1 Inner Products, Norms, and Orthogonality

For vectors

$$x = (x_1, \dots, x_n), \quad y = (y_1, \dots, y_n)$$

in $\mathbb{R}^n$, recall the Euclidean dot product

$$x^\top y = x_1y_1 + \dots + x_ny_n.$$

It contains geometric information that the vector-space operations alone do not provide. In particular,

$$x^\top x = x_1^2 + \dots + x_n^2$$

is the square of the Euclidean length of x , while

$$x^\top y = 0$$

means that x and y are perpendicular.

An inner product abstracts the essential properties of this operation.

Definition 7.1 — Inner product

Let V be a vector space over

$$\mathbb{K} = \mathbb{R} \quad \text{or} \quad \mathbb{K} = \mathbb{C}.$$

An **inner product** on V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{K}$$

such that, for all

$$u, v, w \in V, \quad \alpha, \beta \in \mathbb{K},$$

the following properties hold:

(i) **Linearity in the first argument:**

$$\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle.$$

(ii) **Conjugate symmetry:**

$$\langle u, v \rangle = \overline{\langle v, u \rangle}.$$

(iii) **Positive definiteness:**

$$\langle v, v \rangle \geq 0,$$

with

$$\langle v, v \rangle = 0 \iff v = 0.$$

A vector space equipped with an inner product is called an **inner product space**.

When

$$\mathbb{K} = \mathbb{R},$$

complex conjugation has no effect, so conjugate symmetry becomes ordinary symmetry:

$$\langle u, v \rangle = \langle v, u \rangle.$$

Linearity in the first argument and conjugate symmetry determine how the inner product behaves in the second argument.

Proposition 7.2 — Basic properties of an inner product

For all

$$u, v, w \in V, \quad \alpha, \beta \in \mathbb{K},$$

we have

$$\langle u, \alpha v + \beta w \rangle = \overline{\alpha} \langle u, v \rangle + \overline{\beta} \langle u, w \rangle.$$

In particular,

$$\langle 0, v \rangle = \langle v, 0 \rangle = 0.$$

Proof

By conjugate symmetry and linearity in the first argument,

$$\begin{aligned}\langle u, \alpha v + \beta w \rangle &= \overline{\langle \alpha v + \beta w, u \rangle} \\ &= \overline{\alpha \langle v, u \rangle + \beta \langle w, u \rangle} \\ &= \overline{\alpha} \overline{\langle v, u \rangle} + \overline{\beta} \overline{\langle w, u \rangle} \\ &= \overline{\alpha} \langle u, v \rangle + \overline{\beta} \langle u, w \rangle.\end{aligned}$$

Also, linearity gives

$$\langle 0, v \rangle = \langle 0 + 0, v \rangle = \langle 0, v \rangle + \langle 0, v \rangle,$$

so

$$\langle 0, v \rangle = 0.$$

Conjugate symmetry then gives

$$\langle v, 0 \rangle = 0.$$

Example 7.3 — Examples of inner products

The standard inner product on $\mathbb{R}^n$ is

$$\langle x, y \rangle = x^\top y = \sum_{i=1}^n x_i y_i.$$

More generally, if

$$c_1, \dots, c_n > 0,$$

then

$$\langle x, y \rangle = \sum_{i=1}^n c_i x_i y_i$$

defines another inner product on $\mathbb{R}^n$.

Inner products are not restricted to coordinate vectors. For example, on the vector space $C[a, b]$ of continuous real-valued functions on $[a, b]$,

$$\langle f, g \rangle = \int_a^b f(t)g(t) \, dt$$

defines an inner product.

Thus a vector space can carry different inner products. The underlying linear structure determines which combinations of vectors are allowed; the inner product determines the geometry imposed on that space.

Every inner product induces a notion of length.

Definition 7.4 — Norm induced by an inner product

For

$$v \in V,$$

the **norm** of v induced by the inner product is

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

For the standard inner product on $\mathbb{R}^n$,

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2},$$

so this definition reproduces ordinary Euclidean length.

Proposition 7.5 — Basic properties of the induced norm

For every

$$v \in V, \quad \alpha \in \mathbb{K},$$

we have

$$\|v\| \geq 0,$$

$$\|v\| = 0 \iff v = 0,$$

and

$$\|\alpha v\| = |\alpha| \|v\|.$$

Proof

The first two statements follow immediately from positive definiteness.

For the last statement,

$$\begin{aligned} \|\alpha v\|^2 &= \langle \alpha v, \alpha v \rangle \\ &= \alpha \bar{\alpha} \langle v, v \rangle \\ &= |\alpha|^2 \|v\|^2. \end{aligned}$$

Taking square roots gives

$$\|\alpha v\| = |\alpha| \|v\|.$$

The inner product also generalizes perpendicularity.

Definition 7.6 — Orthogonal vectors

Two vectors

$$u, v \in V$$

are called **orthogonal**, written

$$u \perp v,$$

if

$$\langle u, v \rangle = 0.$$

By conjugate symmetry,

$$u \perp v \iff v \perp u.$$

Orthogonality gives the familiar Pythagorean identity.

Proposition 7.7 — Pythagorean theorem

If

$$u \perp v,$$

then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Proof

Since

$$u \perp v,$$

we have

$$\langle u, v \rangle = \langle v, u \rangle = 0.$$

Therefore

$$\begin{aligned}\|u + v\|^2 &= \langle u + v, u + v \rangle \\ &= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + \|v\|^2.\end{aligned}$$

A particularly important idea is to decompose one vector into a component parallel to another vector and a component orthogonal to it. Let

$$v \neq 0.$$

If we seek

$$u = cv + w, \quad w \perp v,$$

then

$$0 = \langle u - cv, v \rangle = \langle u, v \rangle - c\|v\|^2.$$

Hence the coefficient c is forced to be

$$c = \frac{\langle u, v \rangle}{\|v\|^2}.$$

Proposition 7.8 — Orthogonal decomposition along one direction

Let

$$u, v \in V, \quad v \neq 0.$$

Then

$$u = \frac{\langle u, v \rangle}{\|v\|^2} v + w,$$

where

$$w = u - \frac{\langle u, v \rangle}{\|v\|^2} v$$

satisfies

$$w \perp v.$$

Proof

We compute

$$\begin{aligned} \langle w, v \rangle &= \left\langle u - \frac{\langle u, v \rangle}{\|v\|^2} v, v \right\rangle \\ &= \langle u, v \rangle - \frac{\langle u, v \rangle}{\|v\|^2} \langle v, v \rangle \\ &= 0. \end{aligned}$$

Thus

$$w \perp v.$$

This simple decomposition gives one of the fundamental inequalities of linear algebra.

Theorem 7.9 — Cauchy-Schwarz inequality

For all

$$u, v \in V,$$

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Equality holds if and only if u and v are linearly dependent.

Proof

If

$$v = 0,$$

the result is immediate.

Suppose

$$v \neq 0.$$

By [proposition 7.8](#), write

$$u = \frac{\langle u, v \rangle}{\|v\|^2} v + w, \quad w \perp v.$$

Using [proposition 7.7](#),

$$\begin{aligned} \|u\|^2 &= \left\| \frac{\langle u, v \rangle}{\|v\|^2} v \right\|^2 + \|w\|^2 \\ &= \frac{|\langle u, v \rangle|^2}{\|v\|^2} + \|w\|^2 \\ &\geq \frac{|\langle u, v \rangle|^2}{\|v\|^2}. \end{aligned}$$

Multiplying by $\|v\|^2$ gives

$$|\langle u, v \rangle|^2 \leq \|u\|^2 \|v\|^2,$$

and taking square roots yields

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Equality holds exactly when

$$\|w\| = 0,$$

that is, when

$$u = \frac{\langle u, v \rangle}{\|v\|^2} v.$$

Thus equality holds exactly when u and v are linearly dependent.

Remark 7.10 — Angles in a real inner product space

When V is a real inner product space and

$$u, v \neq 0,$$

the Cauchy–Schwarz inequality guarantees that

$$-1 \leq \frac{\langle u, v \rangle}{\|u\| \|v\|} \leq 1.$$

We can therefore define the angle $\theta \in [0, \pi]$ between u and v by

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \|v\|}.$$

In particular,

$$u \perp v \iff \theta = \frac{\pi}{2}.$$

The Cauchy–Schwarz inequality also implies that the induced norm satisfies the ordinary triangle inequality.

Theorem 7.11 — Triangle inequality

For all

$$u, v \in V,$$

$$\|u + v\| \leq \|u\| + \|v\|.$$

Proof

We have

$$\begin{aligned} \|u + v\|^2 &= \langle u + v, u + v \rangle \\ &= \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} \\ &= \|u\|^2 + \|v\|^2 + 2 \operatorname{Re} \langle u, v \rangle \\ &\leq \|u\|^2 + \|v\|^2 + 2|\langle u, v \rangle| \\ &\leq \|u\|^2 + \|v\|^2 + 2\|u\| \|v\| \\ &= (\|u\| + \|v\|)^2, \end{aligned}$$

where the second inequality follows from [theorem 7.9](#). Taking square roots gives

$$\|u + v\| \leq \|u\| + \|v\|.$$

Thus an inner product recovers the familiar Euclidean notions of length, perpendicularity, and angle, while making them meaningful in general vector spaces. The next step is to construct bases adapted to this geometry: orthonormal bases.

7.2 Orthonormal Bases and the Gram–Schmidt Process

An arbitrary basis gives coordinates for vectors, but finding those coordinates may require solving a system of linear equations. Inner product spaces admit a particularly convenient class of bases for which the coordinates can instead be recovered directly from inner products.

We begin with orthogonal and orthonormal lists.

Definition 7.12 — Orthogonal and orthonormal lists

A list of vectors

$$e_1, \dots, e_m$$

in an inner product space V is called **orthogonal** if

$$\langle e_i, e_j \rangle = 0 \quad \text{whenever } i \neq j.$$

The list is called **orthonormal** if it is orthogonal and

$$\|e_i\| = 1 \quad \text{for every } i.$$

Equivalently,

$$\langle e_i, e_j \rangle = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

An orthonormal list that is also a basis of V is called an **orthonormal basis**.

For example, the standard basis

$$e_1, \dots, e_n$$

of $\mathbb{K}^n$, equipped with the standard inner product, is an orthonormal basis.

Orthogonal vectors behave especially simply under linear combinations.

Proposition 7.13 — Norm of an orthonormal linear combination

If

$$e_1, \dots, e_m$$

is an orthonormal list and

$$a_1, \dots, a_m \in \mathbb{K},$$

then

$$\|a_1 e_1 + \dots + a_m e_m\|^2 = |a_1|^2 + \dots + |a_m|^2.$$

Proof

Using orthonormality,

$$\begin{aligned}\left\|\sum_{i=1}^m a_i e_i\right\|^2 &= \left\langle \sum_{i=1}^m a_i e_i, \sum_{j=1}^m a_j e_j \right\rangle \\ &= \sum_{i=1}^m \sum_{j=1}^m a_i \overline{a_j} \langle e_i, e_j \rangle \\ &= \sum_{i=1}^m |a_i|^2.\end{aligned}$$

All cross terms vanish because

$$\langle e_i, e_j \rangle = 0 \quad \text{for } i \neq j.$$

An immediate consequence is that orthonormal lists can never contain linear redundancy.

Corollary 7.14 — Orthonormal lists are linearly independent

Every orthonormal list is linearly independent.

Proof

Suppose

$$a_1 e_1 + \cdots + a_m e_m = 0.$$

Then by [proposition 7.13](#),

$$0 = \|a_1 e_1 + \cdots + a_m e_m\|^2 = |a_1|^2 + \cdots + |a_m|^2.$$

Hence

$$a_1 = \cdots = a_m = 0.$$

Thus the list is linearly independent.

Corollary 7.15 — An orthonormal list of the right length is a basis

Let V be finite-dimensional. If

$$e_1, \dots, e_n$$

is an orthonormal list and

$$n = \dim V,$$

then

$$e_1, \dots, e_n$$

is an orthonormal basis of V .

Proof

By [corollary 7.14](#), the list is linearly independent. Since it contains exactly

$$\dim V$$

vectors, [proposition 1.21](#) implies that it is a basis of V .

The main computational advantage of an orthonormal basis is that its coordinate coefficients can be read off directly from the inner product.

Theorem 7.16 — Coordinates in an orthonormal basis

Let

$$\mathcal{E} = (e_1, \dots, e_n)$$

be an orthonormal basis of V . Then, for every

$$v \in V,$$

$$v = \sum_{i=1}^n \langle v, e_i \rangle e_i.$$

Thus

$$[v]_{\mathcal{E}} = \begin{pmatrix} \langle v, e_1 \rangle \\ \vdots \\ \langle v, e_n \rangle \end{pmatrix}.$$

Moreover,

$$\|v\|^2 = \sum_{i=1}^n |\langle v, e_i \rangle|^2,$$

and, for all $u, v \in V$,

$$\langle u, v \rangle = \sum_{i=1}^n \langle u, e_i \rangle \overline{\langle v, e_i \rangle}.$$

Proof

Because

$$e_1, \dots, e_n$$

is a basis, there exist unique scalars

$$a_1, \dots, a_n$$

such that

$$v = a_1 e_1 + \dots + a_n e_n.$$

Taking the inner product with e_k gives

$$\begin{aligned}\langle v, e_k \rangle &= \left\langle \sum_{i=1}^n a_i e_i, e_k \right\rangle \\ &= \sum_{i=1}^n a_i \langle e_i, e_k \rangle \\ &= a_k.\end{aligned}$$

Therefore

$$a_k = \langle v, e_k \rangle,$$

which proves

$$v = \sum_{i=1}^n \langle v, e_i \rangle e_i.$$

Applying [proposition 7.13](#) to this representation gives

$$\|v\|^2 = \sum_{i=1}^n |\langle v, e_i \rangle|^2.$$

Finally, writing both u and v in the orthonormal basis gives

$$\begin{aligned}\langle u, v \rangle &= \left\langle \sum_{i=1}^n \langle u, e_i \rangle e_i, \sum_{j=1}^n \langle v, e_j \rangle e_j \right\rangle \\ &= \sum_{i=1}^n \langle u, e_i \rangle \overline{\langle v, e_i \rangle}.\end{aligned}$$

Remark 7.17 — Parseval's identity

The identity

$$\|v\|^2 = \sum_{i=1}^n |\langle v, e_i \rangle|^2$$

is called **Parseval's identity**.

It shows that relative to an orthonormal basis, an abstract n -dimensional inner product space behaves exactly like $\mathbb{K}^n$ with its standard inner product: the quantities

$$\langle v, e_1 \rangle, \dots, \langle v, e_n \rangle$$

play the role of the ordinary coordinates of v .

The usefulness of orthonormal bases raises a natural question:

Given an arbitrary basis, can we construct an orthonormal basis from it?

The idea is simple. Suppose first that

$$v_1, v_2$$

are linearly independent.

Normalize the first vector:

$$e_1 = \frac{v_1}{\|v_1\|}.$$

The vector v_2 generally has a component in the direction e_1 . By [proposition 7.8](#), that component is

$$\langle v_2, e_1 \rangle e_1.$$

Subtracting it gives

$$w_2 = v_2 - \langle v_2, e_1 \rangle e_1,$$

which satisfies

$$w_2 \perp e_1.$$

We then normalize:

$$e_2 = \frac{w_2}{\|w_2\|}.$$

For the next vector v_3 , we remove its components in both of the directions already constructed:

$$w_3 = v_3 - \langle v_3, e_1 \rangle e_1 - \langle v_3, e_2 \rangle e_2.$$

The resulting vector is orthogonal to both e_1 and e_2 .

Continuing in this way gives the Gram–Schmidt process.

Theorem 7.18 — Gram–Schmidt process

Let

$$v_1, \dots, v_m$$

be a linearly independent list in an inner product space V .

Define

$$w_1 = v_1, \quad e_1 = \frac{w_1}{\|w_1\|},$$

and, recursively for

$$k = 2, \dots, m,$$

define

$$w_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j, \quad e_k = \frac{w_k}{\|w_k\|}.$$

Then

$$e_1, \dots, e_m$$

is an orthonormal list. Moreover, for every

$$k = 1, \dots, m,$$

$$\text{Span}(v_1, \dots, v_k) = \text{Span}(e_1, \dots, e_k).$$

Proof

We proceed by induction on k .

For

$$k = 1,$$

we have

$$e_1 = \frac{v_1}{\|v_1\|}.$$

Thus

$$\|e_1\| = 1$$

and

$$\text{Span}(v_1) = \text{Span}(e_1).$$

Now suppose that

$$e_1, \dots, e_{k-1}$$

is orthonormal and that

$$\text{Span}(v_1, \dots, v_{k-1}) = \text{Span}(e_1, \dots, e_{k-1}).$$

First, we show that

$$w_k \neq 0.$$

If

$$w_k = 0,$$

then

$$v_k = \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j \in \text{Span}(e_1, \dots, e_{k-1}).$$

By the induction hypothesis,

$$v_k \in \text{Span}(v_1, \dots, v_{k-1}),$$

contradicting the linear independence of

$$v_1, \dots, v_m.$$

Hence

$$w_k \neq 0,$$

so e_k is well defined and

$$\|e_k\| = 1.$$

Next, for

$$i < k,$$

we have

$$\begin{aligned} \langle w_k, e_i \rangle &= \left\langle v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j, e_i \right\rangle \\ &= \langle v_k, e_i \rangle - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle \langle e_j, e_i \rangle \\ &= \langle v_k, e_i \rangle - \langle v_k, e_i \rangle \\ &= 0. \end{aligned}$$

Thus

$$w_k \perp e_i \quad \text{for every } i < k.$$

Since e_k is a nonzero scalar multiple of w_k ,

$$e_k \perp e_i \quad \text{for every } i < k.$$

Hence

$$e_1, \dots, e_k$$

is orthonormal.

It remains to verify the equality of spans.

From the definition of w_k ,

$$w_k = v_k - \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j.$$

By the induction hypothesis,

$$e_1, \dots, e_{k-1} \in \text{Span}(v_1, \dots, v_{k-1}),$$

so

$$w_k \in \text{Span}(v_1, \dots, v_k).$$

Hence

$$e_k \in \text{Span}(v_1, \dots, v_k),$$

and therefore

$$\text{Span}(e_1, \dots, e_k) \subseteq \text{Span}(v_1, \dots, v_k).$$

Conversely, rearranging the defining equation gives

$$v_k = w_k + \sum_{j=1}^{k-1} \langle v_k, e_j \rangle e_j.$$

Because w_k is a scalar multiple of e_k ,

$$v_k \in \text{Span}(e_1, \dots, e_k).$$

Together with the induction hypothesis, this gives

$$\text{Span}(v_1, \dots, v_k) \subseteq \text{Span}(e_1, \dots, e_k).$$

Thus

$$\text{Span}(v_1, \dots, v_k) = \text{Span}(e_1, \dots, e_k).$$

The result follows by induction.

The construction is worth seeing once explicitly.

Example 7.19 — Gram–Schmidt in $\mathbb{R}^3$

Consider

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

First,

$$\|v_1\| = \sqrt{2},$$

so

$$e_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$$

Next,

$$\langle v_2, e_1 \rangle = \frac{1}{\sqrt{2}}.$$

Therefore

$$\begin{aligned} w_2 &= v_2 - \langle v_2, e_1 \rangle e_1 \\ &= \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{pmatrix}. \end{aligned}$$

Since

$$\|w_2\| = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2},$$

we obtain

$$e_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Thus

$$e_1, e_2$$

is an orthonormal list and

$$\text{Span}(v_1, v_2) = \text{Span}(e_1, e_2).$$

The Gram–Schmidt process immediately gives existence and extension results for orthonormal bases.

Corollary 7.20 — Existence of orthonormal bases

Every finite-dimensional inner product space has an orthonormal basis.

Proof

Choose any basis

$$v_1, \dots, v_n$$

of V . Applying [theorem 7.18](#) produces an orthonormal list

$$e_1, \dots, e_n.$$

Since

$$\text{Span}(e_1, \dots, e_n) = \text{Span}(v_1, \dots, v_n) = V,$$

the resulting list is an orthonormal basis.

Corollary 7.21 — Extension to an orthonormal basis

Let V be finite-dimensional. Every orthonormal list

$$e_1, \dots, e_m$$

in V can be extended to an orthonormal basis of V .

Proof

By [corollary 7.14](#), the list

$$e_1, \dots, e_m$$

is linearly independent.

By the basis extension result, [proposition 3.10](#), we may extend it to a basis

$$e_1, \dots, e_m, v_{m+1}, \dots, v_n$$

of V .

Apply the Gram–Schmidt process to this basis. Because the first m vectors are already orthonormal, Gram–Schmidt leaves them unchanged: for each

$$k \leq m,$$

all projections onto the preceding vectors are zero and

$$\|e_k\| = 1.$$

The resulting orthonormal basis therefore has the form

$$e_1, \dots, e_m, f_{m+1}, \dots, f_n,$$

and hence extends the original orthonormal list.

Remark 7.22 — What Gram–Schmidt preserves

Gram–Schmidt changes the vectors, but it does not change the successive subspaces they generate:

$$\text{Span}(v_1, \dots, v_k) = \text{Span}(e_1, \dots, e_k) \quad \text{for every } k.$$

Thus the procedure does more than produce an orthonormal basis of the same final space. It preserves the entire nested sequence

$$\text{Span}(v_1) \subseteq \text{Span}(v_1, v_2) \subseteq \dots \subseteq \text{Span}(v_1, \dots, v_m).$$

This property will be useful when we study orthogonal complements and orthogonal projections in the next section.

7.3 Orthogonal Complements and Orthogonal Projections

We previously saw that a subspace U of a finite-dimensional vector space V always has a complement:

$$V = U \oplus W.$$

Such a complement is generally not unique.

Once V is equipped with an inner product, however, there is a distinguished choice of complement: the vectors perpendicular to U .

Definition 7.23 — Orthogonal complement

Let U be a subspace of an inner product space V . The **orthogonal complement** of U is

$$U^\perp = \{v \in V : \langle v, u \rangle = 0 \text{ for every } u \in U\}.$$

For example, in $\mathbb{R}^3$, if

$$U = \text{Span} \left(\begin{pmatrix} a \\ b \\ c \end{pmatrix} \right),$$

then

$$U^\perp = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : ax + by + cz = 0 \right\}.$$

Thus the orthogonal complement of a line through the origin is the plane through the origin perpendicular to that line.

Proposition 7.24 — Basic properties of the orthogonal complement

Let U be a subspace of an inner product space V . Then

$$U^\perp$$

is a subspace of V , and

$$U \cap U^\perp = \{0\}.$$

Moreover, if

$$U \subseteq W,$$

then

$$W^\perp \subseteq U^\perp.$$

Proof

First, $0 \in U^\perp$. Suppose

$$v, w \in U^\perp$$

and

$$\alpha, \beta \in \mathbb{K}.$$

For every

$$u \in U,$$

we have

$$\begin{aligned} \langle \alpha v + \beta w, u \rangle &= \alpha \langle v, u \rangle + \beta \langle w, u \rangle \\ &= 0. \end{aligned}$$

Hence

$$\alpha v + \beta w \in U^\perp,$$

so $U^\perp$ is a subspace.

If

$$v \in U \cap U^\perp,$$

then

$$\langle v, v \rangle = 0,$$

and positive definiteness gives

$$v = 0.$$

Thus

$$U \cap U^\perp = \{0\}.$$

Finally, if

$$U \subseteq W$$

and

$$v \in W^\perp,$$

then v is orthogonal to every vector in W , and therefore to every vector in U . Hence

$$v \in U^\perp.$$

Thus

$$W^\perp \subseteq U^\perp.$$

The main structural result is that the orthogonal complement is not merely a complement: it gives a canonical direct-sum decomposition.

Theorem 7.25 — Orthogonal decomposition

Let V be a finite-dimensional inner product space and let

$$U \subseteq V$$

be a subspace. Then

$$V = U \oplus U^\perp.$$

Equivalently, every

$$v \in V$$

can be written uniquely as

$$v = u + w, \quad u \in U, \quad w \in U^\perp.$$

Proof

By [corollary 7.20](#), choose an orthonormal basis

$$e_1, \dots, e_r$$

of U .

For

$$v \in V,$$

define

$$u = \sum_{i=1}^r \langle v, e_i \rangle e_i$$

and

$$w = v - u.$$

Clearly,

$$u \in U.$$

For every

$$j = 1, \dots, r,$$

we have

$$\begin{aligned} \langle w, e_j \rangle &= \left\langle v - \sum_{i=1}^r \langle v, e_i \rangle e_i, e_j \right\rangle \\ &= \langle v, e_j \rangle - \sum_{i=1}^r \langle v, e_i \rangle \langle e_i, e_j \rangle \\ &= 0. \end{aligned}$$

Hence w is orthogonal to every basis vector of U , and therefore to every vector in U . Thus

$$w \in U^\perp.$$

We have shown that

$$V = U + U^\perp.$$

By [proposition 7.24](#),

$$U \cap U^\perp = \{0\}.$$

Therefore, by [corollary 4.10](#),

$$V = U \oplus U^\perp.$$

Uniqueness follows from the direct-sum property.

Remark 7.26 — The orthogonal complement is a canonical complement

Without an inner product, a subspace U may have many different complements W satisfying

$$V = U \oplus W.$$

The inner product singles out one geometrically distinguished complement:

$$U^\perp.$$

Corollary 7.27 — Dimension and double orthogonal complement

Let V be finite-dimensional and let U be a subspace of V . Then

$$\dim U^\perp = \dim V - \dim U$$

and

$$(U^\perp)^\perp = U.$$

Proof

By [theorem 7.25](#),

$$V = U \oplus U^\perp.$$

Hence

$$\dim V = \dim U + \dim U^\perp,$$

which gives

$$\dim U^\perp = \dim V - \dim U.$$

Also, every vector in U is orthogonal to every vector in $U^\perp$, so

$$U \subseteq (U^\perp)^\perp.$$

But

$$\begin{aligned} \dim(U^\perp)^\perp &= \dim V - \dim U^\perp \\ &= \dim U. \end{aligned}$$

Thus the two subspaces have the same dimension, and therefore

$$(U^\perp)^\perp = U.$$

The orthogonal decomposition also gives another proof that orthonormal bases can be extended.

Corollary 7.28 — Extension of an orthonormal basis of a subspace

Let U be a subspace of a finite-dimensional inner product space V . If

$$e_1, \dots, e_r$$

is an orthonormal basis of U , then it can be extended to an orthonormal basis of V .

Proof

By [theorem 7.25](#),

$$V = U \oplus U^\perp.$$

Choose an orthonormal basis

$$f_1, \dots, f_s$$

of $U^\perp$.

Then

$$e_1, \dots, e_r, f_1, \dots, f_s$$

is orthonormal because vectors from U are orthogonal to vectors from $U^\perp$.

Moreover, the list spans

$$U + U^\perp = V.$$

Hence it is an orthonormal basis of V .

The same idea extends naturally to several mutually orthogonal subspaces.

Definition 7.29 — Orthogonal direct sum

Let

$$V_1, \dots, V_m$$

be subspaces of an inner product space V .

They are called **pairwise orthogonal** if

$$\langle v_i, v_j \rangle = 0$$

for every

$$v_i \in V_i, \quad v_j \in V_j, \quad i \neq j.$$

If

$$V = V_1 + \dots + V_m$$

and the subspaces are pairwise orthogonal, we write

$$V = V_1 \perp \dots \perp V_m$$

and call this an **orthogonal direct sum**.

Pairwise orthogonality automatically implies that the sum is direct. Indeed, if

$$v_i = - \sum_{j \neq i} v_j, \quad v_i \in V_i,$$

then taking the inner product with v_i gives

$$\|v_i\|^2 = 0,$$

so

$$v_i = 0.$$

Thus every component in such a decomposition is uniquely determined.

We now use this uniqueness to define projection.

Definition 7.30 — Orthogonal projection

Let V be finite-dimensional and let

$$U \subseteq V$$

be a subspace.

By [theorem 7.25](#), every

$$v \in V$$

can be written uniquely as

$$v = u + w, \quad u \in U, \quad w \in U^\perp.$$

The **orthogonal projection of v onto U** is

$$P_U v := u.$$

Thus

$$P_U : V \rightarrow V$$

extracts the U -component of v .

For a one-dimensional subspace

$$U = \text{Span}(u), \quad u \neq 0,$$

the formula from [proposition 7.8](#) becomes

$$P_U v = \frac{\langle v, u \rangle}{\|u\|^2} u.$$

If u is a unit vector, this simplifies to

$$P_U v = \langle v, u \rangle u.$$

Proposition 7.31 — Algebraic properties of orthogonal projection

Let U be a subspace of a finite-dimensional inner product space V . Then P_U is linear and

$$\operatorname{Im}(P_U) = U, \quad \ker(P_U) = U^\perp.$$

Moreover,

$$P_U^2 = P_U,$$

$$P_U P_{U^\perp} = P_{U^\perp} P_U = 0,$$

and

$$P_U + P_{U^\perp} = I_V.$$

Proof

Write

$$v = u + w, \quad u \in U, \quad w \in U^\perp.$$

Then

$$P_U v = u \quad \text{and} \quad P_{U^\perp} v = w.$$

The uniqueness of the decomposition immediately implies linearity: if

$$v_1 = u_1 + w_1, \quad v_2 = u_2 + w_2,$$

then

$$\alpha v_1 + \beta v_2 = (\alpha u_1 + \beta u_2) + (\alpha w_1 + \beta w_2),$$

so

$$P_U(\alpha v_1 + \beta v_2) = \alpha P_U v_1 + \beta P_U v_2.$$

Because $P_U v \in U$ and every $u \in U$ satisfies

$$P_U u = u,$$

we obtain

$$\operatorname{Im}(P_U) = U.$$

Similarly,

$$P_U v = 0$$

if and only if the U -component of v is zero, that is,

$$v \in U^\perp.$$

Hence

$$\ker(P_U) = U^\perp.$$

Since

$$P_U v \in U,$$

we have

$$P_U(P_U v) = P_U v,$$

and therefore

$$P_U^2 = P_U.$$

Also,

$$P_{U^\perp} v \in U^\perp,$$

so

$$P_U P_{U^\perp} = 0.$$

Similarly,

$$P_{U^\perp} P_U = 0.$$

Finally,

$$v = P_U v + P_{U^\perp} v$$

for every $v \in V$, and hence

$$P_U + P_{U^\perp} = I_V.$$

If we know an orthonormal basis of U , the projection can be written explicitly.

Theorem 7.32 — Projection formula

Let

$$e_1, \dots, e_r$$

be an orthonormal basis of U . Then for every

$$v \in V,$$

$$P_U v = \sum_{i=1}^r \langle v, e_i \rangle e_i.$$

Proof

In the proof of [theorem 7.25](#), we showed that

$$u = \sum_{i=1}^r \langle v, e_i \rangle e_i$$

satisfies

$$u \in U$$

and

$$v - u \in U^\perp.$$

By uniqueness of the orthogonal decomposition,

$$u = P_U v.$$

For Euclidean spaces, the projection formula has a convenient matrix representation.

Proposition 7.33 — Matrix of an orthogonal projection

Let

$$U \subseteq \mathbb{R}^n$$

have orthonormal basis

$$q_1, \dots, q_r,$$

and define

$$Q = \begin{pmatrix} | & & | \\ q_1 & \cdots & q_r \\ | & & | \end{pmatrix}.$$

Then

$$Q^\top Q = I_r$$

and

$$P_U x = Q Q^\top x.$$

Thus the matrix of the orthogonal projection onto U is

$$Q Q^\top.$$

Proof

We have

$$Q^T x = \begin{pmatrix} \langle x, q_1 \rangle \\ \vdots \\ \langle x, q_r \rangle \end{pmatrix}.$$

Therefore

$$QQ^T x = \sum_{i=1}^r \langle x, q_i \rangle q_i.$$

By [theorem 7.32](#), this equals

$$P_U x.$$

In particular,

$$(QQ^T)^T = QQ^T$$

and

$$(QQ^T)^2 = QQ^T.$$

Thus the matrix of an orthogonal projection is symmetric and idempotent.

There is an intrinsic version of this symmetry that will become useful when we study adjoints.

Proposition 7.34 — Symmetry of orthogonal projection

For every

$$\begin{aligned} x, y &\in V, \\ \langle P_U x, y \rangle &= \langle x, P_U y \rangle. \end{aligned}$$

Proof

Write

$$x = u_1 + w_1, \quad y = u_2 + w_2,$$

where

$$u_1, u_2 \in U$$

and

$$w_1, w_2 \in U^\perp.$$

Then

$$P_U x = u_1, \quad P_U y = u_2.$$

Hence

$$\begin{aligned}\langle P_U x, y \rangle &= \langle u_1, u_2 + w_2 \rangle \\ &= \langle u_1, u_2 \rangle,\end{aligned}$$

because

$$u_1 \perp w_2.$$

Similarly,

$$\begin{aligned}\langle x, P_U y \rangle &= \langle u_1 + w_1, u_2 \rangle \\ &= \langle u_1, u_2 \rangle.\end{aligned}$$

Therefore

$$\langle P_U x, y \rangle = \langle x, P_U y \rangle.$$

The orthogonal decomposition also gives a useful geometric inequality.

Theorem 7.35 — Bessel's inequality

Let

$$v_1, \dots, v_m$$

be pairwise orthogonal nonzero vectors in an inner product space V . Then for every

$$y \in V,$$

$$\sum_{k=1}^m \frac{|\langle y, v_k \rangle|^2}{\|v_k\|^2} \leq \|y\|^2.$$

Equality holds if and only if

$$y \in \text{Span}(v_1, \dots, v_m).$$

Proof

Let

$$U = \text{Span}(v_1, \dots, v_m).$$

Because the v_k are pairwise orthogonal,

$$P_U y = \sum_{k=1}^m \frac{\langle y, v_k \rangle}{\|v_k\|^2} v_k.$$

The vectors in this sum are pairwise orthogonal, so

$$\begin{aligned}\|P_U y\|^2 &= \sum_{k=1}^m \left\| \frac{\langle y, v_k \rangle}{\|v_k\|^2} v_k \right\|^2 \\ &= \sum_{k=1}^m \frac{|\langle y, v_k \rangle|^2}{\|v_k\|^2}.\end{aligned}$$

Now

$$y = P_U y + (y - P_U y),$$

where

$$P_U y \in U$$

and

$$y - P_U y \in U^\perp.$$

Hence, by [proposition 7.7](#),

$$\|y\|^2 = \|P_U y\|^2 + \|y - P_U y\|^2 \geq \|P_U y\|^2.$$

Therefore

$$\sum_{k=1}^m \frac{|\langle y, v_k \rangle|^2}{\|v_k\|^2} \leq \|y\|^2.$$

Equality holds if and only if

$$y - P_U y = 0,$$

which is equivalent to

$$y \in U = \text{Span}(v_1, \dots, v_m).$$

Remark 7.36 — Projection as orthogonal decomposition

The central picture of projection is

$$v = P_U v + P_{U^\perp} v,$$

with

$$P_U v \in U, \quad P_{U^\perp} v \in U^\perp.$$

Thus the direct-sum decomposition

$$V = U \oplus U^\perp$$

becomes a geometric decomposition into mutually perpendicular components.

7.4 Adjoints, Self-Adjoint Operators, and the Spectral Theorem

At the end of the previous subsection, we saw that an orthogonal projection satisfies

$$\langle P_U x, y \rangle = \langle x, P_U y \rangle.$$

This suggests a more general question. Given a linear map T , can we transfer T from one side of an inner product to the other?

The resulting transformation is called the adjoint.

Definition 7.37 — Adjoint

Let V and W be finite-dimensional inner product spaces, and let

$$T : V \rightarrow W$$

be linear.

The **adjoint** of T is the linear map

$$T^* : W \rightarrow V$$

satisfying

$$\langle Tv, w \rangle = \langle v, T^*w \rangle$$

for every

$$v \in V, \quad w \in W.$$

The notation T^* should not be confused with the adjugate matrix introduced earlier. The two notions are different.

We first verify that the adjoint always exists and is uniquely determined.

Theorem 7.38 — Existence and uniqueness of the adjoint

Let

$$T : V \rightarrow W$$

be linear between finite-dimensional inner product spaces. Then T has a unique adjoint

$$T^* : W \rightarrow V.$$

Moreover, let

$$\mathcal{E} = (e_1, \dots, e_n)$$

be an orthonormal basis of V and

$$\mathcal{F} = (f_1, \dots, f_m)$$

an orthonormal basis of W . If

$$A = [T]_{\mathcal{F} \leftarrow \mathcal{E}},$$

then

$$[T^*]_{\mathcal{E} \leftarrow \mathcal{F}} = A^*,$$

where

$$A^* := \overline{A}^\top$$

is the conjugate transpose of A .

Proof

Because

$$\mathcal{E} \quad \text{and} \quad \mathcal{F}$$

are orthonormal bases, choose the linear map

$$S : W \rightarrow V$$

whose matrix is

$$[S]_{\mathcal{E} \leftarrow \mathcal{F}} = A^*.$$

Let

$$x = [v]_{\mathcal{E}}, \quad y = [w]_{\mathcal{F}}.$$

Since the bases are orthonormal,

$$\langle v, w \rangle$$

is computed from coordinates by the standard inner product.

Now

$$[Tv]_{\mathcal{F}} = Ax,$$

so

$$\begin{aligned} \langle Tv, w \rangle &= (Ax)^\top \overline{y} \\ &= x^\top A^\top \overline{y}. \end{aligned}$$

On the other hand,

$$[Sw]_{\mathcal{E}} = A^*y,$$

and therefore

$$\begin{aligned} \langle v, Sw \rangle &= x^\top \overline{A^*y} \\ &= x^\top A^\top \overline{y}. \end{aligned}$$

Hence

$$\langle Tv, w \rangle = \langle v, Sw \rangle$$

for every v and w . Thus

$$S = T^*.$$

This proves existence and also gives

$$[T^*]_{\mathcal{E} \leftarrow \mathcal{F}} = A^*.$$

For uniqueness, suppose

$$S_1, S_2 : W \rightarrow V$$

both satisfy the defining identity. Then, for every

$$v \in V, \quad w \in W,$$

$$\langle v, S_1 w \rangle = \langle T v, w \rangle = \langle v, S_2 w \rangle.$$

Hence

$$\langle v, (S_1 - S_2)w \rangle = 0$$

for every $v \in V$.

Taking

$$v = (S_1 - S_2)w$$

gives

$$\|(S_1 - S_2)w\|^2 = 0.$$

Thus

$$S_1 w = S_2 w$$

for every w , so

$$S_1 = S_2.$$

Over $\mathbb{R}$, conjugation has no effect, so the matrix formula simplifies to

$$[T^*]_{\mathcal{E} \leftarrow \mathcal{F}} = [T]_{\mathcal{F} \leftarrow \mathcal{E}}^T.$$

Thus the adjoint is the intrinsic inner-product analogue of the transpose.

Proposition 7.39 — Basic properties of the adjoint

Let

$$S, T : V \rightarrow V$$

be linear and let

$$\alpha \in \mathbb{K}.$$

Then

$$(S + T)^* = S^* + T^*,$$

$$(\alpha T)^* = \bar{\alpha} T^*,$$

$$(ST)^* = T^* S^*,$$

$$(T^*)^* = T,$$

and

$$I_V^* = I_V.$$

Proof

These identities follow from the defining property of the adjoint and its uniqueness.

For example,

$$\begin{aligned}\langle (S + T)v, w \rangle &= \langle Sv, w \rangle + \langle Tv, w \rangle \\ &= \langle v, S^*w \rangle + \langle v, T^*w \rangle \\ &= \langle v, (S^* + T^*)w \rangle.\end{aligned}$$

Hence

$$(S + T)^* = S^* + T^*.$$

Similarly,

$$\begin{aligned}\langle \alpha Tv, w \rangle &= \alpha \langle Tv, w \rangle \\ &= \alpha \langle v, T^*w \rangle \\ &= \langle v, \bar{\alpha} T^*w \rangle,\end{aligned}$$

so

$$(\alpha T)^* = \bar{\alpha} T^*.$$

For the product,

$$\begin{aligned}\langle STv, w \rangle &= \langle Tv, S^*w \rangle \\ &= \langle v, T^*S^*w \rangle.\end{aligned}$$

Therefore

$$(ST)^* = T^* S^*.$$

Finally,

$$\langle T^*v, w \rangle = \overline{\langle w, T^*v \rangle} = \overline{\langle Tw, v \rangle} = \langle v, Tw \rangle,$$

so

$$(T^*)^* = T.$$

The identity

$$I_V^* = I_V$$

follows immediately from

$$\langle I_V v, w \rangle = \langle v, I_V w \rangle.$$

We now focus on operators that coincide with their adjoints.

Definition 7.40 — Self-adjoint operator

Let

$$T : V \rightarrow V$$

be linear on an inner product space.

The operator T is called **self-adjoint** if

$$T^* = T.$$

Equivalently,

$$\langle Tv, w \rangle = \langle v, Tw \rangle$$

for all

$$v, w \in V.$$

Orthogonal projections provide an immediate example. By [proposition 7.34](#),

$$\langle P_U v, w \rangle = \langle v, P_U w \rangle,$$

and hence

$$P_U^* = P_U.$$

The matrix description of self-adjointness is particularly simple when we use an orthonormal basis.

Proposition 7.41 — Matrix of a self-adjoint operator

Let

$$\mathcal{E} = (e_1, \dots, e_n)$$

be an orthonormal basis of V , and let

$$A = [T]_{\mathcal{E} \leftarrow \mathcal{E}}.$$

Then

$$T \text{ is self-adjoint} \iff A^* = A.$$

In particular, over $\mathbb{R}$,

$$T \text{ is self-adjoint} \iff A^\top = A.$$

Proof

By theorem 7.38,

$$[T^*]_{\mathcal{E} \leftarrow \mathcal{E}} = A^*.$$

Therefore

$$T^* = T$$

if and only if

$$A^* = A.$$

Over $\mathbb{R}$,

$$A^* = A^\top,$$

which gives the second statement.

Thus self-adjoint operators on real inner product spaces are precisely the operators represented by symmetric matrices in orthonormal coordinates. Over $\mathbb{C}$, matrices satisfying

$$A^* = A$$

are called **Hermitian**.

A remarkable feature of self-adjoint operators is that their eigenstructure is compatible with the geometry of the inner product.

Theorem 7.42 — Eigenvalues and eigenvectors of self-adjoint operators

Let

$$T : V \rightarrow V$$

be self-adjoint.

Then every eigenvalue of T is real.

Moreover, eigenvectors corresponding to distinct eigenvalues are orthogonal. Equivalently, if

$$\lambda \neq \mu,$$

then

$$E_\lambda(T) \perp E_\mu(T).$$

Proof

Suppose

$$Tv = \lambda v, \quad v \neq 0.$$

Then

$$\begin{aligned}\lambda\|v\|^2 &= \langle \lambda v, v \rangle \\ &= \langle Tv, v \rangle \\ &= \langle v, Tv \rangle \\ &= \langle v, \lambda v \rangle \\ &= \bar{\lambda}\|v\|^2.\end{aligned}$$

Since

$$\|v\|^2 > 0,$$

we obtain

$$\lambda = \bar{\lambda}.$$

Thus λ is real.

Now suppose

$$Tv = \lambda v, \quad Tw = \mu w, \quad \lambda \neq \mu.$$

Since both eigenvalues are real,

$$\begin{aligned}\lambda\langle v, w \rangle &= \langle Tv, w \rangle \\ &= \langle v, Tw \rangle \\ &= \mu\langle v, w \rangle.\end{aligned}$$

Hence

$$(\lambda - \mu)\langle v, w \rangle = 0.$$

Since

$$\lambda \neq \mu,$$

it follows that

$$\langle v, w \rangle = 0.$$

Therefore

$$v \perp w.$$

For a general linear transformation, invariance of a subspace does not usually imply invariance of its orthogonal complement. For a self-adjoint operator, however, it does.

Proposition 7.43 — Orthogonal complements of invariant subspaces

Let

$$T : V \rightarrow V$$

be self-adjoint, and let

$$U \subseteq V$$

be invariant under T .

Then

$$U^\perp$$

is also invariant under T .

Moreover, the restrictions

$$T|_U : U \rightarrow U$$

and

$$T|_{U^\perp} : U^\perp \rightarrow U^\perp$$

are self-adjoint on their respective inner product spaces.

Proof

Let

$$w \in U^\perp.$$

To show that

$$Tw \in U^\perp,$$

take any

$$u \in U.$$

Because U is invariant under T ,

$$Tu \in U.$$

Thus

$$\begin{aligned}\langle Tw, u \rangle &= \langle w, Tu \rangle \\ &= 0,\end{aligned}$$

because

$$w \in U^\perp.$$

Hence

$$Tw \in U^\perp.$$

The restrictions are self-adjoint because for vectors in either subspace the identity

$$\langle Tx, y \rangle = \langle x, Ty \rangle$$

is inherited directly from T .

We can now strengthen the diagonalization theorem from the previous section. A general diagonalizable operator has a basis of eigenvectors; a self-adjoint operator has an *orthonormal* basis of eigenvectors.

Theorem 7.44 — Real spectral theorem

Let V be a finite-dimensional real inner product space and let

$$T : V \rightarrow V$$

be linear.

Then the following statements are equivalent:

- (i) T is self-adjoint;
- (ii) V has an orthonormal basis consisting of eigenvectors of T ;
- (iii) T has a diagonal matrix with respect to some orthonormal basis of V .

Proof

The equivalence

$$(ii) \iff (iii)$$

follows from [theorem 6.33](#).

We first prove

$$(i) \implies (ii)$$

by induction on

$$n = \dim V.$$

If

$$n = 1,$$

the result is immediate.

Now suppose

$$n > 1$$

and that the result holds for all real inner product spaces of dimension smaller than n .

Choose an orthonormal basis of V , and let

$$A = [T]_{\mathcal{E} \leftarrow \mathcal{E}}.$$

By [proposition 7.41](#),

$$A^T = A.$$

Regard the same real matrix A as a complex matrix acting on $\mathbb{C}^n$. By [corollary 6.28](#), it has a complex eigenvalue λ .

Because

$$A^* = A,$$

the corresponding operator on $\mathbb{C}^n$ is self-adjoint. Hence, by [theorem 7.42](#),

$$\lambda \in \mathbb{R}.$$

Let

$$z = x + iy \in \mathbb{C}^n, \quad z \neq 0,$$

be an eigenvector corresponding to λ . Since A and λ are real,

$$A(x + iy) = \lambda(x + iy)$$

implies

$$Ax = \lambda x, \quad Ay = \lambda y.$$

At least one of x and y is nonzero. Thus T has a nonzero real eigenvector.

Normalize such an eigenvector and call it

$$e_1.$$

Let

$$U = \text{Span}(e_1).$$

Then U is invariant under T by [proposition 4.22](#).

By [proposition 7.43](#),

$$U^\perp$$

is also invariant under T , and

$$T|_{U^\perp}$$

is self-adjoint.

By [theorem 7.25](#),

$$V = U \oplus U^\perp,$$

and

$$\dim U^\perp = n - 1.$$

By the induction hypothesis, $U^\perp$ has an orthonormal basis

$$e_2, \dots, e_n$$

consisting of eigenvectors of

$$T|_{U^\perp}.$$

Because

$$e_1 \perp U^\perp,$$

the list

$$e_1, e_2, \dots, e_n$$

is an orthonormal basis of V , and every vector in this basis is an eigenvector of T .

Thus

$$(i) \implies (ii).$$

Finally, suppose (iii) holds. Let $\mathcal{E}$ be an orthonormal basis such that

$$[T]_{\mathcal{E} \leftarrow \mathcal{E}} = D$$

is diagonal.

Because D is a real diagonal matrix,

$$D^\top = D.$$

By [proposition 7.41](#),

$$T = T^*.$$

Hence

$$(iii) \implies (i).$$

Therefore all three statements are equivalent.

The matrix form of the spectral theorem is particularly important.

A real square matrix Q is called **orthogonal** if

$$Q^\top Q = I.$$

For a square matrix this is equivalent to

$$Q^{-1} = Q^\top.$$

Its columns therefore form an orthonormal basis of $\mathbb{R}^n$.

Corollary 7.45 — Orthogonal diagonalization of a symmetric matrix

Let

$$A \in \mathbb{R}^{n \times n}.$$

Then

$$A^\top = A$$

if and only if there exist an orthogonal matrix Q and a real diagonal matrix D such that

$$Q^\top A Q = D.$$

Equivalently,

$$A = QDQ^\top.$$

The columns of Q can be chosen to be an orthonormal basis of eigenvectors of A , and the corresponding diagonal entries of D are the eigenvalues of A .

Proof

Suppose first that

$$A^\top = A.$$

Consider

$$T_A : \mathbb{R}^n \rightarrow \mathbb{R}^n, \quad T_A(x) = Ax,$$

with the standard inner product.

By [proposition 7.41](#), T_A is self-adjoint. Hence [theorem 7.44](#) gives an orthonormal basis

$$q_1, \dots, q_n$$

of $\mathbb{R}^n$ consisting of eigenvectors of A .

Let

$$Q = \begin{pmatrix} | & & | \\ q_1 & \cdots & q_n \\ | & & | \end{pmatrix}.$$

Because the columns of Q are orthonormal,

$$Q^\top Q = I_n,$$

so

$$Q^{-1} = Q^\top.$$

In the basis

$$(q_1, \dots, q_n),$$

the matrix of T_A is diagonal. By the change-of-basis formula, [corollary 3.39](#),

$$D = Q^{-1}AQ = Q^\top AQ.$$

Conversely, suppose

$$Q^\top AQ = D,$$

where Q is orthogonal and D is diagonal. Then

$$A = QDQ^\top.$$

Taking transposes gives

$$A^T = QD^TQ^T.$$

Since

$$D^T = D,$$

we obtain

$$A^T = A.$$

Example 7.46 — Orthogonal diagonalization

Consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

We previously found the eigenvalues

$$3 \quad \text{and} \quad 1,$$

with corresponding eigenvectors

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

These eigenvectors are orthogonal. After normalization, define

$$q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Let

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Then

$$Q^T Q = I_2,$$

and

$$Q^T A Q = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus the diagonalization can be performed by an orthonormal change of coordinates.

Remark 7.47 — Why the spectral theorem is stronger than diagonalization

For a general diagonalizable real matrix, we have

$$A = PDP^{-1}$$

for some invertible matrix P .

For a real symmetric matrix, the spectral theorem gives the stronger representation

$$A = QDQ^T,$$

where

$$Q^{-1} = Q^T.$$

Thus the eigenvectors can be chosen not merely to form a basis, but to form an orthonormal basis.

7.5 Quadratic Forms and Congruence

We now turn from linear transformations to a closely related class of functions. A linear transformation is represented by an expression of the form

$$x \mapsto Ax.$$

A quadratic form is represented instead by a scalar-valued expression of the form

$$x \mapsto x^T Ax.$$

Quadratic forms arise naturally whenever the leading variation of an object is second order. They will later provide the language for curvature and second-order conditions in multivariable optimization.

Throughout this subsection, we work over $\mathbb{R}$.

Definition 7.48 — Quadratic form

A **quadratic form** on $\mathbb{R}^n$ is a function

$$q : \mathbb{R}^n \rightarrow \mathbb{R}$$

of the form

$$q(x_1, \dots, x_n) = \sum_{i=1}^n a_{ii}x_i^2 + 2 \sum_{1 \leq i < j \leq n} a_{ij}x_i x_j,$$

where

$$a_{ij} \in \mathbb{R}.$$

Equivalently, if

$$A = (a_{ij}) \in \mathbb{R}^{n \times n}$$

is symmetric, then

$$q(x) = x^\top Ax.$$

The symmetric matrix A is called the **matrix associated with the quadratic form**.

For example, the quadratic form

$$q(x_1, x_2) = 2x_1^2 + 6x_1x_2 + 4x_2^2$$

has associated symmetric matrix

$$A = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix},$$

because

$$\begin{aligned} x^\top Ax &= \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= 2x_1^2 + 6x_1x_2 + 4x_2^2. \end{aligned}$$

Notice that the coefficient of the cross term $x_i x_j$ is twice the corresponding off-diagonal entry of the symmetric matrix.

Proposition 7.49 — Every quadratic form has a unique symmetric matrix

For every quadratic form

$$q : \mathbb{R}^n \rightarrow \mathbb{R},$$

there exists a unique symmetric matrix

$$A \in \mathbb{R}^{n \times n}$$

such that

$$q(x) = x^\top Ax \quad \text{for every } x \in \mathbb{R}^n.$$

Proof

Existence follows directly from [definition 7.48](#): if

$$q(x) = \sum_{i=1}^n a_{ii}x_i^2 + 2 \sum_{i < j} a_{ij}x_i x_j,$$

define the symmetric matrix A by

$$A_{ii} = a_{ii}, \quad A_{ij} = A_{ji} = a_{ij} \quad (i < j).$$

Then

$$q(x) = x^\top A x.$$

For uniqueness, suppose A and B are symmetric and

$$x^\top A x = x^\top B x \quad \text{for every } x.$$

Set

$$C = A - B.$$

Then C is symmetric and

$$x^\top C x = 0 \quad \text{for every } x.$$

Taking

$$x = e_i$$

gives

$$c_{ii} = 0.$$

Taking

$$x = e_i + e_j, \quad i \neq j,$$

gives

$$0 = (e_i + e_j)^\top C (e_i + e_j) = c_{ii} + 2c_{ij} + c_{jj} = 2c_{ij}.$$

Hence

$$c_{ij} = 0$$

for every i, j . Thus

$$C = 0,$$

and therefore

$$A = B.$$

More generally, even if we begin with a matrix that is not symmetric, only its symmetric part contributes to the quadratic expression.

Proposition 7.50 — Only the symmetric part matters

For every

$$B \in \mathbb{R}^{n \times n}$$

and every

$$x \in \mathbb{R}^n,$$
$$x^\top Bx = x^\top \left(\frac{B + B^\top}{2} \right) x.$$

Proof

Write

$$B = \frac{B + B^\top}{2} + \frac{B - B^\top}{2}.$$

The first matrix is symmetric and the second is skew-symmetric.

Let

$$K = \frac{B - B^\top}{2}.$$

Then

$$K^\top = -K.$$

Because $x^\top Kx$ is a scalar,

$$x^\top Kx = (x^\top Kx)^\top = x^\top K^\top x = -x^\top Kx.$$

Hence

$$x^\top Kx = 0.$$

Therefore

$$x^\top Bx = x^\top \left(\frac{B + B^\top}{2} \right) x.$$

Thus there is no loss of generality in representing quadratic forms by symmetric matrices.

There is also a natural bilinear object associated with a quadratic form.

Proposition 7.51 — Associated symmetric bilinear form

Let

$$q(x) = x^\top Ax, \quad A^\top = A.$$

Define

$$b(x, y) = x^\top Ay.$$

Then b is a symmetric bilinear form and

$$q(x) = b(x, x).$$

Moreover,

$$b(x, y) = \frac{1}{2} [q(x + y) - q(x) - q(y)].$$

Thus the quadratic form uniquely determines its associated symmetric bilinear form.

Proof

Bilinearity follows immediately from matrix multiplication.

Because

$$A^\top = A,$$

we have

$$b(y, x) = y^\top Ax = (y^\top Ax)^\top = x^\top A^\top y = x^\top Ay = b(x, y).$$

Thus b is symmetric.

Also,

$$b(x, x) = x^\top Ax = q(x).$$

Finally,

$$\begin{aligned} q(x + y) &= (x + y)^\top A(x + y) \\ &= x^\top Ax + x^\top Ay + y^\top Ax + y^\top Ay \\ &= q(x) + q(y) + 2x^\top Ay, \end{aligned}$$

where symmetry of A gives

$$x^\top Ay = y^\top Ax.$$

Hence

$$b(x, y) = x^\top Ay = \frac{1}{2} [q(x + y) - q(x) - q(y)].$$

We now ask how a quadratic form changes when we change coordinates.

Let

$$x = Cy,$$

where

$$C \in \mathbb{R}^{n \times n}$$

is invertible. Then

$$\begin{aligned} q(x) &= x^\top Ax \\ &= (Cy)^\top A(Cy) \\ &= y^\top C^\top ACy. \end{aligned}$$

Thus the same quadratic form has matrix

$$C^T A C$$

in the new coordinates.

Definition 7.52 — Congruent matrices

Two real symmetric matrices

$$A, B \in \mathbb{R}^{n \times n}$$

are called **congruent** if there exists an invertible matrix

$$C$$

such that

$$B = C^T A C.$$

Congruence is therefore the matrix relation naturally associated with a change of variables in a quadratic form.

Remark 7.53 — Similarity and congruence

It is important to distinguish congruence from the similarity relation used for linear transformations.

For a linear transformation, a change of basis gives

$$A \mapsto C^{-1} A C.$$

For a quadratic form, a change of variables gives

$$A \mapsto C^T A C.$$

Thus

$$\text{linear transformations} \longleftrightarrow \text{similarity},$$

whereas

$$\text{quadratic forms} \longleftrightarrow \text{congruence}.$$

The basic simplification problem for a quadratic form is therefore to find an invertible change of variables that removes all cross terms.

In matrix language, we seek C such that

$$C^T A C$$

is diagonal.

Because the associated matrix A is symmetric, the spectral theorem gives much more: C can be

chosen to be orthogonal.

Theorem 7.54 — Principal-axis theorem for quadratic forms

Let

$$q(x) = x^T A x, \quad A = A^T \in \mathbb{R}^{n \times n}.$$

Let

$$\lambda_1, \dots, \lambda_n$$

be the eigenvalues of A , counted with algebraic multiplicity.

Then there exists an orthogonal matrix Q such that, under the orthogonal change of variables

$$x = Qy,$$

the quadratic form becomes

$$q(x) = \lambda_1 y_1^2 + \dots + \lambda_n y_n^2.$$

Proof

By [corollary 7.45](#), there exists an orthogonal matrix Q such that

$$Q^T A Q = D,$$

where D is diagonal and its diagonal entries are

$$\lambda_1, \dots, \lambda_n.$$

Set

$$x = Qy.$$

Then

$$\begin{aligned} q(x) &= x^T A x \\ &= (Qy)^T A (Qy) \\ &= y^T Q^T A Q y \\ &= y^T D y \\ &= \lambda_1 y_1^2 + \dots + \lambda_n y_n^2. \end{aligned}$$

The theorem gives a geometric interpretation of the eigenvectors of the associated matrix. The columns of Q form an orthonormal basis of eigenvectors of A . These directions are called the **principal axes** of the quadratic form.

In the original coordinates, cross terms such as

$$x_i x_j$$

describe interaction between coordinate directions. In principal-axis coordinates, these cross terms disappear:

$$q(x) = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2.$$

The behavior of the quadratic form is therefore completely separated across mutually orthogonal directions.

Example 7.55 — Removing a cross term by orthogonal diagonalization

Consider

$$q(x_1, x_2) = 2x_1^2 + 2x_1x_2 + 2x_2^2.$$

Its associated matrix is

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

A has eigenvalues

$$3 \quad \text{and} \quad 1$$

with corresponding orthonormal eigenvectors

$$q_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad q_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Thus, with

$$Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

we have

$$Q^T A Q = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

Under the change of variables

$$x = Qy,$$

that is,

$$x_1 = \frac{y_1 + y_2}{\sqrt{2}}, \quad x_2 = \frac{y_1 - y_2}{\sqrt{2}},$$

the quadratic form becomes

$$q = 3y_1^2 + y_2^2.$$

Thus the cross term in the original coordinates disappears after rotating the coordinate system to the eigenvector directions.

Remark 7.56 — Two notions of diagonalization

The same symmetric matrix A appears in two related but conceptually different diagonalization problems.

As a linear transformation, the spectral theorem gives

$$Q^{-1}AQ = Q^{\top}AQ = D.$$

As the matrix of a quadratic form, the change of variables

$$x = Qy$$

gives

$$A \mapsto Q^{\top}AQ = D.$$

These formulas coincide because Q is orthogonal:

$$Q^{-1} = Q^{\top}.$$

For a general change of coordinates, however, similarity and congruence are different operations.

7.6 Definiteness of Quadratic Forms

Once a real quadratic form has been written in principal-axis coordinates, its sign is particularly easy to understand.

Indeed, if

$$A = A^{\top}$$

and the spectral theorem gives

$$Q^{\top}AQ = D,$$

where the diagonal entries of D are

$$\lambda_1, \dots, \lambda_n,$$

then under the orthogonal change of variables

$$x = Qy$$

we have

$$x^{\top}Ax = \lambda_1 y_1^2 + \dots + \lambda_n y_n^2.$$

Thus the signs of the eigenvalues determine whether the quadratic form is positive, negative, or changes sign.

Definition 7.57 — Definiteness of a quadratic form

Let

$$A = A^T \in \mathbb{R}^{n \times n}.$$

The quadratic form

$$x^T A x$$

is called:

(i) **positive definite** if

$$x^T A x > 0 \quad \text{for every } x \neq 0;$$

(ii) **negative definite** if

$$x^T A x < 0 \quad \text{for every } x \neq 0;$$

(iii) **positive semidefinite** if

$$x^T A x \geq 0 \quad \text{for every } x;$$

(iv) **negative semidefinite** if

$$x^T A x \leq 0 \quad \text{for every } x;$$

(v) **indefinite** if there exist vectors $x, y \in \mathbb{R}^n$ such that

$$x^T A x > 0$$

and

$$y^T A y < 0.$$

The symmetric matrix A is called positive definite, negative definite, positive semidefinite, negative semidefinite, or indefinite, respectively.

We use the standard notation

$$A > 0, \quad A \geq 0, \quad A < 0, \quad A \leq 0$$

for positive definite, positive semidefinite, negative definite, and negative semidefinite matrices, respectively.

Example 7.58 — Definite, semidefinite, and indefinite forms

Consider first

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Then

$$x^T A x = 2x_1^2 + 3x_2^2 > 0$$

for every

$$x \neq 0.$$

Thus

$$A > 0.$$

Next, let

$$B = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then

$$x^\top Bx = 2x_1^2 \geq 0$$

for every x , but

$$\begin{pmatrix} 0 & 1 \end{pmatrix} B \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0.$$

Thus

$$B \geq 0$$

but

$$B \not> 0.$$

Finally, let

$$C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Then

$$\begin{pmatrix} 1 & 0 \end{pmatrix} C \begin{pmatrix} 1 \\ 0 \end{pmatrix} = 1 > 0,$$

whereas

$$\begin{pmatrix} 0 & 1 \end{pmatrix} C \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -1 < 0.$$

Hence C is indefinite.

The distinction between positive definite and positive semidefinite is therefore the distinction between strict and weak positivity.

In particular,

$$A \geq 0$$

allows nonzero directions x for which

$$x^\top Ax = 0,$$

whereas

$$A > 0$$

does not.

The spectral theorem gives the fundamental characterization of all these cases.

Theorem 7.59 — Eigenvalue criterion for definiteness

Let

$$A = A^T \in \mathbb{R}^{n \times n},$$

and let

$$\lambda_1, \dots, \lambda_n$$

be the eigenvalues of A , counted with multiplicity.

Then:

(i)

$$A > 0 \iff \lambda_i > 0 \text{ for every } i;$$

(ii)

$$A \geq 0 \iff \lambda_i \geq 0 \text{ for every } i;$$

(iii)

$$A < 0 \iff \lambda_i < 0 \text{ for every } i;$$

(iv)

$$A \leq 0 \iff \lambda_i \leq 0 \text{ for every } i;$$

(v) A is indefinite if and only if A has at least one positive eigenvalue and at least one negative eigenvalue.

Proof

By [corollary 7.45](#), there exists an orthogonal matrix Q such that

$$Q^T A Q = D,$$

where

$$D$$

is diagonal with diagonal entries

$$\lambda_1, \dots, \lambda_n.$$

For every

$$x \in \mathbb{R}^n,$$

write

$$x = Qy.$$

Because Q is invertible,

$$x \neq 0 \iff y \neq 0.$$

Then

$$\begin{aligned} x^\top Ax &= y^\top Q^\top A Q y \\ &= y^\top D y \\ &= \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2. \end{aligned}$$

Suppose first that

$$\lambda_i > 0 \quad \text{for every } i.$$

If

$$y \neq 0,$$

then at least one y_i is nonzero, and therefore

$$\lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 > 0.$$

Hence

$$A > 0.$$

Conversely, suppose

$$A > 0.$$

If v_i is an eigenvector corresponding to λ_i , then

$$A v_i = \lambda_i v_i.$$

Hence

$$v_i^\top A v_i = \lambda_i \|v_i\|^2.$$

Because

$$v_i \neq 0$$

and A is positive definite,

$$\lambda_i \|v_i\|^2 > 0.$$

Thus

$$\lambda_i > 0.$$

This proves (i).

The proof of (ii) is identical with strict inequalities replaced by weak inequalities.

Statements (iii) and (iv) follow by applying (i) and (ii) to $-A$.

Finally, if A has a positive eigenvalue λ with eigenvector v and a negative eigenvalue μ with eigenvector w , then

$$v^\top A v = \lambda \|v\|^2 > 0$$

and

$$w^T A w = \mu \|w\|^2 < 0.$$

Hence A is indefinite.

Conversely, if all eigenvalues were nonnegative, then (ii) would imply that A is positive semidefinite. If all eigenvalues were nonpositive, then (iv) would imply that A is negative semidefinite. Thus an indefinite symmetric matrix must have eigenvalues of both signs.

Corollary 7.60 — Definiteness and invertibility

Let

$$A = A^T \in \mathbb{R}^{n \times n}.$$

If

$$A > 0 \quad \text{or} \quad A < 0,$$

then A is invertible.

If

$$A \geq 0,$$

then

$$A > 0 \iff A \text{ is invertible.}$$

Similarly, if

$$A \leq 0,$$

then

$$A < 0 \iff A \text{ is invertible.}$$

Proof

By [theorem 7.59](#), a positive or negative definite matrix has no zero eigenvalue and is therefore invertible.

If

$$A \geq 0,$$

all its eigenvalues are nonnegative. Thus A is invertible if and only if none of these eigenvalues is zero, which is equivalent to all of them being strictly positive. By [theorem 7.59](#), this is equivalent to

$$A > 0.$$

The negative semidefinite case is analogous.

Positive semidefinite matrices also have a useful property concerning the directions on which the

quadratic form vanishes.

Proposition 7.61 — Zero directions of a positive semidefinite matrix

Let

$$A = A^T \geq 0.$$

Then

$$x^T Ax = 0 \iff Ax = 0.$$

Proof

If

$$Ax = 0,$$

then clearly

$$x^T Ax = 0.$$

Conversely, by the spectral theorem, write

$$A = QDQ^T,$$

where

$$D$$

has nonnegative diagonal entries

$$\lambda_1, \dots, \lambda_n.$$

Set

$$y = Q^T x.$$

Then

$$x^T Ax = \sum_{i=1}^n \lambda_i y_i^2.$$

If

$$x^T Ax = 0,$$

then every term in this sum is nonnegative, so

$$\lambda_i y_i^2 = 0 \quad \text{for every } i.$$

Hence

$$\lambda_i y_i = 0 \quad \text{for every } i,$$

which means

$$Dy = 0.$$

Therefore

$$Ax = QDQ^T x = QDy = 0.$$

The preceding spectral characterization also gives a particularly simple description under congruence.

Corollary 7.62 — Canonical forms for definite and semidefinite matrices

Let

$$A = A^T \in \mathbb{R}^{n \times n}$$

and let

$$r = \text{Rank}(A).$$

Then:

(i)

$$A > 0$$

if and only if A is congruent to I_n ;

(ii)

$$A < 0$$

if and only if A is congruent to $-I_n$;

(iii)

$$A \geq 0$$

if and only if A is congruent to a diagonal matrix whose first r diagonal entries are 1 and whose remaining diagonal entries are 0;

(iv)

$$A \leq 0$$

if and only if A is congruent to a diagonal matrix whose first r diagonal entries are -1 and whose remaining diagonal entries are 0.

Proof

We prove the positive semidefinite case; the others are analogous.

Suppose

$$A \geq 0.$$

By [theorem 7.59](#), all eigenvalues are nonnegative. After reordering them, write

$$\lambda_1, \dots, \lambda_r > 0, \quad \lambda_{r+1} = \dots = \lambda_n = 0.$$

By the spectral theorem,

$$Q^T A Q$$

is diagonal with these eigenvalues on the diagonal.

Now rescale the first r coordinates by

$$\frac{1}{\sqrt{\lambda_1}}, \dots, \frac{1}{\sqrt{\lambda_r}}.$$

The resulting invertible congruence transformation changes the nonzero diagonal entries to 1, giving a diagonal matrix with r ones and $n - r$ zeros.

Conversely, an invertible change of variables does not change whether the expression

$$x^T A x$$

is nonnegative for every x . Hence any matrix congruent to this diagonal matrix is positive semidefinite.

Remark 7.63 — What semidefiniteness means geometrically

For a positive definite matrix,

$$x^T A x > 0$$

in every nonzero direction.

For a positive semidefinite matrix, the expression is still never negative, but it may be flat along a nontrivial subspace. By [proposition 7.61](#), those flat directions are exactly

$$\ker(A).$$

Thus

$$\dim \ker(A)$$

measures the number of directions in which the quadratic form has zero curvature.

7.7 Inertia and the Leading Principal Minor Criterion

In the previous subsection, we classified a symmetric matrix according to the sign of

$$x^T A x.$$

We now ask which aspects of this sign structure are preserved when we make an arbitrary invertible change of variables.

Recall that if

$$x = Cy, \quad C \text{ invertible,}$$

then

$$x^T Ax = y^T C^T AC y.$$

Thus changes of variables correspond to congruence transformations

$$A \mapsto C^T AC.$$

By the principal-axis theorem, a real symmetric matrix can first be orthogonally diagonalized. Rescaling the nonzero coordinates then shows that every real symmetric matrix is congruent to a particularly simple diagonal matrix.

Proposition 7.64 — Canonical form under congruence

Let

$$A = A^T \in \mathbb{R}^{n \times n}.$$

Then there exists an invertible matrix C such that

$$C^T AC = \text{diag}(\underbrace{1, \dots, 1}_p, \underbrace{-1, \dots, -1}_r, \underbrace{0, \dots, 0}_s),$$

where

$$p + r + s = n.$$

Proof

By [corollary 7.45](#), there exists an orthogonal matrix Q such that

$$Q^T AQ = \text{diag}(\lambda_1, \dots, \lambda_n).$$

Reorder the eigenvalues so that the positive eigenvalues come first, followed by the negative eigenvalues and then the zero eigenvalues:

$$\lambda_1, \dots, \lambda_p > 0,$$

$$\lambda_{p+1}, \dots, \lambda_{p+r} < 0,$$

and

$$\lambda_{p+r+1} = \dots = \lambda_n = 0.$$

For every nonzero eigenvalue, rescale the corresponding coordinate by the reciprocal of the

square root of its absolute value. Thus there is an invertible diagonal matrix S such that

$$S^T \operatorname{diag}(\lambda_1, \dots, \lambda_n) S = \operatorname{diag}(\underbrace{1, \dots, 1}_p, \underbrace{-1, \dots, -1}_r, \underbrace{0, \dots, 0}_s).$$

Taking

$$C = QS$$

gives the desired congruence.

The important fact is that the numbers of positive, negative, and zero terms in this canonical form do not depend on the particular congruence transformation used.

Definition 7.65 — Inertia

Let

$$A = A^T \in \mathbb{R}^{n \times n}.$$

The number of positive entries in a congruence canonical form of A is called the **positive inertia index** of A and is denoted by

$$n_+(A).$$

The number of negative entries is called the **negative inertia index** and is denoted by

$$n_-(A).$$

The number of zero entries is denoted by

$$n_0(A).$$

The triple

$$(n_+(A), n_-(A), n_0(A))$$

is called the **inertia** of A .

The difference

$$n_+(A) - n_-(A)$$

is sometimes called the **signature** of A .

Because the canonical form can be obtained from the eigenvalue decomposition,

$$n_+(A), \quad n_-(A), \quad n_0(A)$$

are precisely the numbers of positive, negative, and zero eigenvalues of A , counted with multiplicity.

We now show that these numbers really are invariants under congruence.

Theorem 7.66 — Sylvester's law of inertia

Let

$$A, B \in \mathbb{R}^{n \times n}$$

be symmetric matrices. If A and B are congruent, then

$$n_+(A) = n_+(B),$$

$$n_-(A) = n_-(B),$$

and

$$n_0(A) = n_0(B).$$

Thus the numbers of positive, negative, and zero squares in the canonical form are independent of the change of variables used to obtain that form.

Proof

Suppose

$$B = C^T A C$$

for some invertible matrix C . Then

$$y^T B y = (C y)^T A (C y).$$

Thus the invertible map

$$y \mapsto C y$$

preserves the sign of the corresponding quadratic expressions.

We first characterize the positive inertia index geometrically.

Suppose A has canonical form

$$D = \text{diag}(\underbrace{1, \dots, 1}_p, \underbrace{-1, \dots, -1}_r, \underbrace{0, \dots, 0}_s).$$

On the p -dimensional subspace

$$U_+ = \text{Span}(e_1, \dots, e_p),$$

we have

$$y^T D y = y_1^2 + \dots + y_p^2 > 0$$

for every nonzero

$$y \in U_+.$$

Hence there exists a p -dimensional subspace on which the quadratic expression is positive definite.

We claim that no subspace of dimension greater than p can have this property.

Let W be a subspace such that

$$y^T D y > 0 \quad \text{for every nonzero } y \in W.$$

Consider the projection

$$\pi : W \rightarrow \mathbb{R}^p$$

onto the first p coordinates.

If

$$\pi(y) = 0,$$

then

$$y_1 = \cdots = y_p = 0,$$

and therefore

$$y^T D y = -y_{p+1}^2 - \cdots - y_{p+r}^2 \leq 0.$$

The assumed positivity on W therefore implies

$$y = 0.$$

Hence π is injective on W , so

$$\dim W \leq p.$$

Thus p is intrinsically characterized as the largest possible dimension of a subspace on which the quadratic expression is positive definite. Because an invertible congruence transformation maps subspaces bijectively and preserves the sign of the quadratic expression, this number is invariant under congruence.

Therefore

$$n_+(A) = n_+(B).$$

Applying the same argument to $-A$ and $-B$ gives

$$n_-(A) = n_-(B).$$

Finally,

$$n_0(A) = n - n_+(A) - n_-(A),$$

and similarly for B . Hence

$$n_0(A) = n_0(B).$$

Corollary 7.67 — Congruence classification of real symmetric matrices

Two real symmetric matrices of the same size are congruent if and only if they have the same inertia.

Proof

If the matrices are congruent, they have the same inertia by [theorem 7.66](#).

Conversely, if two symmetric matrices have the same inertia

$$(p, r, s),$$

then by [proposition 7.64](#) both are congruent to

$$\text{diag}(\underbrace{1, \dots, 1}_p, \underbrace{-1, \dots, -1}_r, \underbrace{0, \dots, 0}_s).$$

Hence they are congruent to each other.

Remark 7.68 — Inertia and definiteness

The sign classifications from the previous subsection can now be read directly from the inertia:

$$A > 0 \iff (n_+(A), n_-(A), n_0(A)) = (n, 0, 0),$$

$$A \geq 0 \iff n_-(A) = 0,$$

$$A < 0 \iff (n_+(A), n_-(A), n_0(A)) = (0, n, 0),$$

and

$$A \text{ is indefinite} \iff n_+(A) > 0 \text{ and } n_-(A) > 0.$$

The eigenvalue criterion gives a conceptually transparent test for definiteness. There is also a useful criterion involving only determinants of successively larger submatrices.

Definition 7.69 — Leading principal minors

Let

$$A = (a_{ij}) \in \mathbb{R}^{n \times n}.$$

For

$$k = 1, \dots, n,$$

let

$$A_k = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix}$$

be the upper-left $k \times k$ submatrix of A .

The determinants

$$\Delta_k := \det(A_k), \quad k = 1, \dots, n,$$

are called the **leading principal minors** of A .

Thus

$$\begin{aligned} \Delta_1 &= a_{11}, \\ \Delta_2 &= \det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \end{aligned}$$

and

$$\Delta_n = \det(A).$$

Theorem 7.70 — Sylvester's criterion

Let

$$A = A^\top \in \mathbb{R}^{n \times n}.$$

Then

$$A \succ 0$$

if and only if

$$\Delta_1 > 0, \quad \Delta_2 > 0, \quad \dots, \quad \Delta_n > 0.$$

Proof

We first prove necessity.

Suppose

$$A \succ 0.$$

For every

$$k = 1, \dots, n,$$

consider the leading principal submatrix A_k .

For any nonzero

$$z = \begin{pmatrix} z_1 \\ \vdots \\ z_k \end{pmatrix} \in \mathbb{R}^k,$$

define

$$x = \begin{pmatrix} z_1 \\ \vdots \\ z_k \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^n.$$

Then

$$x^\top A x = z^\top A_k z.$$

Because

$$A > 0$$

and

$$x \neq 0,$$

we have

$$z^\top A_k z > 0.$$

Thus

$$A_k > 0.$$

By [corollary 7.62](#), there exists an invertible matrix B_k such that

$$B_k^\top A_k B_k = I_k.$$

Taking determinants gives

$$\det(B_k)^2 \det(A_k) = 1.$$

Hence

$$\det(A_k) > 0.$$

Therefore

$$\Delta_k > 0 \quad \text{for every } k.$$

We now prove sufficiency by induction on n .

For

$$n = 1,$$

the result is immediate.

Suppose the result holds for symmetric matrices of size $n - 1$, and let

$$A = \begin{pmatrix} A_{n-1} & a \\ a^\top & a_{nn} \end{pmatrix}$$

be a symmetric $n \times n$ matrix satisfying

$$\Delta_1, \dots, \Delta_n > 0.$$

The first $n - 1$ leading principal minors of A_{n-1} are positive. Hence, by the induction hypothesis,

$$A_{n-1} > 0.$$

By [corollary 7.62](#), there exists an invertible matrix B such that

$$B^\top A_{n-1} B = I_{n-1}.$$

Define

$$C = \begin{pmatrix} B & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$C^\top A C = \begin{pmatrix} I_{n-1} & c \\ c^\top & d \end{pmatrix}$$

for some

$$c \in \mathbb{R}^{n-1}, \quad d \in \mathbb{R}.$$

Now define

$$R = \begin{pmatrix} I_{n-1} & -c \\ 0 & 1 \end{pmatrix}.$$

A direct multiplication gives

$$R^\top \begin{pmatrix} I_{n-1} & c \\ c^\top & d \end{pmatrix} R = \begin{pmatrix} I_{n-1} & 0 \\ 0 & d - c^\top c \end{pmatrix}.$$

Hence A is congruent to

$$\text{diag}(1, \dots, 1, \gamma), \quad \gamma = d - c^\top c.$$

Taking determinants,

$$\det(A)$$

and γ have the same sign, because all congruence matrices used above are invertible and

$$\det(C^\top A C) = \det(C)^2 \det(A).$$

Since

$$\Delta_n = \det(A) > 0,$$

we obtain

$$\gamma > 0.$$

Therefore A is congruent to I_n . By [corollary 7.62](#),

$$A > 0.$$

The corresponding criterion for negative definiteness follows immediately.

Corollary 7.71 — Leading principal minor criterion for negative definiteness

Let

$$A = A^T \in \mathbb{R}^{n \times n}.$$

Then

$$A < 0$$

if and only if

$$(-1)^k \Delta_k > 0 \quad \text{for } k = 1, \dots, n.$$

Equivalently,

$$\Delta_1 < 0, \quad \Delta_2 > 0, \quad \Delta_3 < 0, \quad \dots$$

with alternating signs.

Proof

We have

$$A < 0 \iff -A > 0.$$

The k th leading principal minor of $-A$ is

$$\det(-A_k) = (-1)^k \det(A_k) = (-1)^k \Delta_k.$$

The result therefore follows from [theorem 7.70](#).

Remark 7.72 — Leading principal minors and semidefiniteness

The strict criterion in [theorem 7.70](#) does *not* extend to positive semidefiniteness merely by replacing

$$\Delta_k > 0$$

with

$$\Delta_k \geq 0.$$

For example,

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

has

$$\Delta_1 = \Delta_2 = \Delta_3 = 0,$$

yet

$$x^\top Ax = -x_2^2 - x_3^2$$

is not positive semidefinite.

For a real symmetric matrix, positive semidefiniteness is instead equivalent to the nonnegativity of *all* principal minors, not only the leading principal minors.

Example 7.73 — Using Sylvester's criterion

Consider

$$A = \begin{pmatrix} 1 & t & -1 & 0 \\ t & 4 & 2 & 0 \\ -1 & 2 & 4 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Its leading principal minors are

$$\Delta_1 = 1,$$

$$\Delta_2 = \det \begin{pmatrix} 1 & t \\ t & 4 \end{pmatrix} = 4 - t^2,$$

$$\Delta_3 = \det \begin{pmatrix} 1 & t & -1 \\ t & 4 & 2 \\ -1 & 2 & 4 \end{pmatrix} = -4(t-1)(t+2),$$

and

$$\Delta_4 = -12(t-1)(t+2).$$

By [theorem 7.70](#), A is positive definite precisely when all four quantities are positive.

The first condition is automatic. The remaining conditions reduce to

$$4 - t^2 > 0$$

and

$$-4(t - 1)(t + 2) > 0.$$

Thus

$$-2 < t < 1.$$

Remark 7.74 — Eigenvalues versus principal minors

For a real symmetric matrix, we now have two complementary tests for positive definiteness:

$$A > 0 \iff \text{all eigenvalues of } A \text{ are positive,}$$

and

$$A > 0 \iff \Delta_1, \dots, \Delta_n > 0.$$

The eigenvalue criterion follows naturally from the spectral theorem.

Sylvester's criterion avoids computing eigenvalues and reduces the question instead to a sequence of determinant calculations.